



AKTU DSTL

DISCRETE MATHEMATICS

ONE SHOT UNIT-I

SET THEORY & RELATIONS + POSET & LATTICES

COMPLETE REVISION

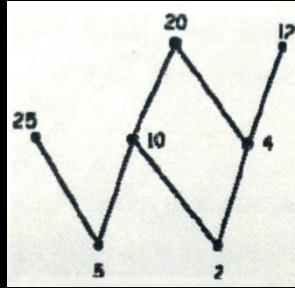


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Section – A

b. Find the Maximal elements and minimal elements form the following Hasse's diagram.



g. Draw the Hasse's diagram of the POSET $(L, \subseteq)$, where

$L = \{S_0, S_1, S_2, S_3, S_4, S_5, S_6, S_7\}$, where the sets are given by

$$\begin{aligned} S_0 &= \{a, b, c, d, e, f\}, & S_1 &= \{a, b, c, d, e\}, & S_2 &= \{a, b, c, e, f\} & S_3 &= \{a, b, c, e\} \\ S_4 &= \{a, b, c\}, & S_5 &= \{a, b\}, & S_6 &= \{a, c\}, & S_7 &= \{a\} \end{aligned}$$

Section – B

a. Identify whether the each of the following relations defined on the set $X = \{1, 2, 3, 4\}$ are reflexive, symmetric, transitive and/or antisymmetric?

(i) $R_1 = \{(1, 1), (1, 2), (2, 1)\}$

(ii) $R_2 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$

(iii) $R_3 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$

Section – C

5. Attempt any one part of the following

(a) Define complemented lattice and then show that in a distributive lattice, if an element has a complement then this complement is unique.

Section – A

- a. Let $A = \{1, 2, 3, 4, 5, 6\}$ be the set and $R = \{(1, 1)(1, 5)(2, 2)(2, 3)(2, 6)(3, 2)(3, 3)(3, 6)(4, 4)(5, 1)(5, 5)(6, 2)(6, 3)(6, 6)\}$ be the relation defined on set A . Find Equivalence classes induced by R
- e. Differentiate complemented lattice and distributed lattice.

Section – C

3. Attempt any one part of the following

(a) State Principle of Duality. Let A denote the set of real numbers and a relation R is defined on A such that $(a, b) R (c, d)$ if and only if $a^2 + b^2 + c^2 + d^2$. Justify that R is an equivalence relation.

7. Attempt any one part of the following

(a) Define Modular Lattice. Justify that if ' a ' and ' b ' are the elements in a bounded distributive lattice and ' a ' has complement a' . then

$$(i) a \vee (a' \wedge b) = a \vee b \quad (ii) a \wedge (a' \vee b) = a \wedge b$$

(b) (i) Justify that $(D_{36}, \leq)$ is lattice.

(ii) Let L_1 be the lattice defined as D_6 and L_2 be the lattice $(P(S), \leq)$, where $P(S)$ be the power set defined on set $S = \{a, b\}$. Justify that the two lattices are isomorphic.

Section – A

b. Let R be a relation on set A with cardinality n . Write down the number of reflexive and symmetric relation on set A .

e. Let $A = \{1, 2, 3, 4, 6, 8, 9, 12, 18, 24\}$ be ordered by the relation ' a divides b '. Find the Hasse diagram .

f. If L be a lattice, then for every a and b in L prove that $a \wedge b = a$ if and only if $a \leq b$.

Section – B

c. Prove that in any lattice the following distributive inequalities hold.

(i) $a \wedge (b \vee c) \geq (a \wedge b) \vee (a \wedge c)$

(ii) $a \vee (b \wedge c) \leq (a \vee b) \wedge (a \vee c)$

Section – A

b. How many symmetric and reflexive relations are possible form a set A containing ' n ' elements?

Section – C

3. Attempt any one part of the following

(b) Is the "divides" relation on the set of positive integers transitive? What is the reflexive and symmetric closure of the relation? $R = \{(a, b) | a > b\}$ on the set of positive integers?

5. Attempt any one part of the following

(a) Let $(L, \vee, \wedge, \leq)$ be a distributive lattice and $a, b \in L$. If $a \wedge b = a \wedge c$ and $a \vee b = a \vee c$ then show that $b = c$.

Section – A

a. Find the power set of each of these sets, where a and b are distinct elements.

- (i) $\{a\}$ (ii) $\{a, b\}$ (iii) $\{\emptyset, \{\emptyset\}$ (iv) $\{a, \{a\}\}$

b. Draw the Hasse diagram for D_{30} .

Section – B

a. Define a lattice. For any a, b, c, d in a lattice $(A, \leq)$ if $a \leq b$ and $c \leq d$ then show that $a \vee c \leq b \vee d$ and $a \wedge c \leq b \wedge d$.

Section – C

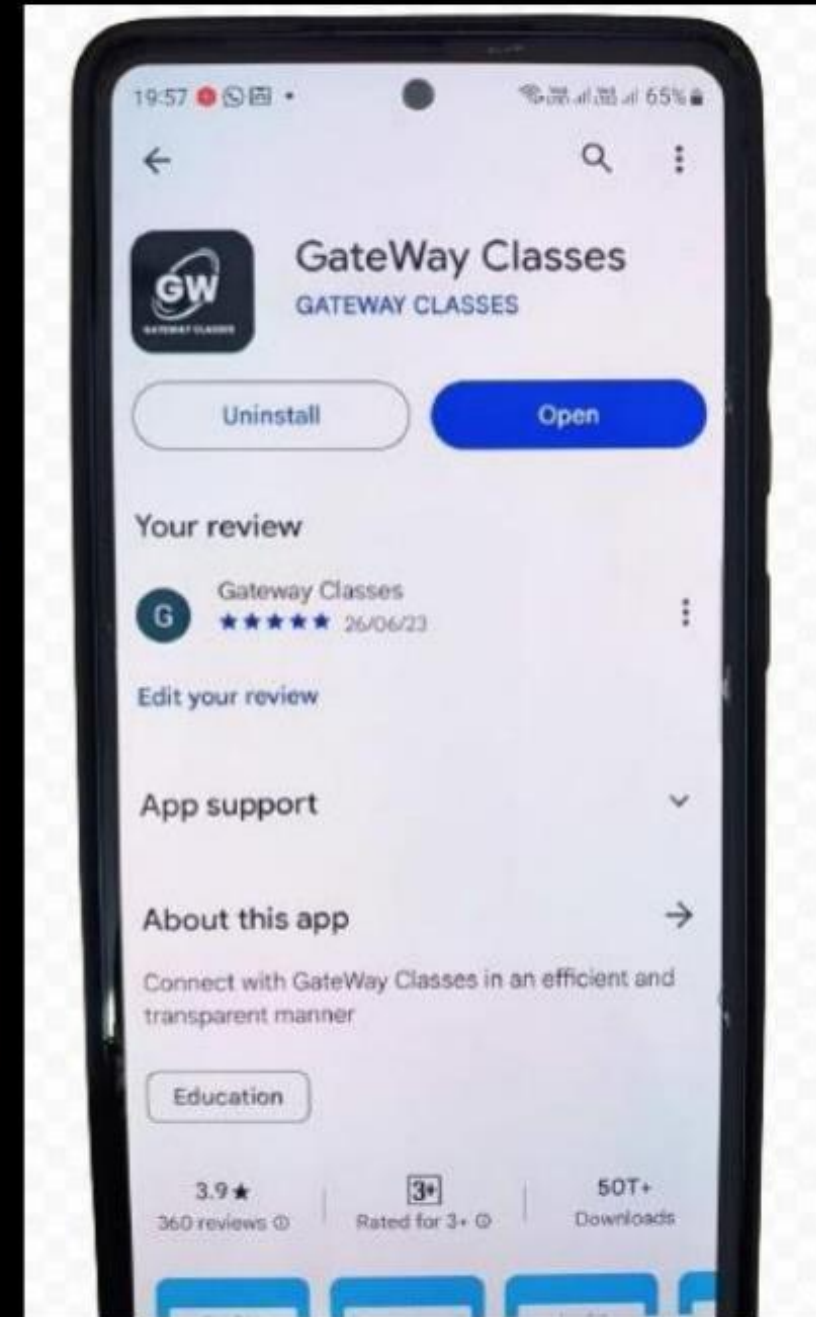
5. Attempt any one part of the following

- (a) Show that the inclusion relation $\subseteq$ is a partial ordering on the power set of a set S . Draw the Hasse diagram for inclusion on the set $P(S)$, where $S = \{a, b, c, d\}$. Also Determine whether $(P(S), \subseteq)$ is a lattice.

**B.Tech -II-Year (4th Sem)
Courses will be available
in
March Last Week**

B.Tech IV-Sem

- CS, IT & CS Allied
- EC & EC Allied
- ME & ME Allied
- EE & EE Allied



DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattices

Lec-1

Today's Target

- What is set?
- Standard sets
- Number system

SET

SET: A Set is a Well-defined collection of objects

Well-defined collection: Collection does not change person to person

Not Well-defined collection: Collection can be change person to person

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$$A = \{a, e, i, o, u\}$$

Name of Set

Elements or member or objects of set

Sets are usually denoted by capital letters, A, B, C, X, Y, Z etc.

The elements of a set are represented by small letters a, b, c, y, z etc.

NOTE

(i) a belongs to A

$$a \in A$$

(ii) b does not belong to A

$$b \notin A$$

Where ϵ = epsilon

Some Standard Sets

N: The set of all natural numbers

W: Set of whole numbers

Z or I: The set of all integers

Q: The set of all rational numbers

R: The set of real numbers

C: The set of all complex number

I⁺ or Z⁺: The set of positive integers

Q⁺: The set of positive rational numbers.

R⁺: The set of positive real numbers.

$$N = \{1, 2, 3, 4, \dots\}$$

$$W = \{0, 1, 2, 3, \dots\}$$

$$I = \{\dots, -3, -2, -1, 0, 1, 2, \dots\}$$

COMPLEX NUMBER (C)

Real Number

Imaginary Number

$$\sqrt{-1}, \sqrt{-2}, \sqrt{-3}, \dots$$

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations
(ii) POSET & Lattices

Lec-2

Today's Target

- *Representation of Set*
- *Types of sets*

Representation of Set

Roster Form
Or
Tabular Form

Set – builder Form
or
Rule Method

Roster Form or Tabular Form

In roster form, all the elements are being separated by commas and are enclosed with in curly brackets { }

Examples

(1) The set of all natural numbers which divide 42

$$N = \{1, 2, 3, 6, 7, 14, 21, 42\}$$

(2) The set of all prime number less than 20

$$A = \{2, 3, 5, 7, 11, 13, 17, 19\}$$

(3) The set of odd natural number .

$$N = \{1, 3, 5, 7, \dots\}$$

(4) The set of Binary digits.

$$A = \{0, 1\}$$

Note: (i) All infinite sets can not be written in roster form

Example: The set of real number can not be written in roster form.

(ii) In roster form the order in Which the elements of a set are written is not important .

Example:

$$A = \{1, 2, 3\} \quad B = \{2, 1, 3\}$$

Here A and B
are same set

(iii) In roster form, an element is not generally repeated, i.e all elements are taken as distinct

Example:

$$A = \{1, 2, 3\}$$

$$B = \{1, 2, 3, 2, 1\}$$

Here set A and set B
are same

Set – Builder Form

In Set – builder form, all the elements of a set possess a single common property which is not possessed by any element outside the set .

$$N = \{1, 2, 3, \dots\}$$

$$A = \{x : p(x)\}$$

such that

Where $P(x)$ is the property by which $x \in A$

Note:- A vertical bar is also used in place of colon (:)

Examples

(i) The set of all natural number less than 10 .

$$N = \{x : x \in N \text{ and } x < 10\}$$

(ii) The set of Real number .

$$R = \{x : x \text{ is a real number}\}$$

(iii) The set of natural number .

$$N = \{x : x \text{ is a natural number}\}$$

1. Empty Set / Null Set / Void Set

- A Set having no element in it.
- It is denoted by $\{ \}$ or ϕ

Examples

$$\textcircled{1} A = \{x : x \in \mathbb{N}, 4 < x < 5\} = \phi$$

$$\textcircled{2} A = \{x : x^2 = 9, x \in \text{Even number}\} = \phi$$

2. Singleton Set

A Set containing exactly one element is called a singleton set.

Examples

$$\textcircled{1} A = \{5\}$$

$$\textcircled{2} B = \{x : x^2 = 4, x \in \mathbb{N}\} = \{2\}$$

Cardinal Number of a Set

OR

Cardinality of a set

The number of distinct elements containing in any set A is called cardinal number of set A

It is denoted by $n(A)$ or $|A|$

Examples

$$\textcircled{1} \quad A = \{2, 4, 6, 8\}$$

$$n(A) = 4$$

$$\textcircled{2} \quad A = \{\emptyset\}$$

$$n(A) = 1$$

$$B = \{1, 2, 3, 2, 4, 1\}$$

$$n(B) = 4$$

Finite and infinite Set

3. Finite Set - A Set is said to be finite if it contains finite number of elements.

Examples ① $A = \{1, 2, 3\}$

② The set students in a class

4. Infinite Set - A Set is said to be infinite if it contains infinite number of elements.

Examples

(a) The Set of natural number

$$N = \{1, 2, 3, 4, \dots\}$$

(b) The Set of integers

$$Z = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

5. Equal Set : Two Sets A and B are said to be equal, if they have exactly the same elements and we Write

$$A = B$$

Examples

$$A = \{1, 4, 6, 7\}$$

$$B = \{4, 1, 6, 7\}$$

6. Equivalent Set : Two finite Sets A and B are said to be equivalent if

$$n(A) = n(B)$$

Examples

$$A = \{2, 4, 6, 8\}$$

$$B = \{a, b, c, d\}$$

Subset and Super Set

7. Subset : a set A is said to be a subset of set B , if every element of A is also an element of B

$$A \subseteq B$$

Examples

$$A = \{2, 4, 6\}$$

$$B = \{1, 2, 3, 4, 5, 6\}$$

$$A \subseteq B$$

Note: (i) If A is not a subset of B

$$A \not\subseteq B$$

(ii) Null set is the subset of all set

$$\phi \subseteq A$$

(iii) Every set is a subset of itself

$$A \subseteq A$$

The total number of subset of a finite set containing n elements = 2^n

Q.1 Write down all possible subsets of

(i) $A = \{a, b\}$

(ii) $B = \{1, 2, 3\}$

(i) $A = \{a, b\}$

subsets of A

$\{a\}, \{b\}, \{\}, \{a, b\}$

(ii) $B = \{1, 2, 3\}$

subsets of B

$\{\}, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\}$

8. Super Set : If $A \subseteq B$, the B is called a superset of A and we write

$$B \supseteq A$$

Examples

$$A = \{a, b, c\}$$

$$B = \{a, b, c, d\}$$

$$A \subseteq B$$

$$B \supseteq A$$

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9. Proper subset: If $A \subseteq B$ and $A \neq B$, then A is called a proper subset of B and we write

$$A \subset B$$

Examples

$$A = \{1, 2\}$$

$$B = \{1, 2, 3\}$$

Note (1) $N \subset W \subset Z \subset Q \subset R$

(2) Set A is not proper subset of A

The total number of proper subset of a finite set containing n elements $= 2^n - 1$

10. Power Set

➤ The collection of all subsets of a set A is called the power set of A

➤ It is denoted by $P(A)$

In $P(A)$, every element is a set

$$m = 3$$

$$\text{If } n(A) = m \text{ then } n[P(A)] = 2^m$$

Q.1. Write down the power set of $A = \{1, 2, 3\}$

$$P(A) = \left\{ \{\}, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\} \right\}$$

$$n[P(A)] = 8$$

11. Universal Set:- Collection of all elements under consideration is called universal set

Example

$$A = \{1, 2, 3\}$$

$$B = \{2, 3, 4, 5\}$$

$$C = \{4, 5, 6, 7\}$$

$$U = \{1, 2, 3, 4, 5, 6, 7\}$$

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DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations
(ii) POSET & Lattices

Lec- 3

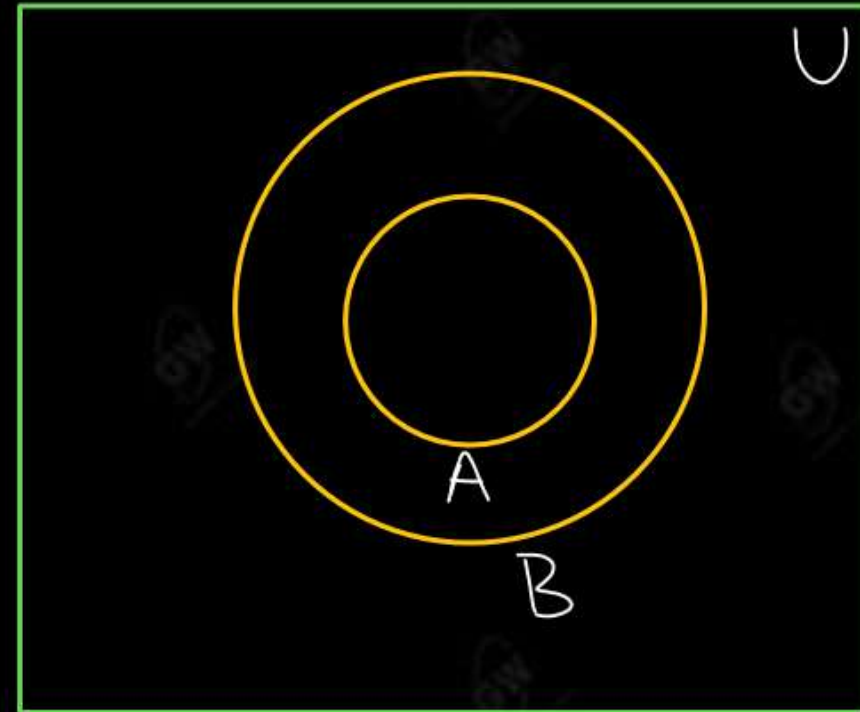
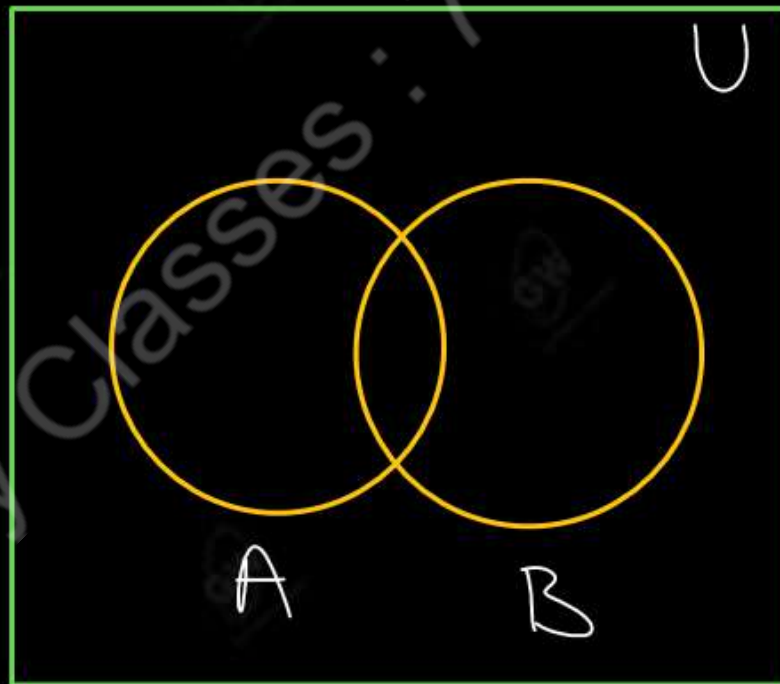
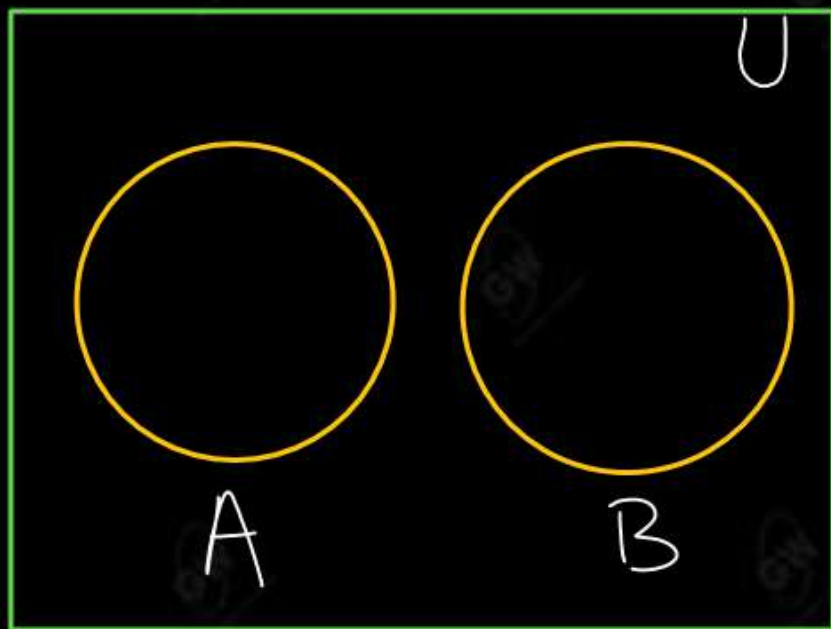
Today's Target

Operations on Sets

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Venn Diagram

- Most of the relationships between sets can be represented by means of diagram which are known as Venn Diagram
- The universal set is represented by the interior of a rectangle and its subsets are represented by circular areas drawn within the rectangle.



1. Union of Set:- The union of two sets A and B is the set of all elements which are either in A or in B or in both A and B .

It is denoted by $A \cup B$

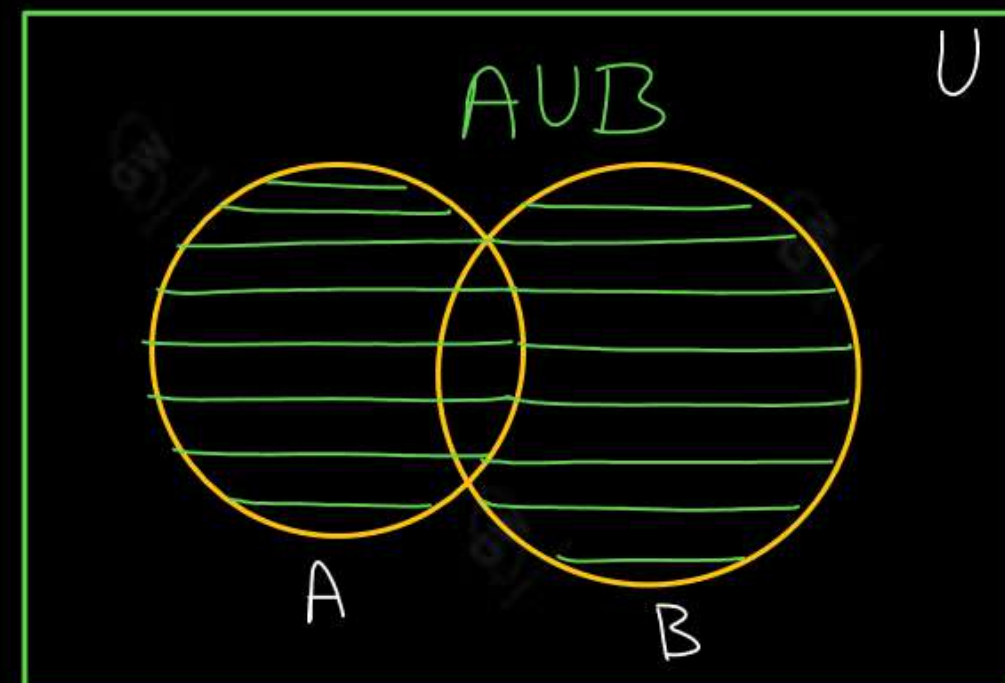
$$A \cup B = \{x: x \in A \text{ or } x \in B\}$$

Note:

(1) The common elements being taken only once

$$(2) x \in A \cup B \Rightarrow x \in A \text{ or } x \in B$$

$$(3) x \notin A \cup B \Rightarrow x \notin A \text{ and } x \notin B$$



Example: Let $A = \{1, 2, 3, 4\}$ and $B = \{2, 4, 6, 8\}$ Find $A \cup B$.

$$A \cup B = \{1, 2, 3, 4, 6, 8\}$$

Example: If $A = \{x : x \text{ is a prime number less than } 15\}$

and $B = \{x : x \in \mathbb{N}, x \text{ is a factor of } 12\}$ Find $A \cup B$.

$$A = \{2, 3, 5, 7, 11, 13\} \text{ and } B = \{1, 2, 3, 4, 6, 12\}$$

$$A \cup B = \{1, 2, 3, 4, 5, 6, 7, 11, 12, 13\}$$

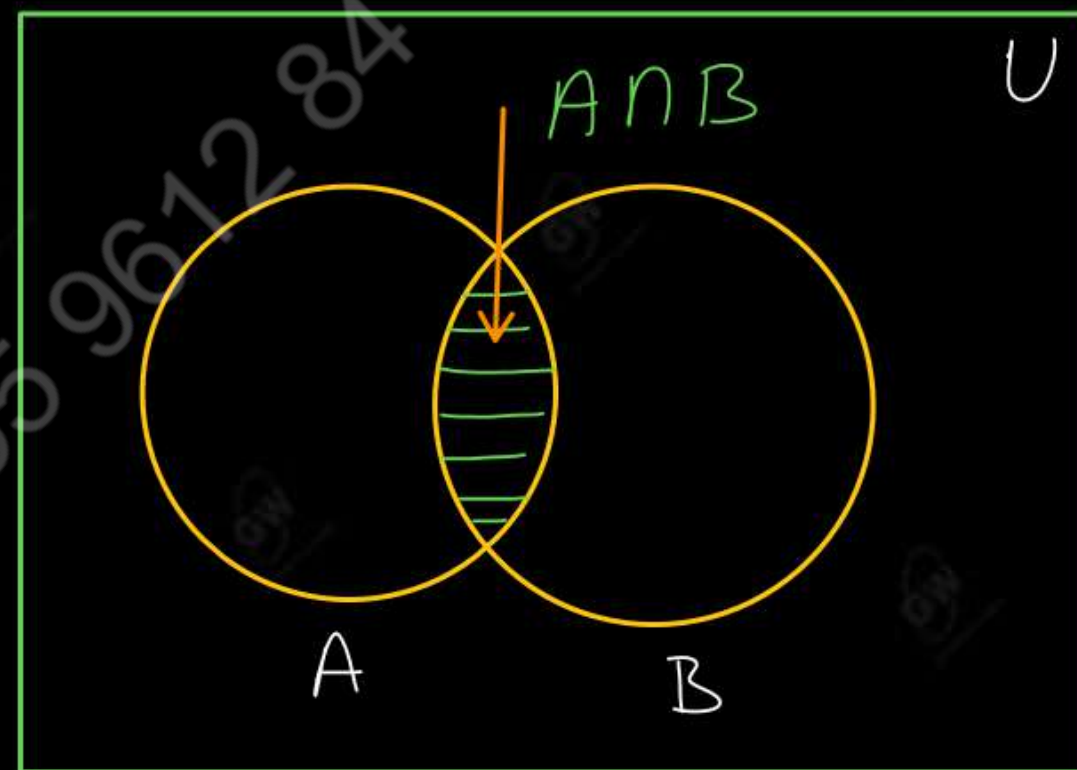
2. Intersection of sets:- The intersection of two sets A and B is the set of all elements which are common to both A and B .

➤ It is denoted by $A \cap B$

➤ $A \cap B = \{x : x \in A \text{ and } x \in B\}$

Note: (1) $x \in A \cap B \Rightarrow x \in A \text{ and } x \in B$

(2) $x \notin A \cap B \Rightarrow x \notin A \text{ or } x \notin B$



Example :- If $A = \{2, 4, 6, 8, 10\}$ and $B = \{1, 2, 3, 4\}$ Find $A \cap B$.

$$A \cap B = \{2, 4\}$$

3. Disjoint Sets:- Two sets A and B are said to be disjoint if

$$A \cap B = \emptyset$$

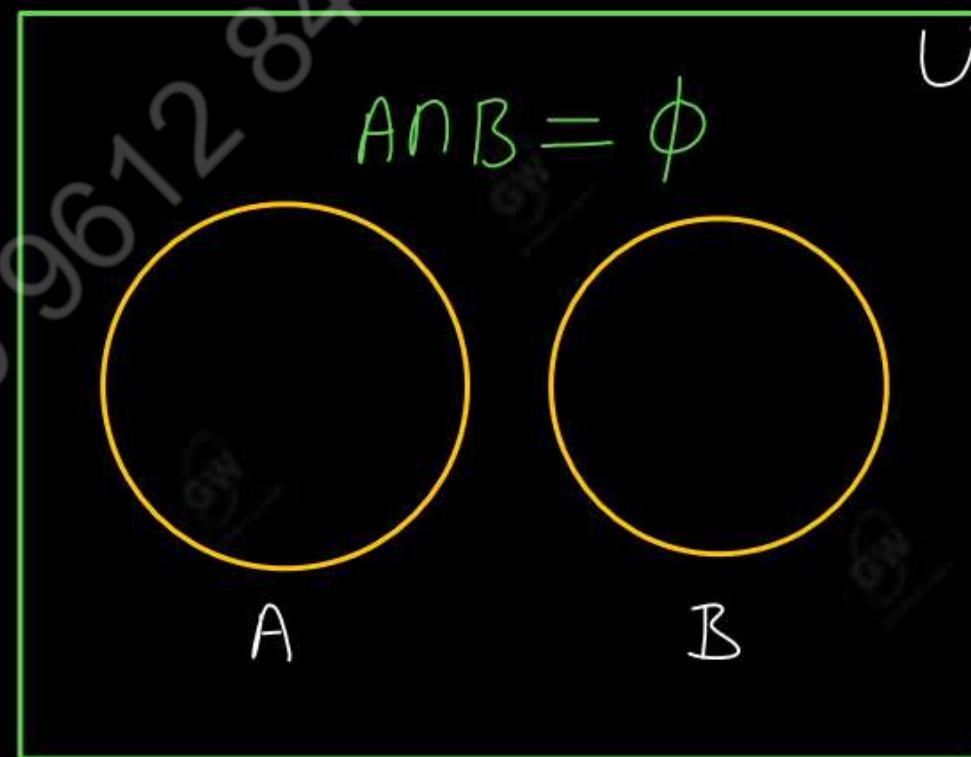
Example :-

If $A = \{2, 4, 6, 8, 10\}$
and $B = \{1, 3, 5, 7, 9\}$

$$A \cap B = \emptyset$$

Here, A and B are
disjoint sets

Null set



4. Difference of Sets : For any two sets A and B

(i) $A - B =$

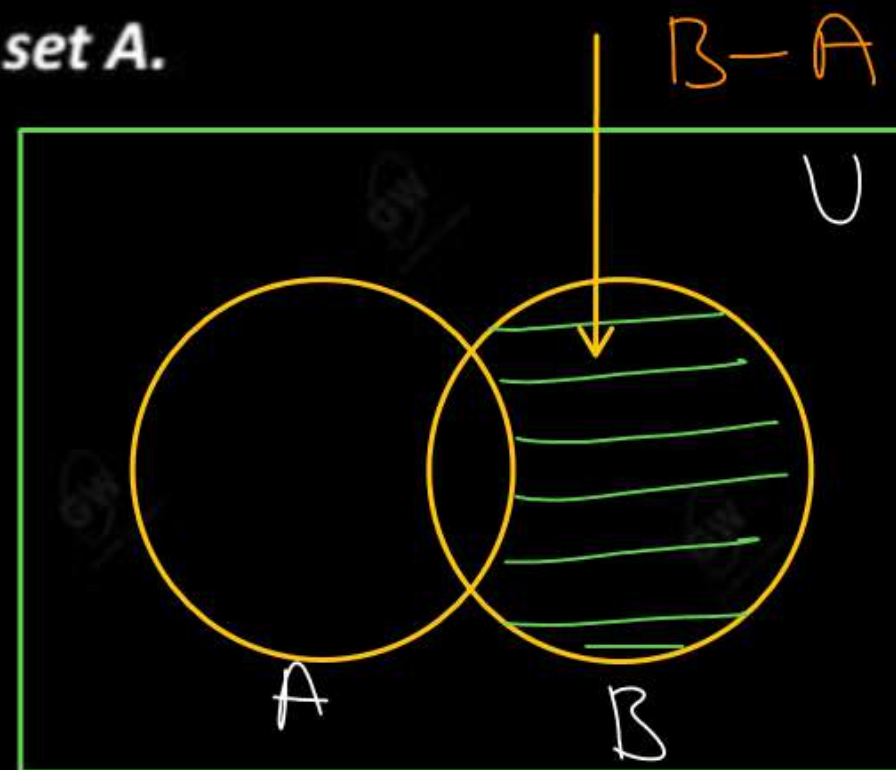
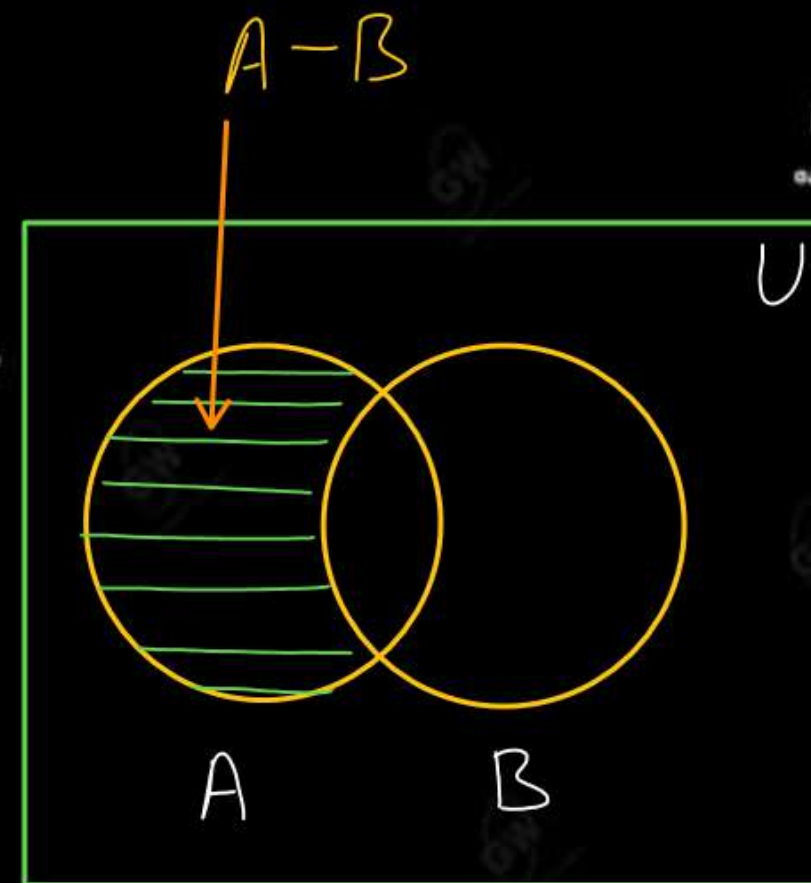
It contains all the elements of set A which are not present in set B.

$$A - B = \{x : x \in A \text{ and } x \notin B\}$$

(ii) $B - A$

It contains all the elements of set B which are not present in set A.

$$B - A = \{x : x \in B \text{ and } x \notin A\}$$



Example:- If $A = \{3, 5, 6, 8, 9\}$ and $B = \{5, 8\}$ Find $A - B$

$$A - B = \{3, 6, 9\}$$

Example :- If $A = \{x : x \in N, x \text{ is a factor of } 12\}$ and

$B = \{x : x \in N, x \text{ is a factor of } 15\}$ then Find (i) $A - B$ (ii) $B - A$.

$$A = \{1, 2, 3, 4, 6, 12\} \text{ and } B = \{1, 3, 5, 15\}$$

$$A - B = \{2, 4, 6, 12\}$$

$$B - A = \{5, 15\}$$

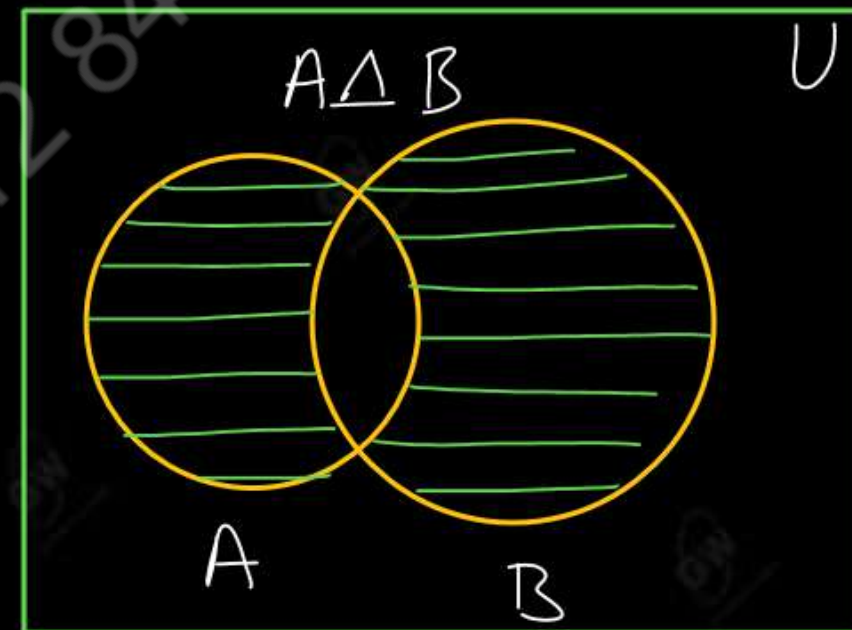
(5) Symmetric difference of sets

The symmetric difference of any two sets A and B is denoted by $A \triangle B$ and defined as

$$A \triangle B = (A - B) \cup (B - A)$$

OR

$$A \triangle B = (A \cup B) - (A \cap B)$$



Example:- If $A = \{1, 2, 3, 6\}$ and $B = \{3, 6, 9, 12\}$, then find $A \triangle B$.

$$A - B = \{1, 2\}$$

$$B - A = \{9, 12\}$$

$$A \triangle B = \{1, 2, 9, 12\}$$

$$A \cup B = \{1, 2, 3, 6, 9, 12\}$$

$$A \cap B = \{3, 6\}$$

$$A \triangle B = \{1, 2, 9, 12\}$$

(6) Complement of set :- Let $\cup$ be the universal set and A be any subset of $\cup$.

The complement of A , denoted by A' or A^c is the set of elements which belongs to $\cup$ but which does not belongs to A .

$$A' = \cup - A$$

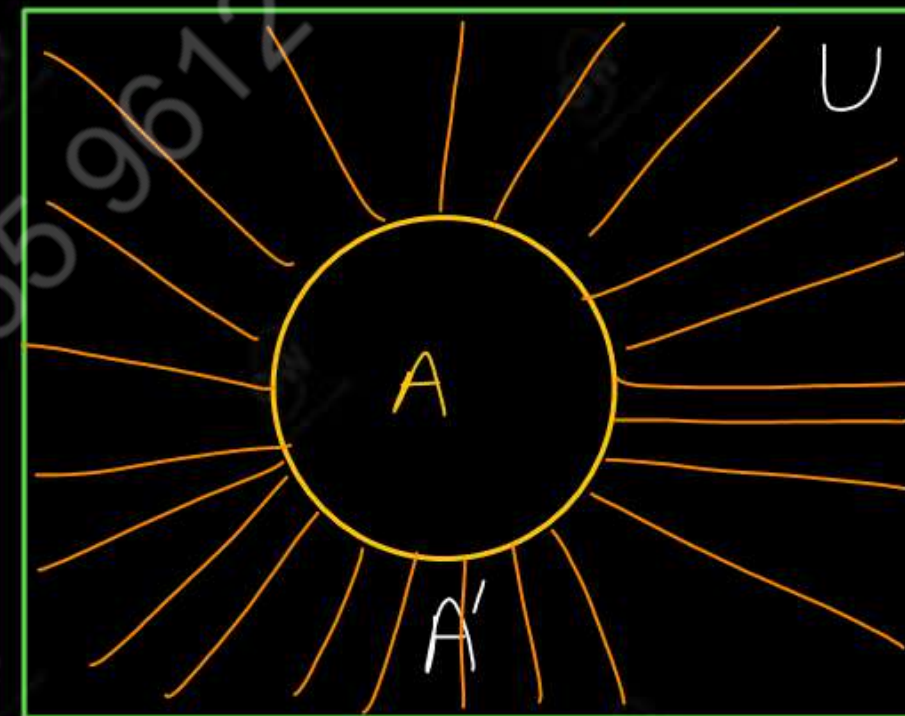
$$A' = \{x : x \in \cup \text{ and } x \notin A\}$$

$$x \in A^c \Rightarrow x \notin A$$

Note:- $A \cap A^c = \emptyset$ ✓

$$A \cup A^c = \cup$$
 ✓

$$(A^c)^c = A$$
 ✓



Example:- If $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ and

$$A = \{1, 3, 5, 7, 9, 12\}$$

Find

(i) A^c

(ii) $A \cap A^c$

(iii) $A \cup A^c$

(iv) $(A^c)^c$

(i) $A^c = \{2, 4, 6, 8, 10, 11\}$

(ii) $A \cap A^c = \phi$

(iii) $A \cup A^c = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

(iv) $(A^c)^c = ?$

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations
(ii) POSET & Lattices

Lec-4

Today's Target

➤ **Algebra of sets**

Revise

Operations on Sets

1. Union of Set:

$$A \cup B = \{x: x \in A \text{ or } x \in B\}$$

Note:

$$(a) x \in A \cup B \Rightarrow x \in A \text{ or } x \in B$$

$$(b) x \notin A \cup B \Rightarrow x \notin A \text{ and } x \notin B$$

2. Intersection of sets:

$$A \cap B = \{x: x \in A \text{ and } x \in B\}$$

Note:

$$(a) x \in A \cap B \Rightarrow x \in A \text{ and } x \in B$$

$$(b) x \notin A \cap B \Rightarrow x \notin A \text{ or } x \notin B$$

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3. Difference of Sets : For any two sets A and B

$$A - B = \{x : x \in A \text{ and } x \notin B\}$$

$$B - A = \{x : x \in B \text{ and } x \notin A\}$$

$$x \in (A - B) = x \in A \text{ and } x \notin B$$

$$x \in (B - A) = x \in B \text{ and } x \notin A$$

4. Symmetric difference of sets

$$A \Delta B = (A - B) \cup (B - A)$$

OR

$$A \Delta B = (A \cup B) - (B \cap A)$$

$$A' \text{ or } A^c$$

5. Complement of set :-

$$A' = U - A$$

$$A' = \{x : x \in U \text{ and } x \notin A\}$$

$$x \in A^c \Rightarrow x \notin A$$

$$x \in A \Rightarrow x \notin A^c$$

Note:- $A \cap A^c = \emptyset$

$$A \cup A^c = U$$

$$(A^c)^c = A$$

1. Idempotent Laws

$$(a) A \cup A = A$$

$$(b) A \cap A = A$$

2. Commutative Laws

$$(a) A \cup B = B \cup A$$

$$(b) A \cap B = B \cap A$$

3. Associative Laws

$$(a) (A \cup B) \cup C = A \cup (B \cup C)$$

$$(b) (A \cap B) \cap C = A \cap (B \cap C)$$

4. Distributive Laws

$$(a) A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$(b) A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

5. DeMorgan's Laws

$$(a) (A \cup B)' = A' \cap B'$$

$$(b) (A \cap B)' = A' \cup B'$$

6. Identity Laws

(a) $A \cup \phi = A$

(b) $A \cap U = A$

(c) $A \cup U = U$

(d) $A \cap \phi = \phi$

$$A = \{1, 2, 3\}$$

$$B = \{1, 2, 3\}$$

$$A \subseteq B$$

$$B \subseteq A$$

$$\boxed{A = B}$$

7. Involution Laws

$(A')' = A$

8. Complement Laws

(a) $A \cup A' = U$

(b) $A \cap A' = \phi$

(a) $U' = \phi$

(b) $\phi' = U$

9. Absorption Laws

(a) $A \cup (A \cap B) = A$

(b) $A \cap (A \cup B) = A$

Methods

✓ ① By definition

② By Venn diagram X

✓ ③ For any two sets A and B

$$A \subseteq B \text{ — ①}$$

$$B \subseteq A \text{ — ②}$$

$$\Rightarrow A = B$$

5. DE Morgan's Laws: For any two sets A and B, Prove that

(a) $(A \cup B)' = A' \cap B'$

Proof:

Let $x \in (A \cup B)'$

$\Rightarrow x \notin (A \cup B)$

$\Rightarrow x \notin A$ and $x \notin B$

$\Rightarrow x \in A'$ and $x \in B'$

$\Rightarrow x \in A' \cap B'$

$\Rightarrow (A \cup B)' \subseteq A' \cap B' \text{ ——— ①}$

conversely if

$x \in A' \cap B'$

$\Rightarrow x \in A'$ and $x \in B'$

$\Rightarrow x \notin A$ or $x \notin B$

$\Rightarrow x \notin (A \cup B)$

$\Rightarrow x \in (A \cup B)'$

$\Rightarrow A' \cap B' \subseteq (A \cup B)' \text{ ——— ②}$

From ① and ②

$(A \cup B)' = A' \cap B'$

(b) $(A \cap B)' = A' \cup B'$

Proof

$$A' \cup B' = \{x : x \in A' \text{ or } x \in B'\}$$

$$= \{x : x \notin A \text{ and } x \notin B\}$$

$$= \{x : x \notin (A \cap B)\}$$

$$= \{x : x \in (A \cap B)'\}$$

$$= (A \cap B)'$$

$$A' \cup B' = (A \cap B)'$$

$$(A \cap B)' = A' \cup B'$$

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations
(ii) POSET & Lattices

Lec-5

Today's Target

- Question Based on Set theory
- PYQ
- DPP

Q.1 Let $A = \{a, b\}$ then calculate $A \cup P(A)$, where $P(A)$ is a power set of A .

$$A = \{a, b\}$$

subset of A

$$\emptyset, \{a\}, \{b\}, \{a, b\}$$

$$P(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$$

$$A \cup P(A) = \{a, b, \emptyset, \{a\}, \{b\}, \{a, b\}\}$$

Q.2 For any set A and B , show that $A - B = A \cap B^c$.

V. Imp

To Prove

$$A - B = A \cap B^c$$

$$\text{Let } x \in (A - B)$$

$$\Rightarrow x \in A \text{ and } x \notin B$$

$$\Rightarrow x \in A \text{ and } x \in B^c$$

$$\Rightarrow x \in A \cap B^c$$

$$\Rightarrow A - B \subseteq A \cap B^c \quad \text{--- (1)}$$

Conversely if

$$x \in A \cap B^c$$

$$\Rightarrow x \in A \text{ and } x \in B^c$$

$$\Rightarrow x \in A \text{ and } x \notin B$$

$$\Rightarrow x \in (A - B)$$

$$\Rightarrow A \cap B^c \subseteq A - B \quad \text{--- (2)}$$

From (1) and (2)

$$A - B = A \cap B^c$$

Q.3 Show that for any two sets A and B : $A - (A \cap B) = A - B$

AKTU-2021-2022

To Prove

$$A - (A \cap B) = A - B$$

LHS

$$A - (A \cap B)$$

$$= A \cap (A \cap B)^c \quad \{A - B = A \cap B^c\}$$

$$= A \cap (A^c \cup B^c)$$

$$= (A \cap A) \cup (A \cap B^c) \quad (\text{By Distributive law})$$

$$= \phi \cup (A \cap B^c)$$

$$= A \cap B^c$$

$$= A - B \quad \{ \because A - B = A \cap B^c \}$$

$$\boxed{LHS = RHS}$$

Hence proved

Q.4 If A, B, C are three sets then show that $A \cup (B - A) = A \cup B$

To Prove

$$A \cup (B - A) = A \cup B$$

LHS

$$A \cup (B - A)$$

$$= A \cup (B \cap A^c) \left\{ \because B - A = B \cap A^c \right\}$$

$$= (A \cup B) \cap (A \cup A^c)$$

$$= (A \cup B) \cap U$$

$$= A \cup B$$

$$\boxed{LHS = RHS}$$

Hence proved

Q.5 If A, B, C are three sets then prove that $A - (B \cup C) = \overline{(A - B)} \cap \overline{(A - C)}$

To Prove

$$A - (B \cup C) = (A - B) \cap (A - C)$$

LHS

$$A - (B \cup C)$$

$$= A \cap (B \cup C)'$$

$$= A \cap (B' \cap C') \quad \left\{ \begin{array}{l} \because A - B = A \cap B' \\ \text{By De Morgan's law} \end{array} \right\}$$

$$= A \cap A \cap (B' \cap C') \quad \left\{ \begin{array}{l} \text{By Idempotent law} \end{array} \right\}$$

$$= (A \cap B') \cap (A \cap C')$$

$$= (A - B) \cap (A - C) \quad \left\{ \because A - B = A \cap B' \right\}$$

$$LHS = RHS$$

Hence proved

Q.6 If A, B, C are three sets then show that $A - (B \cap C) = (A - B) \cup (A - C)$.

AKTU-2021-2022

To Prove

$$A - (B \cap C) = (A - B) \cup (A - C)$$

LHS

$$A - (B \cap C)$$

$$= A \cap (B \cap C)'$$

{ $A - B = A \cap B'$ }

$$= A \cap (B' \cup C')$$

{ By Demorgan's law }

$$= (A \cap B') \cup (A \cap C')$$

{ By distributive law }

$$= (A - B) \cup (A - C)$$

{ $\because A - B = A \cap B'$ }

$$LHS = RHS$$

Hence proved

Q.7 If A, B, C are three sets then show that $(A \cap B) \cup (A - B) = A$

To Prove

$$(A \cap B) \cup (A - B) = A$$

LHS

$$\begin{aligned} & (A \cap B) \cup (A - B) \\ &= (A \cap B) \cup (A \cap B') \quad \{ \because A - B = A \cap B' \} \\ &= A \cap (B \cup B') \quad \{ \text{By Distributive law} \} \\ &= A \cap U \quad \{ \because A \cup A' = U \} \end{aligned}$$

$$\begin{aligned} &= A \cap U \\ &= A \end{aligned}$$

$$\boxed{LHS = RHS}$$

Hence proved

Q.8 Prove that $A \subseteq B$, then $B' \subseteq A'$

Given

$$A \subseteq B$$

$$\text{If } x \in A \implies x \in B$$

$$x \notin B \implies x \notin A$$

$$x \in B' \implies x \in A'$$

$$\implies B' \subseteq A'$$

Hence proved

Q.1 State De Morgan's law and Absorption law **AKTU-2021-22**

Q.2 Prove for any two sets A and B $(A \cup B)' = A' \cap B'$ **AKTU-2014-15**

Q.3 Let $A = \{a, \phi\}$ then calculate $A \cup P(A)$, where $P(A)$ is a power set of A **AKTU-2022-23**

Q.4 List the subsets of the set $A = \{0, 1, 2\}$

Q.5 List the subset of the set $B = \{1, \{2, 3\}\}$

Q.6 If $A = \{1, 2, 3, 4\}$, $B = \{2, 3, 4, 5\}$ and $C = \{3, 4, 5, 6\}$, find $A - B$, $B - C$, $A - (B - C)$, $(A - B) - C$

Q.7 If $A = \{1, 2, 3, 4\}$, $B = \{2, 4, 6, 8\}$ and $C = \{3, 4, 5, 6\}$ find **(i)** $(A \cap B) \cap C$ **(ii)** $A \cap (B \cap C)$

Q.8 Prove that $A \cap (B - C) = (A \cap B) - (A \cap C)$

Q.9 Prove that $A \cap (B - C) = (A \cap B) - (A \cap C)$

Q.10 Prove that $A - (B - C) = (A - B) \cup (A \cap C)$

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattices

Lec-6

Today's Target

- *Ordered Pair*
- *Cartesian product of two sets*
- *Relation or Binary Relation, Relation on a set*
- *Domain and Range of relation*
- *Operation on relation*

Ordered Pair:- An ordered pair consist of two elements grouped together in a particular order

Example :- $(1, 4)$

In the ordered pair (a, b)

(i) a is called first element

(ii) b is called second element

Universal set

Cartesian product of two sets :- The set of all possible ordered pairs (a, b) , where $a \in A$ and $b \in B$ is called the Cartesian product of set A and B and is denoted by $A \times B$

usually

$$A \times B \neq B \times A$$

Note :- If $n(A) = m$ and $n(B) = n$ then

$$n(A \times B) = mn$$

Example :- If $A = \{1, 2\}$ and $B = \{a, b, c\}$ then find

(i) $A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$

$$n(A \times B) = 6$$

(ii) $B \times A = \{(a, 1), (a, 2), (b, 1), (b, 2), (c, 1), (c, 2)\}$

$$n(B \times A) = 6$$

(iii) $A \times A = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$

$$n(A \times A) = 4$$

(iv) $B \times B = \{(a, a), (a, b), (a, c), (b, a), (b, b), (b, c), (c, a), (c, b), (c, c)\}$

$$n(B \times B) = 9$$

Relation or Binary relation :- Let A and B be two non-empty sets, then any subset R of the Cartesian product $A \times B$ is called a relation from set A to set B

$$R \subseteq A \times B$$

$$R = \{(x, y) : x \in A, y \in B \text{ and } x R y\}$$

Note :-

(i) If $(x, y) \in R$ then we say that ' x is related to y ' and we write $x R y$

(ii) If $(x, y) \notin R$ then we say that ' x is not related to y ' and we write $x \not R y$

(iii) If $n(A) = m$ and $n(B) = n$, then $n(A \times B) = mn$

Number of relations from A to B = Total number of subsets of $A \times B = 2^{mn}$

Example : If $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ and $B = \{1, 2, 3, 4, 5\}$ and

$R = \{(a, b) : a \in A, b \in B \text{ and } a = 2b\}$ then find R

Given

$$a = 2b$$

$$R = \{(2, 1), (4, 2), (6, 3), (8, 4)\}$$

Gateway Classes : 7455 9612 84

Relation on a set:- A Relation R from set A to the set A itself is called a relation on set A

$$R \subseteq A \times A$$

$$R = \{(x, y): x \in A, y \in B \text{ and } x R y\}$$

Example :- If $A = \{1, 2, 3\}$. Find relation on A defined by "is less than equal to"

$$A = \{1, 2, 3\}, \quad A = \{1, 2, 3\}$$

$$A \times A = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$$

$$1 = 1 \Rightarrow (1, 1) \in R$$

$$1 < 2 \Rightarrow (1, 2) \in R$$

$$1 < 3 \Rightarrow (1, 3) \in R$$

$$2 \not< 1 \Rightarrow (2, 1) \notin R$$

$$2 = 2 \Rightarrow (2, 2) \in R$$

$$2 < 3 \Rightarrow (2, 3) \in R$$

$$3 \not< 1 \Rightarrow (3, 1) \notin R$$

$$3 \not< 2 \Rightarrow (3, 2) \notin R$$

$$3 = 3 \Rightarrow (3, 3) \in R$$

$$R = \{ (1,1) (1,2) (1,3) (2,2) (2,3) (3,3) \}$$

Gateway Classes : 7455 9612 84

Domain and Range of Relation :-

Let $R = \{(x, y) : x \in A, y \in B \text{ and } x R y\}$

Domain :- The set of all first elements of the ordered pairs belonging to R is called the domain of R .
it is denoted by $\text{Dom}(R)$ or $d(R)$

$$d(R) = \{x : x \in A \text{ and } (x, y) \in R\}$$

Range :- The set of all second elements of the ordered pairs belonging to R is called the range of R .
it is denoted by $\text{Ran}(R)$ or $r(R)$

$$r(R) = \{y : y \in B \text{ and } (x, y) \in R\}$$

Example :- Let $A = \{2, 4, 6\}$ and $B = \{1, 4, 5, 6\}$ then find out the relation from A to B defined by "is less than or equal to". Find out the domain and range of the relation

$$A \times B = \{(2, 1) (2, 4) (2, 5) (2, 6) (4, 1) (4, 4) (4, 5) (4, 6) (6, 1) (6, 4) (6, 5) (6, 6)\}$$

Relation from A to B "is less than or equal to"

$$2 \not\leq 1 \Rightarrow (2, 1) \notin R$$

$$2 < 4 \Rightarrow (2, 4) \in R$$

$$2 < 5 \Rightarrow (2, 5) \in R$$

$$2 < 6 \Rightarrow (2, 6) \in R$$

$$4 \not\leq 1 \Rightarrow (4, 1) \notin R$$

$$4 = 4 \Rightarrow (4, 4) \in R$$

$$4 < 5 \Rightarrow (4, 5) \in R$$

$$4 < 6 \Rightarrow (4, 6) \in R$$

$$6 \not\leq 1 \Rightarrow (6, 1) \notin R$$

$$6 \not\leq 4 \Rightarrow (6, 4) \notin R$$

$$6 \leq 5 \Rightarrow (6, 5) \notin R$$

$$6 = 6 \Rightarrow (6, 6) \in R$$

$$R = \{(2, 4) (2, 5) (2, 6) (4, 4) (4, 5) (4, 6) (6, 6)\}$$

$$d(R) = \{2, 4, 6\}$$

$$r(R) = \{4, 5, 6\}$$

Operation on relation

- (a) Complement of Relation
- (b) Inverse relation
- (5) Intersection and Union of Relation

Complement of a Relation

Consider a relation R from set A to set B . The complement of relation R denoted by $\bar{R}$ or R' is a relation from A to B such that

$$\bar{R} = \{(a, b) : (a, b) \notin R\}$$

Example : Let R be a relation from X to Y , where $X = \{1, 2, 3\}$ and $Y = \{8, 9\}$

and $R = \{(1, 8), (2, 8), (1, 9), (3, 9)\}$ find the complement of relation R

$$R = \{(1, 8), (2, 8), (1, 9), (3, 9)\}$$

$$X \times Y = \left\{ \underset{x}{(1, 8)}, \underset{x}{(1, 9)}, \underset{x}{(2, 8)}, \underset{x}{(2, 9)}, (3, 8), (3, 9) \right\}$$

Universal set

$$\overline{R} = \{(2, 9), (3, 8)\}$$

Gateway Classes : 7455961284

Let R be relation from a set A to B . The inverse of relation R denoted by R^{-1} is a relation from B to A such that

$$R^{-1} = \{(b, a) : (a, b) \in R\}$$

Thus, to find R^{-1} we write in reverse order all ordered pairs belonging to R .

Example: Find the R^{-1} to the relation R on A defined " $x + y$ " divisible by 2.

For $A = \{1, 2, 3, 4, 6\}$.

$$R = \{(1, 1) (1, 3) (2, 2) (2, 4) (2, 6) (3, 1) (3, 3) (4, 2) (4, 4) (4, 6) (6, 2) (6, 4) (6, 6)\}$$

$$R^{-1} = \{(1, 1) (3, 1) (2, 2) (4, 2) (6, 2) (1, 3) (3, 3) (2, 4) (4, 4) (6, 4) (2, 6) (4, 6) (6, 6)\}$$

Intersection and Union of Relation

If R and S are the two relations then intersection of R and S denoted by $R \cap S$ and the union of R and S denoted by $R \cup S$ are two new relation that can be formed from R and S .

Thus $R \cup S = \{(x, y): xRy \text{ or } xSy\}$

$$R \cap S = \{(x, y): xRy \text{ and } xSy\}$$

Example : Let $R = \{(1, 1), (2, 2), (3, 3), (4, 4), (3, 4), (4, 3)\}$

and $S = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 1)\}$. Find $R \cup S$ and $R \cap S$

$$R \cup S = \{(1, 1), (2, 2), (3, 3), (4, 4), (3, 4), (4, 3), (1, 2), (2, 1)\}$$

$$R \cap S = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$$

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattices

Lec-7

Today's Target

➤ **Properties of Relation**

Properties of Relation (Types of Relation)

1. Void Relation:- A Relation R in a set A is called a void relation or an empty relation, if no element of A is related to any element of A and we denote such a relation by ϕ

Thus

$$R = \phi$$

where $\phi \subseteq A \times A$

Example :- Consider a relation R on the set $A = \{1, 2, 3, 4\}$ and defined by $R = \{(a, b) : a - b = 12\}$

$$R = \{ \} = \phi$$

2. Universal Relation:- A Relation R in a set A is called a universal relation, if each element of A is related to every element of A

Thus

$$R = A \times A$$

where $A \times A \subseteq A \times A$

Example :- Consider a relation R on the set $A = \{2, 3, 4\}$ and $R = \{(a, b) : a + b < 9\}$

$$A \times A = \{(2, 2) (2, 3) (2, 4) (3, 2) (3, 3) (3, 4) (4, 2) (4, 3) (4, 4)\}$$

$$R = \{(2, 2) (2, 5) (2, 4) (3, 5) (3, 3) (3, 7) (4, 2) (4, 3) (4, 8)\}$$

3. Identify Relation:- A Relation R in a set A is called an identify relation, if every element of A is related to it self only .

Example :- Consider the relation R on set $A = \{1, 2, 3, 4\}$ and $R = \{(a, b) : a \div b = 1\}$

$$R = \{(1, 1) (2, 2) (3, 3) (4, 4)\}$$

4. Reflexive Relation:- A relation R on set A is reflexive if every element of A is related to itself.

Thus R is Reflexive $\Leftrightarrow (a, a) \in R$ for all $a \in A$

aRa for all $a \in A$

Example : Consider the relation R on the set $A = \{1, 2, 3, 4\}$ and $R = \{(a, b) : a, b \in A, (a \div b) \in \text{Integer}\}$

$$R = \{(1, 1) (2, 2) (3, 3) (4, 4) (2, 1) (3, 1) (4, 1) (4, 2)\}$$

Note :- Every identify relation is reflexive but every reflexive relation is not identify relation.

5. Irreflexive Relation:- A Relation R on Set A is irreflexive if

$$(a, a) \notin R \text{ for all } a \in A$$

Example:- Let $A = \{1, 2\}$ and $R = \{(1, 2), (2, 1)\}$

$\rightarrow a \not R a \quad \forall a \in A$

6. Non-reflexive Relation:- A Relation R on a Set A is non-reflexive if R is neither reflexive nor irreflexive.

Example:- Let $A = \{1, 2\}$ and $R = \{(1, 2), (2, 1), (1, 1)\}$

^{Imp}
7. Symmetric Relation:- A relation R on Set A is symmetric relation if a is related to b then b is also related to a .

Thus R is symmetric if $(a, b) \in R \Rightarrow (b, a) \in R$ for all $a, b \in A$

$$aRb \Rightarrow bRa \quad \forall a, b \in A$$

Example 8 :- Consider the relation R_1 and R_2 on set $A = \{2, 4, 5, 6\}$.

$$R_1 = \{(2, 4), (4, 2), (4, 5), (5, 4), (6, 6)\} \text{ and } R_2 = \{(2, 4), (2, 6), (6, 2), (5, 4), (4, 5)\}$$

For R_1

$$(2, 4) \in R_1 \Rightarrow (4, 2) \in R_1$$

$$(4, 5) \in R_1 \Rightarrow (5, 4) \in R_1$$

$$(6, 6) \in R_1 \Rightarrow (6, 6) \in R_1$$

Hence R_1 is symmetric

For R_2

$$(2, 4) \in R_2 \Rightarrow (4, 2) \notin R_2$$

Hence R_2 is not symmetric

8. Asymmetric Relation:- A Relation R on set A is asymmetric if $(a, b) \in R$ then $(b, a) \notin R$ for $a \neq b$.

Example:- Let $A = \{1, 2, 3\}$ and $R = \{(1, 2), (1, 1), (2, 3), (3, 1)\}$ is a asymmetric relation.

$$(1, 2) \in R \Rightarrow (2, 1) \notin R$$

$$(2, 3) \in R \Rightarrow (3, 2) \notin R$$

$$(3, 1) \in R \Rightarrow (1, 3) \notin R$$

Hence R is Asymmetric Relation

9. Antisymmetric Relation:- A Relation R is said to be antisymmetric if $(a, b) \in R$ and $(b, a) \in R$ for $a = b$

Example:- The relation $R = \{(1, 1), (3, 3)\}$ is Antisymmetric Relation on $A = \{1, 2, 3\}$

$$A \times A = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$$

$$R = \{(1, 1), (3, 3)\}$$

10. Transitive Relation:- The Relation R on set A is transitive relation if a is related to b , b is related to c then a is also related to c

Imp

Thus R is transitive if $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$ for all $a, b, c \in A$

Thus R is transitive if aRb and $bRc \Rightarrow aRc \quad \forall a, b, c \in A$

Example:- Consider a relation R on set $A = \{1, 2, 3, 4\}$ and $R = \{(a, b) : a < b\}$

$$A \times A = \{(1, 1)(1, 2)(1, 3)(1, 4)(2, 1)(2, 2)(2, 3)(2, 4)(3, 1)(3, 2)(3, 3)(3, 4)(4, 1)(4, 2)(4, 3)(4, 4)\}$$

$$R = \{(1, 2)(1, 3)(1, 4)(2, 3)(2, 4)(3, 4)\}$$

$$(1, 2) \in R \text{ and } (2, 3) \in R \Rightarrow (1, 3) \in R$$

$$(1, 2) \in R \text{ and } (2, 4) \in R \Rightarrow (1, 4) \in R$$

$$(1, 3) \in R \text{ and } (3, 4) \in R \Rightarrow (1, 4) \in R$$

$$(2, 3) \in R \text{ and } (3, 4) \in R \Rightarrow (2, 4) \in R$$

Hence R is transitive

VIMP (7) Marks

Equivalence Relation:- A Relation R on set A is said to be an equivalence relation if and only if it is Reflexive, Symmetric and transitive simultaneously.

NOTE: A relation R in a set A is called

- (i) Reflexive if $(a, a) \in R$ for all $a \in A$
- (ii) Symmetric if $(a, b) \in R \Rightarrow (b, a) \in R$ for all $a, b \in A$
- (iii) Transitive if $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$ for all $a, b, c \in A$

DISCRETE STRUCTURES & THEORY OF LOGICS

(Discrete Mathematics)

UNIT -1 : (i) Set Theory & Relations
(ii) POSET & Lattices

Lec-8

Today's Target

- *Equivalence relation*
- PYQ
- DPP

Equality of Relation Or Equivalence Relation

Let A be a non-empty set and R be a relation defined on set A , then R is said to be equivalence relation if it is

(i) **Reflexive:** $(a, a) \in R \quad \forall a \in A$

$$aRa \quad \forall a \in A$$

(ii) **Symmetric:** If $(a, b) \in R \Rightarrow (b, a) \in R \quad \forall a, b \in A$

$$aRb \Rightarrow bRa \quad \forall a, b \in A$$

(iii) **Transitive:** If $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R \quad \forall a, b, c \in A$

$$aRb \text{ and } bRc \Rightarrow aRc \quad \forall a, b, c \in A$$

Equivalence class

Let R be an equivalence relation on a non-empty set A . The equivalence class of an element $a \in A$ is the set of elements of A which are related to ' a '

It is denoted by $[a]$ or $\bar{a}$

✓ $[a] = \text{equivalence class of } a$

$$[a] = \{x : x \in A, x R a\}$$

Gateway Classes :: 7455 9612 84

Q.1. Let $A = \{a, b, c\}$ and let $R = \{(a, a), (b, b), (c, c), (a, b), (b, a)\}$ Where R is clearly an equivalence relation. Find the equivalence classes of the elements of A .

Equivalence classes

$$[a] = \{a, b\}$$

$$[b] = \{a, b\}$$

$$[c] = \{c\}$$

Gateway Classes: 7455 9612 84

Q.2. Let $A = \{0, 1, 2, 3, 4\}$. Show that the relation $R = \{(0, 0), (0, 4), (1, 1), (1, 3), (2, 2), (3, 1), (3, 3), (4, 0), (4, 4)\}$ is an equivalence relation. Also find the distinct equivalence Classes of R .

$$A = \{0, 1, 2, 3, 4\}$$

$$R = \{(0, 0), (1, 1), (2, 2), (3, 3), (4, 4), (0, 4), (1, 3), (3, 1), (4, 0)\}$$

Reflexive:

since $(0, 0) \in R$

$$(1, 1) \in R$$

$$(2, 2) \in R$$

$$(3, 3) \in R$$

$$(4, 4) \in R$$

$$\Rightarrow (a, a) \in R \quad \forall a \in A$$

$$\Rightarrow a R a \quad \forall a \in A$$

so, R is Reflexive

Symmetric

$$\text{since } (0, 4) \in R \Rightarrow (4, 0) \in R$$

$$(1, 3) \in R \Rightarrow (3, 1) \in R$$

$$(3, 1) \in R \Rightarrow (1, 3) \in R$$

$$(4, 0) \in R \Rightarrow (0, 4) \in R$$

$$\Rightarrow (a, b) \in R \Rightarrow (b, a) \quad \forall a, b \in A$$

$$\Rightarrow a R b \Rightarrow b R a \quad \forall a, b \in A$$

so, R is symmetric

Transitive:

$$(0, 4) \in R \text{ and } (4, 0) \in R \Rightarrow (0, 0) \in R$$

$$(0, 4) \in R \text{ and } (4, 4) \in R \Rightarrow (0, 4) \in R$$

$$(1, 3) \in R \text{ and } (3, 1) \in R \Rightarrow (1, 1) \in R$$

$$(1, 3) \in R \text{ and } (3, 3) \in R \Rightarrow (1, 3) \in R$$

$$(4, 0) \in R \text{ and } (4, 4) \in R \Rightarrow (4, 4) \in R$$

$$\Rightarrow (a, b) \in R \text{ and } (b, c) \in R \Rightarrow$$

$$(a, c) \in R \quad \forall a, b, c \in A$$

$$\Rightarrow a R b \text{ and } b R c \Rightarrow a R c \quad \forall a, b, c \in A$$

so, R is transitive

Equivalence classes

$$[0] = \{0, 4\}$$

$$[1] = \{1, 3\}$$

$$[2] = \{2\}$$

$$[3] = \{1, 3\}$$

$$[4] = \{0, 4\}$$

Q.3 Let $X = \{1, 2, 3, \dots, 7\}$ and $R = \{(x, y) : x - y \text{ is divisible by } 3\}$. Show that R is an equivalence relation.

R is an equivalence Relation if

(i) Reflexive: For any $x \in X$

$$x - x = 0, \text{ which is divisible by } 3$$

$$\Rightarrow (x, x) \in R \quad \forall x \in X$$

so, R is Reflexive

(ii) Symmetric: Let $x, y \in X$, such that

$$(x, y) \in R$$

$$\Rightarrow x - y \text{ is divisible by } 3$$

$$\Rightarrow x - y = 3n, \quad n \in \mathbb{Z}$$

$$\Rightarrow -(x - y) = -3n$$

$$\Rightarrow y - x = -3n$$

$$\Rightarrow (y, x) \in R$$

So, R is symmetric

Transitive:

Let $x, y, z \in X$ such that

$$(x, y) \in R \text{ and } (y, z) \in R$$

$\Rightarrow (x - y)$ is divisible by 3 and $(y - z)$ is divisible by 3

$$\Rightarrow x - y = 3n_1 \text{ and } y - z = 3n_2$$

$$\Rightarrow x - \cancel{y} + \cancel{y} - z = 3n_1 + 3n_2$$

$$\Rightarrow x - z = 3(n_1 + n_2)$$

$$\Rightarrow (x, z) \in R$$

So, R is transitive

Hence

R is an equivalence

Relation

Q4: Let $A = \{1, 2, 3, 4, 6, 7, 8, 9\}$ and let R be the relation on $A \times A$ defined by $(a, b) R (c, d)$ if $a + d = b + c$ for all $(a, b), (c, d) \in A \times A$. Prove that

(a) R is an equivalence relation.

(b) Find $[(2, 5)]$.

R is an equivalence Relation if

(1) Reflexive: For any $(a, a) \in A \times A$

$$(a, a) R (a, a) \text{ if } a + a = a + a (\text{true})$$

So, R is Reflexive

Symmetric:

Let $(a, b), (c, d) \in A \times A$ such

that $(a, b) R (c, d)$

$$\Rightarrow a + d = b + c$$

$$\Rightarrow b + c = a + d$$

$$\Rightarrow c + b = d + a$$

$$\Rightarrow (c, d) R (a, b)$$

So, R is symmetric

Transitive:

Let $(a, b), (c, d), (e, f) \in A \times A$ such that

$$(a, b) R (c, d) \text{ and } (c, d) R (e, f)$$

$$\Rightarrow a + d = b + c \text{ and } c + f = d + e$$

$$\Rightarrow a + \cancel{d} + \cancel{c} + f = b + \cancel{c} + \cancel{d} + e$$

$$\Rightarrow a + f = b + e$$

$$\Rightarrow (a, b) R (e, f)$$

$$(a, b) R (c, d)$$

$$a + d = b + c$$

So, R is transitive

Hence

R is an equivalence Relation

Equivalence class

$$[(2, 5)] = \left\{ (1, 4), (2, 5), (3, 6), \right. \\ \left. (4, 7), (5, 8), (6, 9) \right\}$$

Q.5: Let $A = R \times R$ (R be a set of Real numbers) and define the following relation on A

$(a, b) R (c, d) \Leftrightarrow a^2 + b^2 = c^2 + d^2$. Verify that R is an equivalence relation

R is an equivalence Relation if

(i) Reflexive: For any $(a, a) \in A$

$$(a, a) R (a, a) \text{ if } a^2 + a^2 = a^2 + a^2 \text{ (True)}$$

So, R is Reflexive

(ii) Symmetric

Let $(a, b), (c, d) \in A$ such that

$$(a, b) R (c, d)$$

$$\Rightarrow a^2 + b^2 = c^2 + d^2$$

$$\Rightarrow c^2 + d^2 = a^2 + b^2$$

$$\Rightarrow (c, d) R (a, b)$$

So, R is Symmetric

(iii) Transitive:

Let $(a, b), (c, d), (e, f) \in A$

such that

$$(a, b) R (c, d) \text{ and } (c, d) R (e, f)$$

$$\Rightarrow a^2 + b^2 = c^2 + d^2 \text{ and } c^2 + d^2 = e^2 + f^2$$

$$\Rightarrow a^2 + b^2 + \cancel{c^2 + d^2} = \cancel{c^2 + d^2} + e^2 + f^2$$

$$\Rightarrow a^2 + b^2 = e^2 + f^2$$

$$\Rightarrow (a, b) R (e, f)$$

So, R is transitive

Hence, R is an equivalence Relation

NOTE: Two integers a and b are said to be congruence modulo m if $a-b$ is divisible by m and we write $a \equiv b \pmod{m}$

Thus, $a \equiv b \pmod{m} \iff (a-b)$ is divisible by m

Q.6: Let A be the set of all integers and a relation R is defined as

$R = \{(x, y) : x \equiv y \pmod{m}, m \text{ divide } (x - y)\}$ where m is a positive integer.

Prove that R is an equivalence relation.

R is an equivalence Relation if

(I) Reflexive: For any $x \in A$

$x - x = 0$ which is divisible by m

$\Rightarrow x \equiv x \pmod{m}$

$\Rightarrow x R x \forall x \in A$

So, R is Reflexive

(II) Symmetric:

Let $x, y \in A$ such that

$(x, y) \in R$

$\Rightarrow x - y$ is divisible by m

$\Rightarrow x \equiv y \pmod{m}$

$$\Rightarrow x - y = mn$$

$n \in \mathbb{I}$

$$\Rightarrow -(x - y) = -mn$$

$$\Rightarrow (y - x) = -mn$$

$$\Rightarrow (y, x) \in R$$

$$\Rightarrow y \equiv x \pmod{m}$$

So, R is symmetric

Transitive

Let $x, y, z \in A$ such that

$$(x, y) \in R \text{ and } (y, z) \in R$$

$(x - y)$ is divisible by m and $(y - z)$ is divisible by m

$$x \equiv y \pmod{m} \text{ and } y \equiv z \pmod{m}$$

$$\Rightarrow x - y = mn_1 \text{ and } y - z = mn_2$$

$$\Rightarrow \cancel{x} - \cancel{y} + \cancel{y} - z = mn_1 + mn_2$$

$$\Rightarrow x - z = m(n_1 + n_2)$$

$$\Rightarrow (x - z) \text{ is divisible by } m$$

$$\Rightarrow x \equiv z \pmod{m}$$

$$\Rightarrow (x, z) \in R$$

So, R is transitive

Hence

R is an equivalence Relation

Gateway Classes: 7455 9612 84

Topic : Equivalence Relation

Q.1. Let $N = \{1, 2, 3, \dots\}$ and a relation is defined in $N \times N$ as follows: (a, b) is related to (c, d) iff $ad = bc$. Then show that whether R is an equivalence relation or not.

Q.2. Let R be a binary relation defined as $R = \{(a, b) \in \mathbb{R}^2 : (a - b) \leq 3\}$

Determine whether R is an equivalence relation.

Q.3. Let S be the set of all points in a plane. Let R be a relation such that for any two points a and b
 $(a, b) \in R$ is not an equivalence relation.

DISCRETE STRUCTURES & THEORY OF LOGICS

(Discrete Mathematics)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattices

Lec-9

Today's Target

- *Important Theorem on Equivalence relation*
- *Composition of relation*
- *PYQ*
- *DPP*

Equivalence Relation

Let A be a non-empty set and R be a relation defined on set A , then R is said to be equivalence relation if it is

(i) **Reflexive:** $(a, a) \in R \quad \forall a \in A$

i.e. $aRa \quad \forall a \in A$

(ii) **Symmetric:** $(a, b) \in R \Rightarrow (b, a) \in R \quad \forall a, b \in A$

i.e. $aRb \Rightarrow bRa \quad \forall a, b \in A$

(iii) **Transitive:** $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R \quad \forall a, b, c \in A$

i.e. aRb and $bRc \Rightarrow aRc \quad \forall a, b, c \in A$

Theorem-1

If R is an equivalence relation on A , then prove that R^{-1} is also an equivalence relation on A .

Given

Since R is an equivalence relation

- (i) R is Reflexive
- (ii) R is symmetric
- (iii) R is transitive

To Prove

R^{-1} is also an equivalence relation

Proof:

(i) Reflexive: For any $n \in A$

$$(n, n) \in R \quad \left\{ \because R \text{ is reflexive} \right\}$$

$$\Rightarrow (n, n) \in R^{-1} \quad \forall n \in A$$

So, R^{-1} is reflexive

(II) Symmetric:

Let $x, y \in A$ such that

$$(x, y) \in R \Rightarrow (y, x) \in R$$

$\left\{ \because R \text{ is symmetric} \right\}$

$$\Rightarrow (y, x) \in R^{-1} \Rightarrow (x, y) \in R^{-1}$$

$$\Rightarrow (x, y) \in R^{-1} \Rightarrow (y, x) \in R^{-1}$$

So, R^{-1} symmetric

Transitive: Let $x, y, z \in A$ such that

$$(x, y) \in R \text{ and } (y, z) \in R \Rightarrow (x, z) \in R$$

$\left\{ \because R \text{ is transitive} \right\}$

$$(y, x) \in R^{-1} \text{ and } (z, y) \in R^{-1} \Rightarrow (z, x) \in R^{-1}$$

$$(z, y) \in R^{-1} \text{ and } (y, x) \in R^{-1} \Rightarrow (z, x) \in R^{-1}$$

So, R^{-1} is transitive Relation

Hence

R^{-1} is also an equivalence Relation

Theorem-2

If R_1 and R_2 are two equivalence relations on set A , then prove that $R_1 \cap R_2$ is also an equivalence relation.

Given

R_1 and R_2 are equivalence relation

(i) R_1 and R_2 are reflexive

(ii) R_1 and R_2 are symmetric

(iii) R_1 and R_2 are transitive

To Prove

$R_1 \cap R_2$ is also an equivalence Relation

Proof

(i) Reflexive For any $x \in A$

$(x, x) \in R_1$ { $\because R_1$ is reflexive }

$(x, x) \in R_2$ { $\because R_2$ is reflexive }

$\Rightarrow (x, x) \in R_1 \cap R_2 \quad \forall x \in A$

so, R is reflexive

(II) Symmetric: Let $x, y \in A$ such that

$$(x, y) \in R_1 \Rightarrow (y, x) \in R_1 \quad \left\{ R \text{ is symmetric} \right\}$$

$$(x, y) \in R_2 \Rightarrow (y, x) \in R_2 \quad \left\{ \begin{array}{l} \text{"} \\ \text{"} \end{array} \right\}$$

$$\Rightarrow (x, y) \in R_1 \cap R_2 \Rightarrow (y, x) \in R_1 \cap R_2 \quad \forall x, y \in A$$

So, $R_1 \cap R_2$ is symmetric

(III) Transitive:

Let $x, y, z \in A$ such that

$$(x, y) \in R_1 \text{ and } (y, z) \in R_1 \Rightarrow (x, z) \in R_1$$

$$(x, y) \in R_2 \text{ and } (y, z) \in R_2 \Rightarrow (x, z) \in R_2$$

$\left\{ \begin{array}{l} \therefore R_1 \text{ and } R_2 \text{ are transitive} \end{array} \right\}$

$$\Rightarrow (x, y) \in R_1 \cap R_2 \text{ and } (y, z) \in R_1 \cap R_2$$

$$\Rightarrow (x, z) \in R_1 \cap R_2$$

So, $R_1 \cap R_2$ is transitive

Hence, $R_1 \cap R_2$ is also an equivalence relation

If R_1 and R_2 are two equivalence relations on set A , then prove that $R_1 \cup R_2$ is not necessarily an equivalence relation.

Example: Let $A = \{1, 2, 3\}$ and Equivalence relation. $R_1 = \{(1, 1)(2, 2)(3, 3)\}, (1, 2), (2, 1)$ and $R_2 = \{(1, 1)(2, 2)(3, 3)(2, 3), (3, 2)\}$, Show that $R_1 \cup R_2$ is not an equivalence relation.

Proof : $R_1 \cup R_2 = \{(1, 1)(2, 2)(3, 3)(1, 2)(2, 1)(2, 3)(3, 2)\}$

since $(3, 2) \in R_1 \cup R_2$ and $(2, 1) \in (R_1 \cup R_2)$

Bwt $(3, 1) \notin R_1 \cup R_2$

So, $R_1 \cup R_2$ is not an equivalence Relation

Composition of Relations or Composite Relation

Let A , B and C be three non-empty sets.

Let R is a relation from A to B i.e $R \subseteq A \times B$

and S is a relation from B to C . i.e $S \subseteq B \times C$

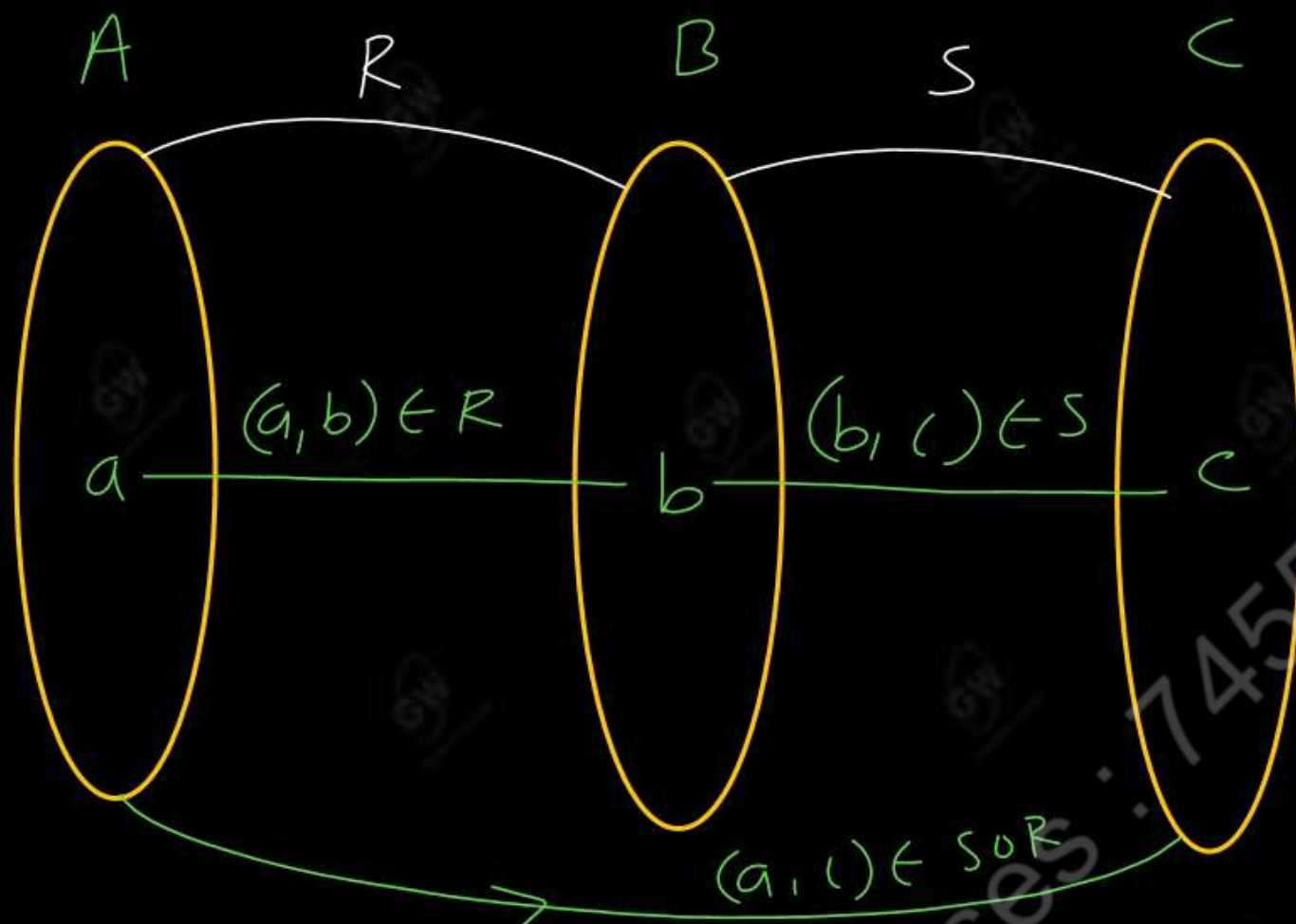
Then the composition of Relation of R and S denoted by SoR , is a relation from A to C and defined as

$$SoR = \{(a, c) \in A \times C : \text{for some } b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in S\}$$

Where $a \in A$, $b \in B$, $c \in C$

Hence we can say

$$(a, b) \in R, (b, c) \in S \Rightarrow (a, c) \in SoR$$



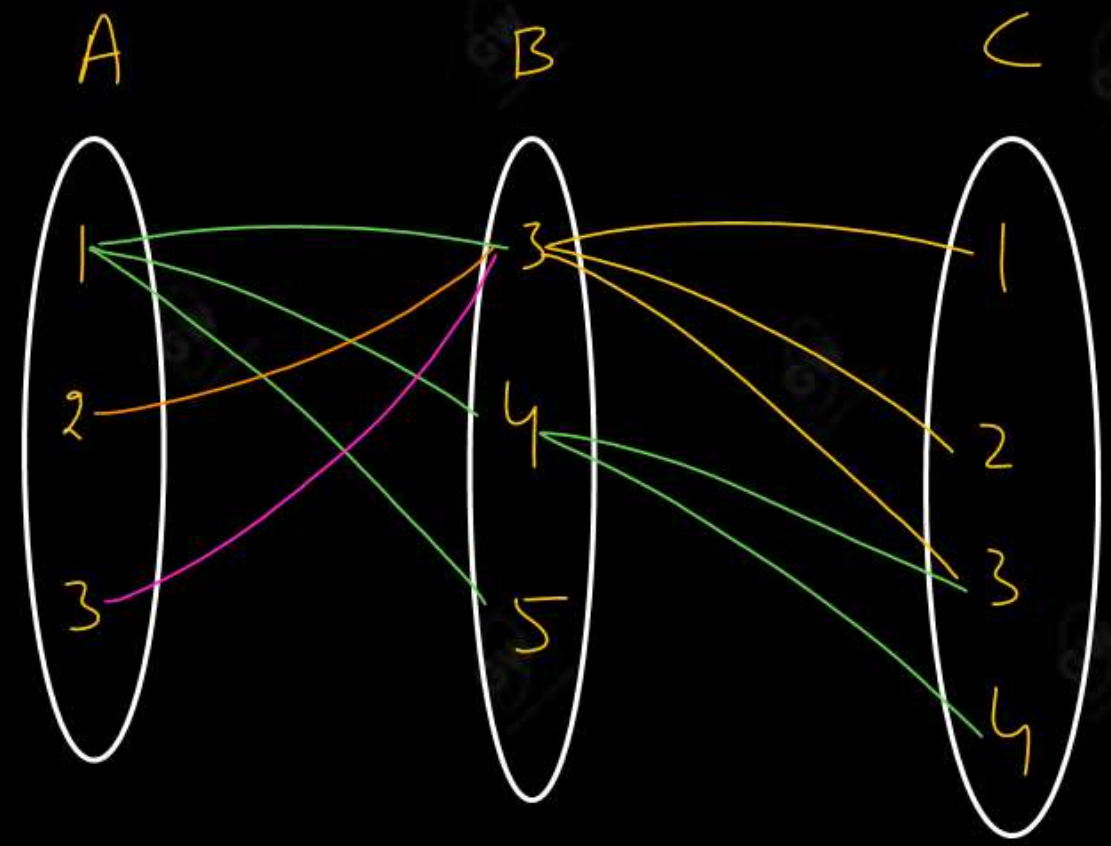
Note: (i) If R and S are relation from A to B and C to D , then SoR is not defined unless $B=C$

(ii) The composition of R with R : $R \circ R = R^2$

(iii) The composition of R^2 with R : $R^2 \circ R = R^3 = R \circ R \circ R$

(iv) The composition of R with SoR : $R \circ (SoR)$

Q.1. Let $A = \{1, 2, 3\}$ $B = \{3, 4, 5\}$, $C = \{1, 2, 3, 4\}$ and Let $R = \{(1, 3) (1, 4) (1, 5) (2, 3) (3, 3)\}$
 $S = \{(3, 1) (3, 2) (3, 3) (4, 3) (4, 4)\}$. compute $S \circ R$, $R \circ S$



$$R = \{(1, 3) (1, 4) (1, 5) (2, 3) (3, 3)\}$$

$$S = \{(3, 1) (3, 2) (3, 3) (4, 3) (4, 4)\}$$

$$S \circ R = \{(1, 1) (1, 2) (1, 3) (1, 4) (2, 1) (2, 2) (2, 3) (3, 1) (3, 2) (3, 3)\}$$

$$S \circ R = \{(1, 1) (1, 2) (1, 3) (1, 4) (2, 1) (2, 2) (2, 3) (3, 1) (3, 2) (3, 3)\}$$

Q.2 Let $R = \{(1, 2)(3, 4)(2, 2)\}$ and $S = \{(4, 2)(2, 5)(3, 1)(1, 3)\}$

Find (i) SoR (ii) RoS (iii) RoR (iv) $RoRoR$ (v) $Ro(SoR)$ (vi) $(RoS)oR$

(i) SoR

$$R = \{(1, 2)(3, 4)(2, 2)\}$$

$$S = \{(4, 2)(2, 5)(3, 1)(1, 3)\}$$

$$SoR = \{(1, 5)(3, 2)(2, 5)\}$$



(iii) RoS

$$S = \{(4, 2)(2, 5)(3, 1)(1, 3)\}$$

$$R = \{(1, 2)(3, 4)(2, 2)\}$$

$$RoS = \{(4, 2)(3, 2)(1, 4)\}$$



(III) $R \circ R$

$$R = \{(1,2)(3,4)(2,2)\}$$

$$R = \{(1,2)(3,4)(2,2)\}$$

$$R \circ R = \{(1,2)(2,2)\}$$

$$R^2 = \{(1,2)(2,2)\}$$

(IV) $R \circ R \circ R = R^2 \circ R$

$$R = \{(1,2)(3,4)(2,2)\}$$

$$R^2 = \{(1,2)(2,2)\}$$

$$R \circ R \circ R = \{(1,2)(2,2)\}$$

$$R^3 = \{(1,2)(2,2)\}$$

(V) $R \circ (S \circ R)$

$$S \circ R = \{ \quad \}$$

$$R = \{ \quad \}$$

(VI) $(R \circ S) \circ R$

$$R = \{ \quad \}$$

$$R \circ S = \{ \quad \}$$

Theorem:- Let R be a relation from the set A to the Set B and S be a relation from the Set B to Set C , then

$$(SOR)^{-1} = R^{-1} \circ S^{-1} \text{ (Reversal law in composite relation)}$$

Proof Let $(c, a) \in (SOR)^{-1}$

$$\Rightarrow (a, c) \in S \circ R$$

$\Rightarrow$ For $b \in B$ such that

$$\Rightarrow (a, b) \in R \text{ and } (b, c) \in S$$

$$\Rightarrow (b, a) \in R^{-1} \text{ and } (c, b) \in S^{-1}$$

$$\Rightarrow (c, b) \in S^{-1} \text{ and } (b, a) \in R^{-1}$$

$$\Rightarrow (c, a) \in R^{-1} \circ S^{-1}$$

Hence

$$(SOR)^{-1} = R^{-1} \circ S^{-1}$$

Hence proved

Q.3. Let $A = \{a, b\}$

$$\checkmark R = \{a, a\}(b, a)(b, b)$$

$$\checkmark S = \{a, b\}(b, a)(b, b)$$

Then verify $(SOR)^{-1} = R^{-1} \circ S^{-1}$

LHS

$$SOR = \{(a, b)(b, b)(b, a)\}$$

$$(SOR)^{-1} = \{(b, a)(b, b)(a, b)\}$$

⌞ ①

RHS

$$S^{-1} = \{(b, a)(a, b)(b, b)\}$$

$$R^{-1} = \{(a, a)(a, b)(b, b)\}$$

$$R^{-1} \circ S^{-1} = \{(b, a)(b, b)(a, b)\}$$

⌞ ②

From ① and ②

$$(SOR)^{-1} = R^{-1} \circ S^{-1}$$

Topic : Composition of relation

Q.1. Let $R = \{(1, 1)(2, 1)(3, 2), (4, 3)\}$, compute R^2, R^3, R^4 .

Q.2. Let $A = \{1, 2, 3\}$ $B = \{p, q, r\}$ and $C = \{x, y, z\}$

and let $R = \{(1, p)(1, r)(2, q)(3, q)\}$

and $S = \{(p, y)(q, x)(r, z)\}$ compute $S \circ R$

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattices

Lec-10

Today's Target

- **Total Number of relation**
- **Partial Order Relation**

$$A \times A = n^2$$

Total Number of relations

Let A be a set with n elements i.e $n(A) = n$, then the total number of

✓ (i) Relation = 2^{n^2}

✓ (ii) Reflexive relation = 2^{n^2-n} or $2^{n(n-1)}$

✓ (iii) Irreflexive relation = 2^{n^2-n} or $2^{n(n-1)}$

✓ (iii) Symmetric relation = $2^{\frac{n(n+1)}{2}}$

✓ (iv) Asymmetric relation = $3^{\frac{n(n-1)}{2}}$

✓ (v) Anti-symmetric relation = $2^n \times 3^{\frac{n(n-1)}{2}}$

✓ (vi) Reflexive and symmetric = $2^{\frac{n^2+n}{2}}$

$$A = \{1, 2, 3\}$$

$$A \times A = \{(1, 1) (1, 2) (1, 3)$$

$$(2, 1) (2, 2) (2, 3)$$

$$(3, 1) (3, 2) (3, 3)\}$$

$$n^2 - n$$

A relation R on set A is called partial order relation If it is

(i) Reflexive : $(a, a) \in R \quad \forall a \in A$

(ii) Anti symmetric : $(a, b) \in R \text{ and } (b, a) \in R \Rightarrow a = b \quad \forall a, b \in A$

(iii) Transitive : $(a, b) \in R \text{ and } (b, c) \in R \Rightarrow (a, c) \in R \quad \forall a, b, c \in A$

Q.1. Identify the partial order relation from the following Relation on a set $A = \{1, 2, 3, 4\}$

$\times$ (i) $R_1 = \phi$

$\times$ (ii) $R_2 = A \times A$
 Reflexive $\checkmark$
 Anti-symmetric $\times$ $(2, 1) \in R$ and $(1, 2)$ but $2 \neq 1$
 Transitive $\checkmark$

$\checkmark$ (iii) $R_3 = \{(1, 1)(2, 2)(3, 3)(4, 4)(1, 2)(2, 3)(1, 3)\}$
 R $\checkmark$
 A $\checkmark$
 T $\checkmark$

$\checkmark$ (iv) $R_4 = \{(1, 1)(2, 2)(3, 3)(4, 4)\}$
 R $\checkmark$
 A $\checkmark$
 T $\checkmark$

$\times$ (v) $R_5 = \{(1, 1)(2, 2)(3, 3)(4, 4)(1, 2)(2, 1)\}$
 R $\checkmark$
 A $\times$

Q.2. Let N be the set of natural numbers, prove that the relation of "less than or equal to" is a partial order relation on N .

$$R = \{(a, b) : a \leq b \text{ and } a, b \in N\}$$

R is a partial order relation if

(I) Reflexive: For Any $a \in N$

Since $a \leq a$ (True)

$$(a, a) \in R \quad \forall a \in N$$

So, R is a reflexive Relation

(II) Anti-Symmetric

Let $a, b \in N$ such that

$$(a, b) \in R \text{ and } (b, a) \in R$$

$$\Rightarrow a \leq b \text{ and } b \leq a$$

$$\Rightarrow a = b$$

So, R is Anti-symmetric Relation

(III) Transitive: Let $a, b, c \in \mathbb{N}$ such that

$$(a, b) \in R \text{ and } (b, c) \in R$$

$$\Rightarrow a \leq b \text{ and } b \leq c$$

$$\Rightarrow a \leq c$$

$$\Rightarrow (a, c) \in R \quad \forall a, b, c \in \mathbb{N}$$

= So, it is Transitive

Hence, given relation is
a partial order relation

Q.3. Which of the following is not a partial order relation ?

~~(i) $R_1 = \{(a, b) : a, b \in \mathbb{Z}, a < b\}$~~

~~(ii) $R_2 = \{(a, b) : a, b \in \mathbb{Z}, a \leq b\}$~~

~~(iii) $R_3 = \{(a, b) : a, b \in \mathbb{Z}, \frac{b}{a} \leq \mathbb{Z}\}$~~

(i) $R_1 = \{(a, b) : a, b \in \mathbb{Z}, a < b\}$

Reflexive : For any $a \in \mathbb{Z}$

$a < a$ (False)

$\Rightarrow (a, a) \notin R$

$\Rightarrow$ so, R is not reflexive

Hence, it is not a partial order relation

Topic : Partial Order Relation

Q.1. Let $A = \{1, 2, 3\}$. Show that $R = \{(1, 1)(2, 2)(3, 3)(1, 2)(2, 3)(1, 3)\}$ is a partial order relation.

Q.2. Show that the relation $R = \{(x, y) : x \geq y\}$ where $x, y \in \mathbb{Z}$ is a partial order relation

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattices

Lec-11

Today's Target

- *Digraphs of Relation*
- **Matrix Representation of Relation**

OR

Directed Graph of Relation

If A is finite set and R is a relation on A , then we can also represent R pictorially as

- (i) Draw a small circle for each element of A and label the circle with corresponding elements of A , these circles are called nodes or vertices
- (ii) Draw an arrow, called an edge, from vertex

$$a_i \text{ to } a_j \Leftrightarrow a_i R a_j$$

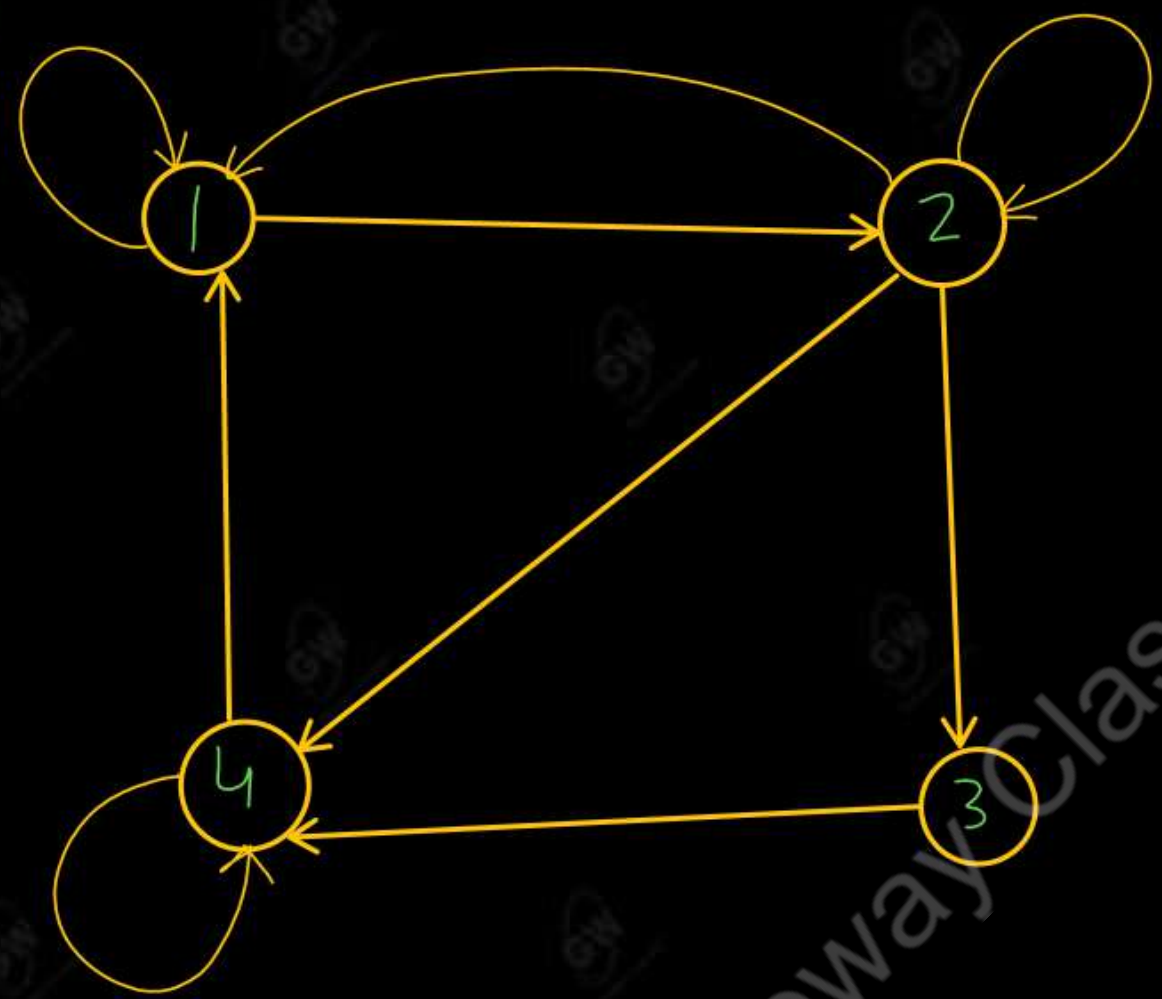
The resulting pictorial representation of R is called a Directed graph or Digraph of R .

Gateway Classes : 7455 9612 84

Q.1- Let $A = \{1, 2, 3, 4\}$ $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (2, 3), (2, 4), (3, 4), (4, 1), (4, 4)\}$

Construct the digraph of R

solution

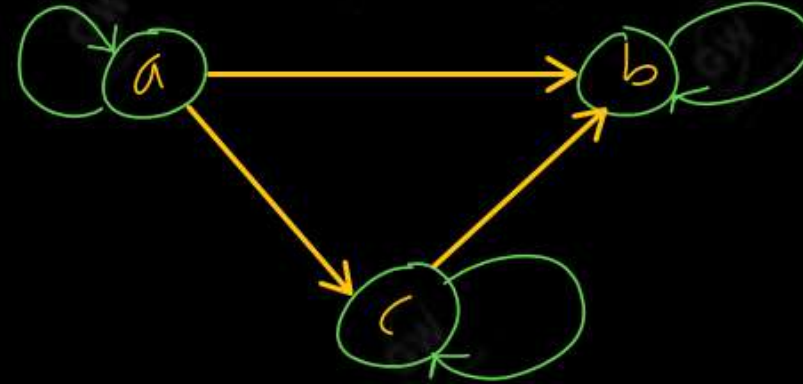


It is digraph of the given Relation

Gatewayclasses: 7455 9612 84

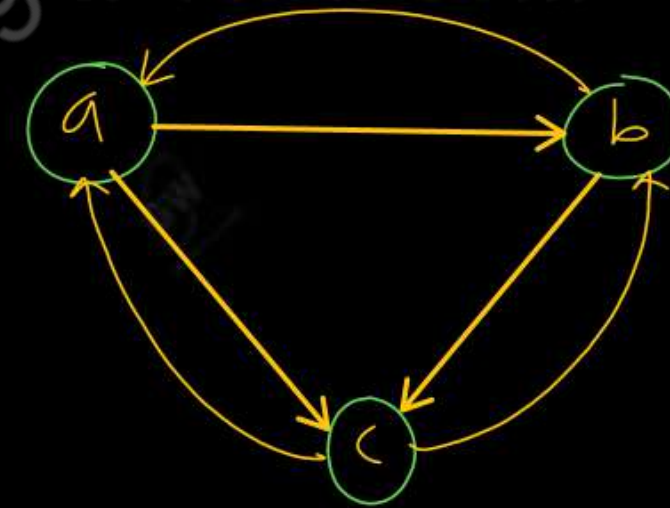
(1) Reflexive -

A Relation R is reflexive if there is a loop at every Vertex of the directed graph



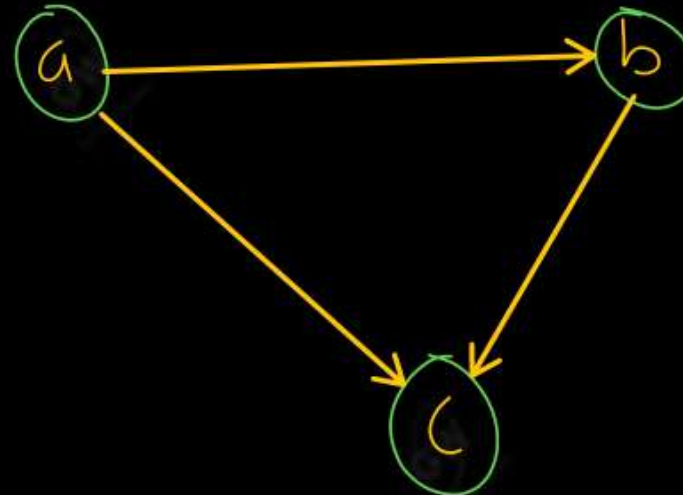
(2) Symmetric -

A Relation R is symmetric, if for every edge between distinct vertices in its digraph there is an edge in the opposite direction.



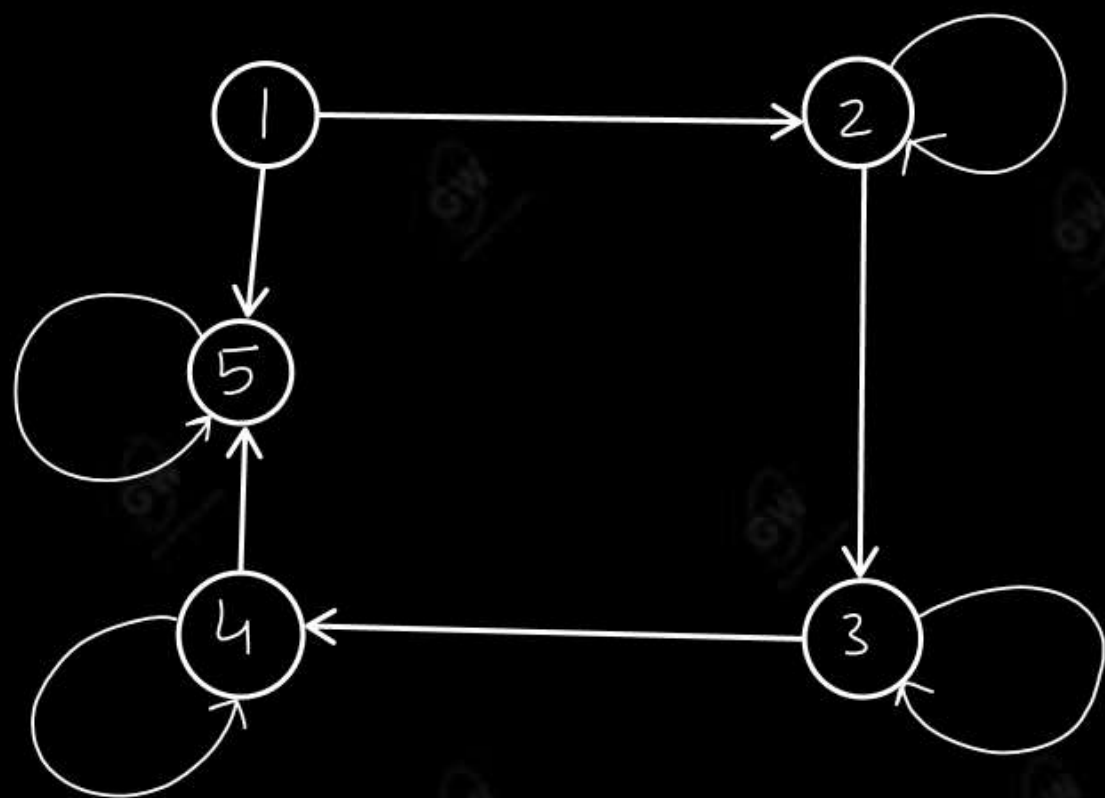
(3) Transitive -

A Relation is transitive if there is an edge from vertex a to vertex b and from vertex b to vertex c , then there is an edge from vertex a to vertex c .



Q.2- Find the relation by the following figures

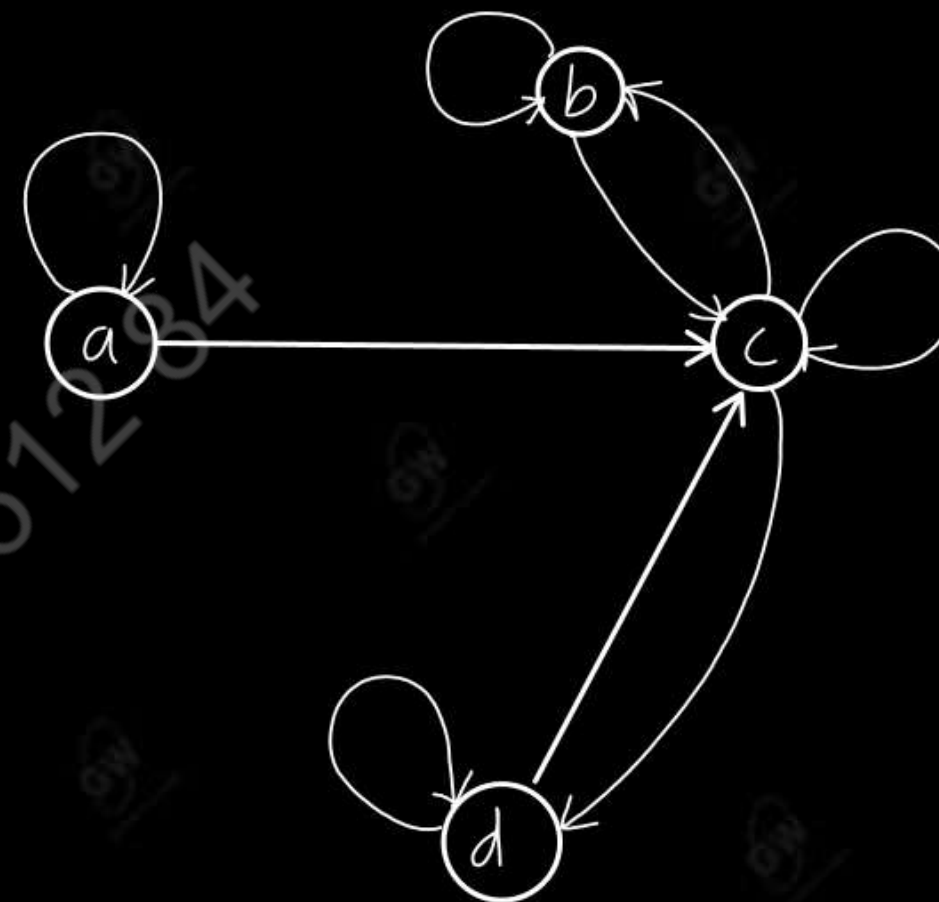
(i)



Solution

$$R = \{ (1,2), (2,2), (2,3), (3,3), (3,4), (4,4), (4,5), (5,5), (1,5) \}$$

(ii)



solution

$$R = \{ (a,a), (a,c), (c,c), (c,b), (b,c), (b,b), (c,d), (d,c), (d,d) \}$$

Matrix Representation of Relation

Let $A = \{a_1, a_2, a_3 \text{ --- } a_m\}$ and $B = \{b_1, b_2, b_3 \text{ --- } b_n\}$ are finite sets containing m and n elements respectively and let R be a relation from A to B

Then R can be represented by the $m \times n$ matrix

$$M_R = [m_{ij}]$$

Where

$$m_{ij} = \begin{cases} 1 & \text{if } (a_i, b_j) \in R \\ 0 & \text{if } (a_i, b_j) \notin R \end{cases}$$

The matrix M_R is called matrix of Relation R

Gateway Classes :: 7455 9612 84

Q.3- Let R be the relation from the set $A = \{1, 3, 4\}$ on itself and defined by

$R = \{(1, 1), (1, 3), (3, 1), (3, 3), (4, 4)\}$ then find relation

matrix

Given $A = \{1, 3, 4\}$

$$R = \{(1, 1), (1, 3), (3, 1), (3, 3), (4, 4)\}$$

$$M_R = \begin{matrix} & \begin{matrix} 1 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

Q.4- Let $A = \{a, b, c, d\}$ and let $M_R = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$ Find R

solution

$$A = \{a, b, c, d\}$$

$$M_R = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

$$R = \{(a, a), (a, b), (b, c), (b, d), (c, c), (c, d), (d, a), (d, c)\}$$

Q.5- Let $A = \{1, 2, 3, 4\}$ and let R be a relation on A whose matrix is $M_R = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$

show that R is transitive.

Sol

$$A = \{1, 2, 3, 4\}$$

$$M_R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \end{matrix}$$

First Method

$$R = \{ \underbrace{(1,1)}_{\checkmark} \underbrace{(1,2)}_{\checkmark} \underbrace{(1,3)}_{\checkmark} \underbrace{(1,4)}_{\checkmark} \underbrace{(3,1)}_{\checkmark} \underbrace{(3,2)}_{\checkmark} \underbrace{(3,3)}_{\checkmark} \underbrace{(3,4)}_{\checkmark} \underbrace{(4,2)}_{\checkmark} \}$$

Second Method

R is transitive if

$$M_R^2 + M_R = M_R$$

$$M_R^2 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_R^2 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_R^L + M_R$$

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_R^L + M_R = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = M_R$$

Q.6- Let $P = \{2, 3, 4, 5\}$. Consider the relation R and S on P defined by

$$R = \{(2, 2), (2, 3), (2, 4), (2, 5), (3, 4), (3, 5), (4, 5), (5, 4)\} \text{ and}$$

$$S = \{(2, 3), (2, 5), (3, 4), (3, 5), (4, 2), (4, 3), (4, 5), (5, 2), (5, 5)\}$$

Find the matrices of the above relations

Use matrices to find the composition relation of R and S

$$M_R = \begin{matrix} & \begin{matrix} 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

$$M_S = \begin{matrix} & \begin{matrix} 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

$$SOR = M_R \times M_S$$

Gateway Classes :: 7455 9612 84

Q.1. Let $A = \{a_1, a_2, a_3\}$ and $B = \{b_1, b_2, b_3, b_4\}$ which ordered pairs are in the relation R represented by the matrix

$$M_R = \begin{matrix} & \begin{matrix} b_1 & b_2 & b_3 & b_4 \end{matrix} \\ \begin{matrix} a_1 \\ a_2 \\ a_3 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

Q.2. Let $A = \{1, 2, 3, 4, 8\}$, $B = \{1, 4, 6, 9\}$. Let $a R b$ iff a divides b . Find the relation matrix

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattices

Lec-12

Today's Target

- *Closure of Relation*
- *Warshall's Algorithm*
- *Univ PYQs*
- *DPPs*

Closure of Relation

Let R be a relation on a set A . R may or may not have some property P , such as reflexivity, symmetry or transitivity.

If we add some pairs then we have the desired property, the smallest relation S on A that contains R and possesses the desired property P is called Closure of relation R with respect to that property.

$$R \subseteq S$$

Set A



Relation R (does not satisfy some property P)



Add some order pairs



New Relation S

Where S is called closure of Relation

Reflexive
Property

Symmetric
property

Transitive
property

Closer of Relation (S)

Reflexive closure

$$R^r$$

Symmetric closure

$$R^s$$

Transitive closure

$$R^+$$

(i) Reflexive closure

The reflexive closure R^r of a relation R is the *smallest* reflexive relation that contains R as a subset.

Reflexive closure is obtained by taking union of R and I_A .

$$R^r = R \cup I_A$$

Where $I_A = \{(a, a) : a \in A\}$
= Identity Relation

$$A = \{1, 2, 3\}$$

$$I_A = \{(1, 1), (2, 2), (3, 3)\}$$

Q.1 Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 1)(1, 2)(2, 3)\}$ be the relation on A. Find the reflexive closure of R.

Given

$$A = \{1, 2, 3, 4\}$$

$$R = \{(1, 1), (1, 2), (2, 3)\}$$

$$I_A = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$$

$$R^x = R \cup I_A$$

$$R^x = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 3)\}$$

(ii) Symmetric closure

The Symmetric closure R^s is the smallest symmetric relation that contains R as a subset.

The symmetric closure is obtained by taking the union of R and R^{-1}

$$R^s = R \cup R^{-1}$$

$$R^{-1} = \{(b, a) : (a, b) \in R\}$$

$$R = \{(1, 2) (1, 3)\}$$

$$R^{-1} = \{(2, 1) (3, 1)\}$$

Q.2 If $R = \{(1, 2)(4, 3)(2, 2)(2, 1)(3, 1)\}$ be a relation on $A = \{1, 2, 3, 4\}$. Find the symmetric closure.

Given $A = \{1, 2, 3, 4\}$

$$R = \{(1, 2), (4, 3), (2, 2), (2, 1), (3, 1)\}$$

$$R^{-1} = \{(2, 1), (3, 4), (2, 2), (1, 2), (1, 3)\}$$

$$R^s = R \cup R^{-1}$$

$$R^s = \{(1, 2), (4, 3), (2, 2), (2, 1), (3, 1), (3, 4), (1, 3)\}$$

Transitive closure

The relation obtained by adding the least number of ordered pairs to ensure transitivity is called the transitive closure of the relation.

The transitive closure of relation, R is denoted by R^+

Consider a relation, R on Set $A = \{a_1, a_2, a_3 \dots \dots \dots a_m\}$

Then transitive closure of R

$$R^+ = R \cup R^2 \cup R^3 \dots \dots \dots \cup R^m$$

$$R^2 = R \circ R$$

$$R^3 = R^2 \circ R$$

$$R^4 = R^3 \circ R$$

$$R^5 = R^4 \circ R$$

Q.3 Let $R = \{(1, 2)(2, 3)(3, 1)\}$ defined on $A = \{1, 2, 3\}$ find the transitive closure of R .

Given

$$A = \{1, 2, 3\}$$

$$R = \{(1, 2)(2, 3)(3, 1)\}$$

$$R^+ = R \cup R^2 \cup R^3$$

$$R^2 = R \circ R$$

$$R^2 = \{(1, 2)(2, 3)(3, 1)\} \circ \{(1, 2)(2, 3)(3, 1)\}$$

$$R^2 = \{(1, 3)(2, 1)(3, 2)\}$$

$$R^3 = R^2 \circ R$$

$$R^3 = \{(1, 2)(2, 3)(3, 1)\} \circ \{(1, 3)(2, 1)(3, 2)\}$$

$$R^3 = \{(1, 1)(2, 2)(3, 3)\}$$

$$R^+ = \{(1,2)(2,3)(3,1)\} \cup \{(1,3)(2,1)(3,2)\} \cup \{(1,1)(2,2)(3,3)\}$$

$$R^+ = \{(1,2)(2,3)(3,1)(1,3)(2,1)(3,2)(1,1)(2,2)(3,3)\}$$

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A more efficient method to calculate transitive closure of relation

Step-1: Let n be the number of elements in a given Set A

To find transitive closure of Relation R on A , maximum n warshall sets can be find.

$w_0, w_1, w_2 \dots \dots \dots w_n$ (Here $w_0 = M_R$)

Step-2: Procedure to calculate w_k from w_{k-1}

- (i) Copy 1 from w_{k-1} in w_k with same position
- (ii) In K^{th} row and K^{th} column of w_{k-1} , see K^{th} column and check the position of 1 and put below C
See K^{th} row and check the position of 1 and put below R
- (iii) Mark entries in w_k as 1 for (C, R) of w_k if there are not already 1 .

Step-3: Stop when w_n is obtained and it the required transitive closure.

Q.4 Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 4)(2, 1)(2, 3)(3, 1)(3, 4)(4, 3)\}$ be a relation on A.

Find transitive closure using Warshall's Algorithm.

Given

$$A = \{1, 2, 3, 4\}$$

$$R = \{(1, 4)(2, 1)(2, 3)(3, 1)(3, 4)(4, 3)\}$$

$$n(A) = 4$$

$$M_R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

$$W_0 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

From First column and First Row

$$W_1 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

$$\begin{matrix} C & R \\ \{2, 3\} & \{4\} \end{matrix}$$

$$C \times R = \{(2, 4)(3, 4)\}$$

From second column and second Row

$$w_2 = 2$$

	1	2	3	4
1	0	0	0	1
2	1	0	1	1
3	1	0	0	1
4	0	0	1	0

C { } R {1, 3, 4}

CXR = { }

From Third column and Third Row

$$w_3 = 2$$

	1	2	3	4
1	0	0	0	1
2	1	0	1	1
3	1	0	0	1
4	1	0	1	1

C {2, 4} R {1, 4}

CXR = { (2, 1) (2, 4) (4, 1) (4, 4) }

From Fourth column and Fourth Row

$$w_4 = 2$$

	1	2	3	4
1	1	0	1	1
2	1	0	1	1
3	1	0	1	1
4	1	0	1	1

C {1, 2, 3, 4} R {1, 3, 4}

CXR = { (1, 1) (1, 3) (1, 4) (2, 1) (2, 3) (2, 4) (3, 1) (3, 3) (3, 4) (4, 1) (4, 3) (4, 4) }

Transitive closure

$$R^+ = \{ (1, 1) (1, 3) (1, 4) (2, 1) (2, 3) (2, 4) (3, 1) (3, 3) (3, 4) (4, 1) (4, 3) \}$$

Topic : Closure of Relation and Marshall's Algorithm

Q.1. Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 2)(2, 3)(3, 4)\}$ be a relation on set A . Find the transitive closure.

Q.2. Let $A = \{4, 6, 8, 10\}$ and $R = \{(4, 4)(4, 10)(6, 6)(6, 8)(8, 10)\}$ is a relation on set A . Determine the transitive closure of R using Marshall's Algorithm.

Q.3. Let $R = \{(1, 2)(2, 3)(3, 1)\}$ defined on $A = \{1, 2, 3\}$ find the transitive closure of R using Marshall's Algorithm.

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattices

Today's Target

- ✓ ➤ Partial order relation
- ✓ ➤ Partial order set or POSET
- ✓ ➤ Graphical representation of POSET - Hasse Diagram.
- ✓ ➤ Hasse Diagram from Directed graph.
- ✓ ➤ Easiest way to draw Hasse diagram

Partial order relation

A relation R on set A is called partial order relation if it is

(i) **Reflexive** $(a, a) \in R \quad \forall a \in A$

(ii) **Anti-symmetric** $(a, b) \in R \text{ and } (b, a) \in R \Rightarrow a = b \quad \forall a, b \in A$

(iii) **Transitive** $(a, b) \in R \text{ and } (b, c) \in R \Rightarrow (a, c) \in R \quad \forall a, b, c \in A$

Partial order set or POSET

If A is any non-empty set and R is a partial order relation on set A , then the order pair (A, R) is called Partial order set or POSET

POSET is denoted by (A, R) Or $(A, \leq)$

A partial order $\leq$ on a set A can be represented by means of a diagram known as Hasse Diagram of $(A, \leq)$

i.e Hasse diagram is a graphical representation of POSET

Hasse Diagram from directed graph or digraph

Step 1:- Start with directed Graph

Step 2:- Remove self loop

Reason : Self loops are quite obvious and they must be present for all vertices because R is reflexive, so we can remove them to simplify diagram.

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Step 3:- Remove all transitive edges.

Reason: Transitive edges are quite obvious and they must be present because R is transitive. So, we can remove all transitive edges to Simply diagram.

Step 4:- Remove circle by dot

Step-5: Arrange each edge so that its initial vertex is below its terminal vertex i.e. all edges are pointing upwards.

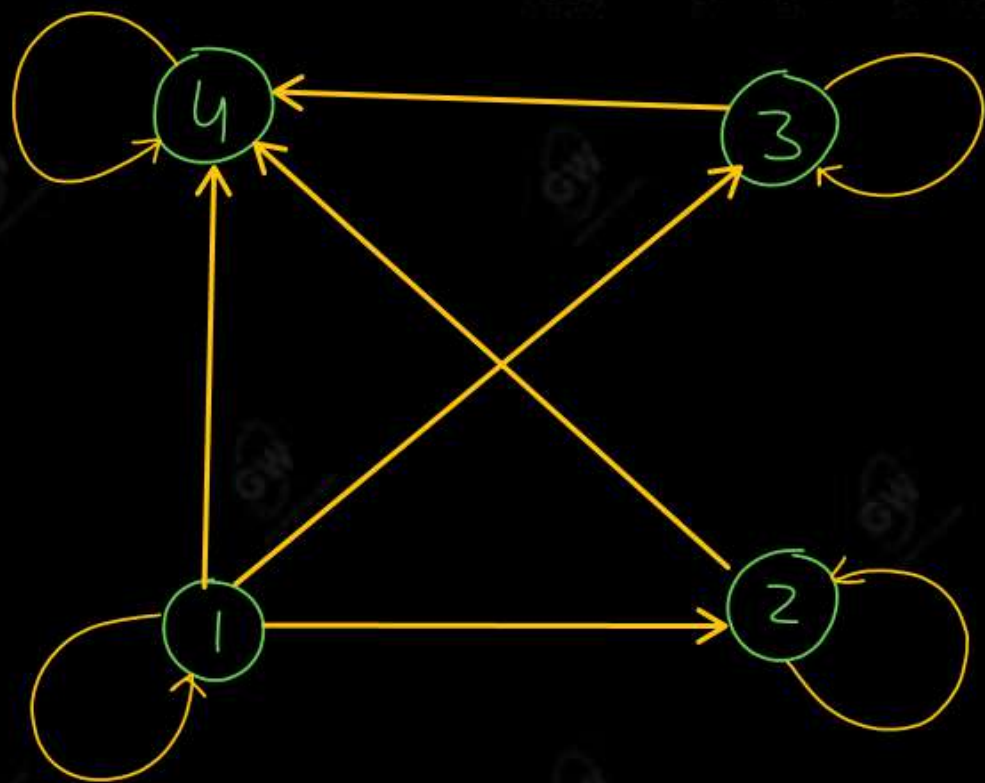
Step-6: Remove all the arrows the directed graph.

Reason: since all edges are pointing upwards, so there is no need to show the directions in the graph.

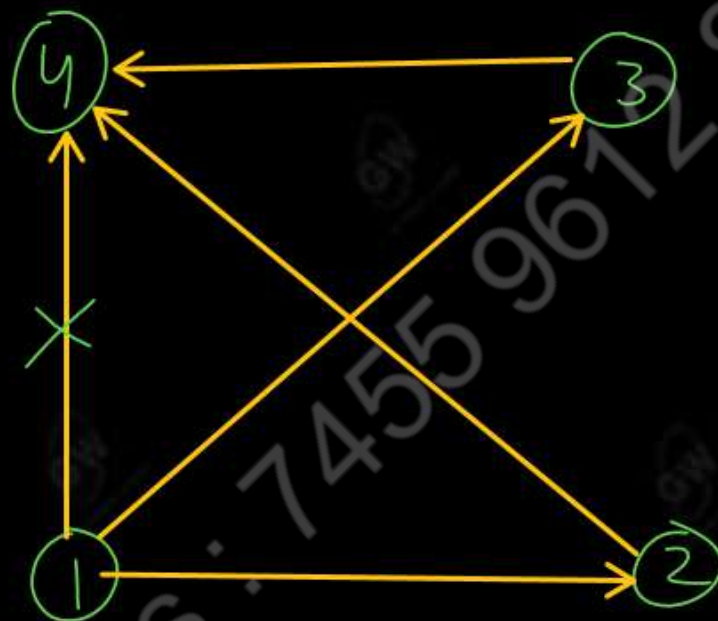
Q.1 Determine the Hasse diagram of the relation R on set $A = \{1, 2, 3, 4\}$

(A, R)

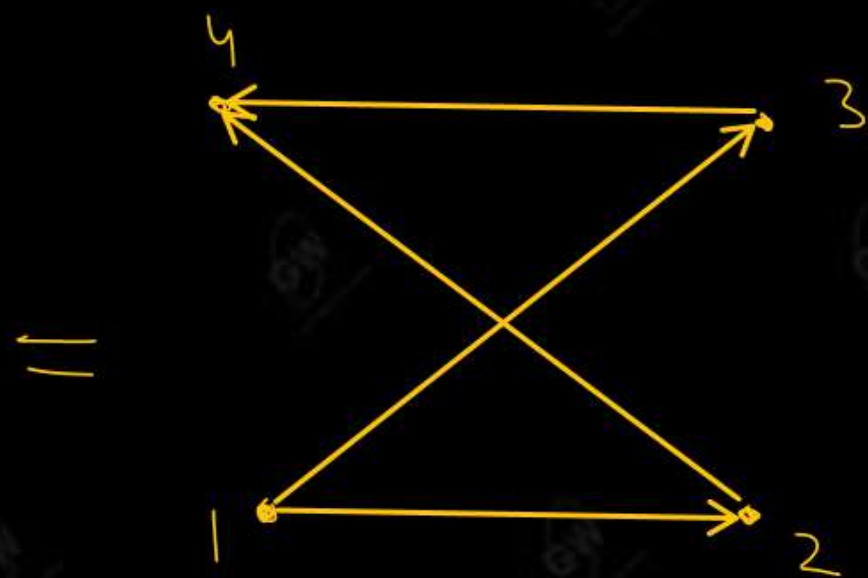
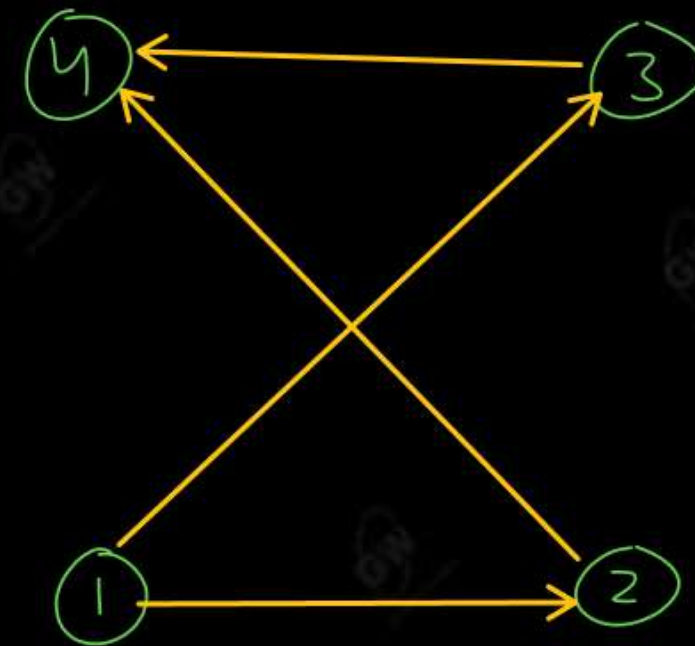
$$R = \{(1, 1), (1, 2), (2, 2), (2, 4), (1, 3), (3, 3), (3, 4), (1, 4), (4, 4)\}$$



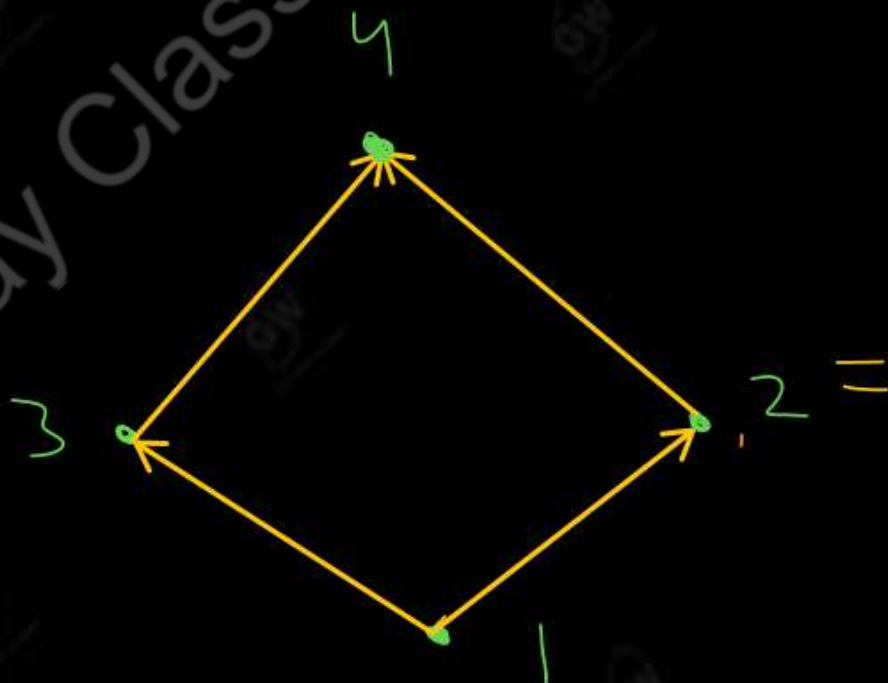
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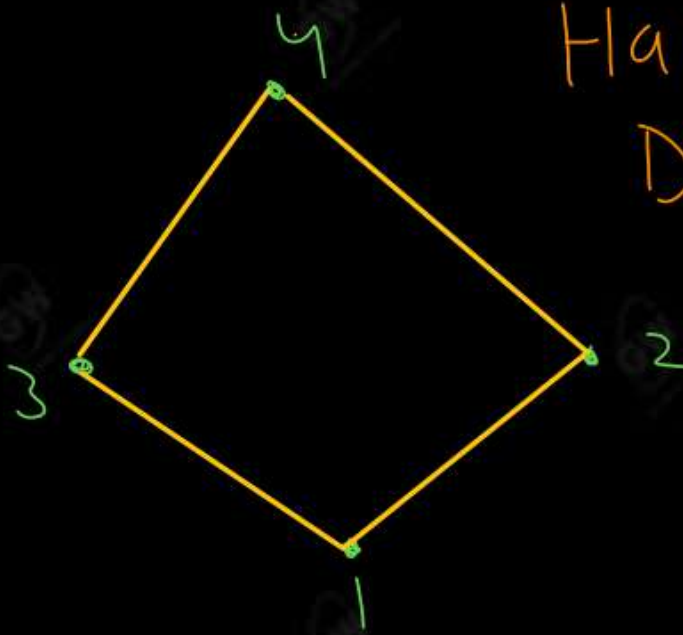
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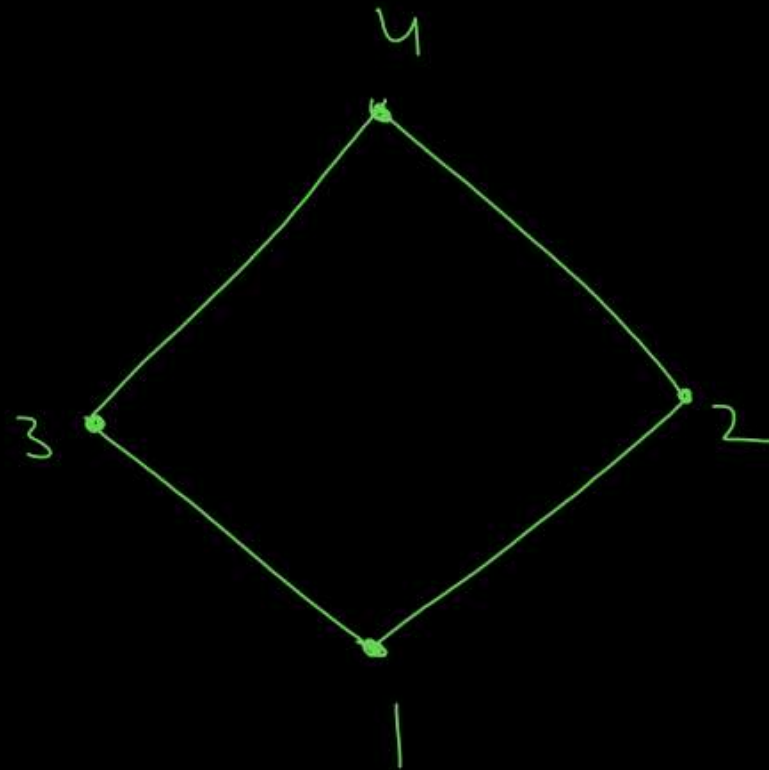
Hasse
Diagram

Easy Method

$$R = \{(1,1) (1,2) (2,2) (2,4) (1,3) (3,3) (3,4) (1,4) (4,4)\}$$

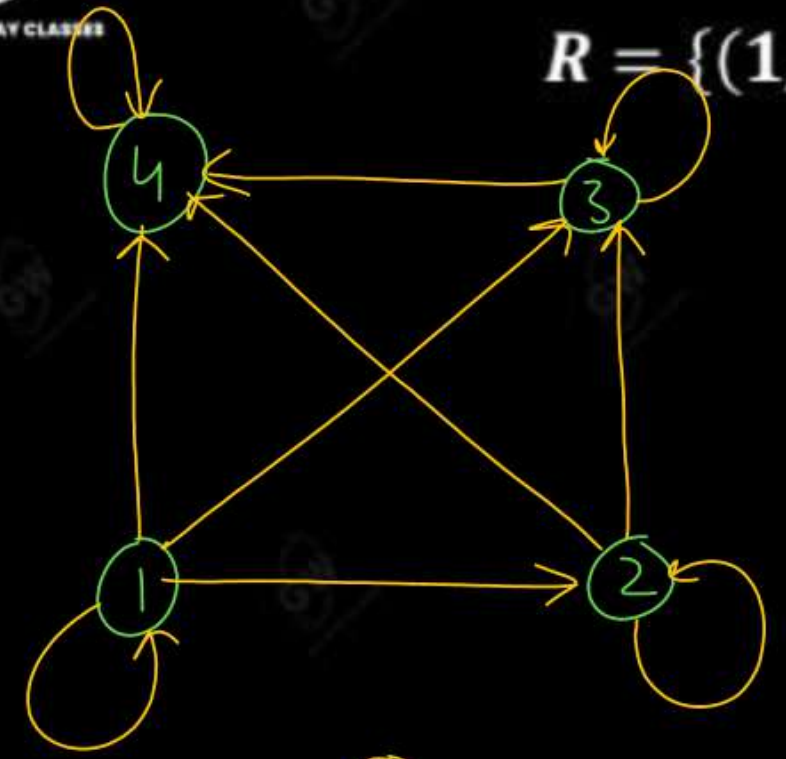
$$R = \{(1,2) (2,4) (1,3) (3,4) \cancel{(1,4)}\}$$

$$R = \{ \underset{\checkmark}{(1,2)} \underset{\checkmark}{(2,4)} \underset{\checkmark}{(1,3)} (3,4) \}$$

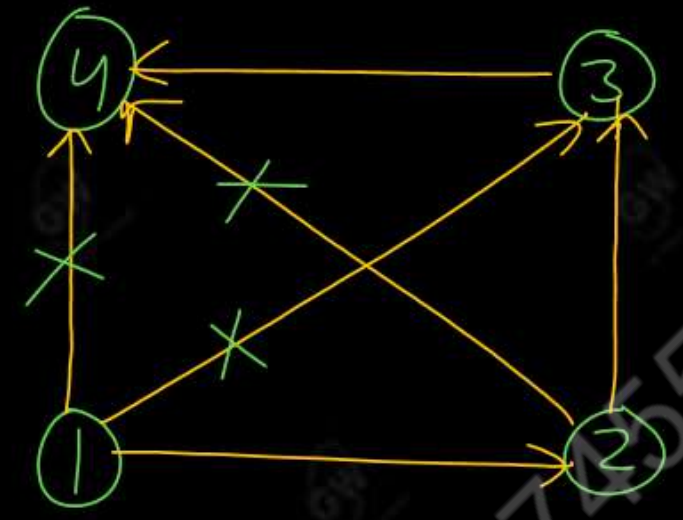


Q.2 Determine the Hasse diagram of the relation R on set $A = \{1, 2, 3, 4\}$ (A, R)

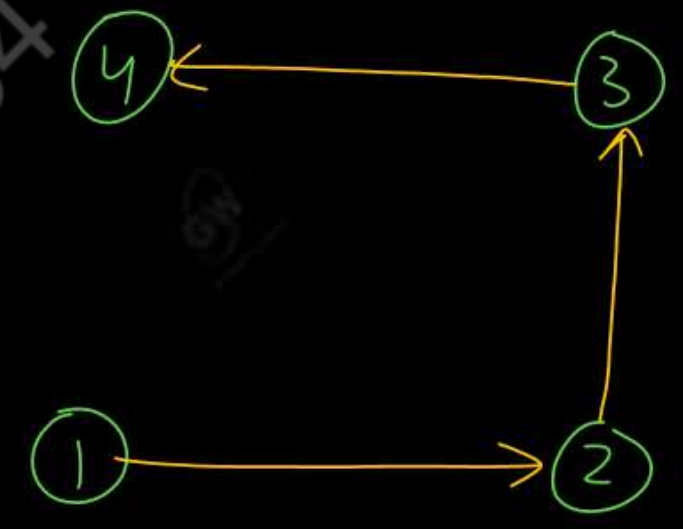
$$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$$



=



=



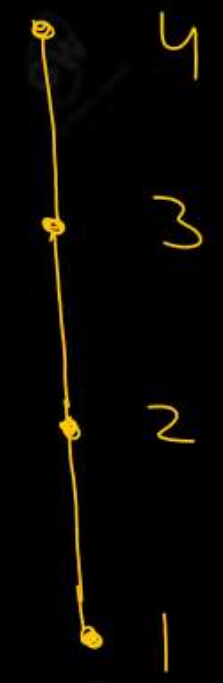
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Hasse Diagram

Easy Method

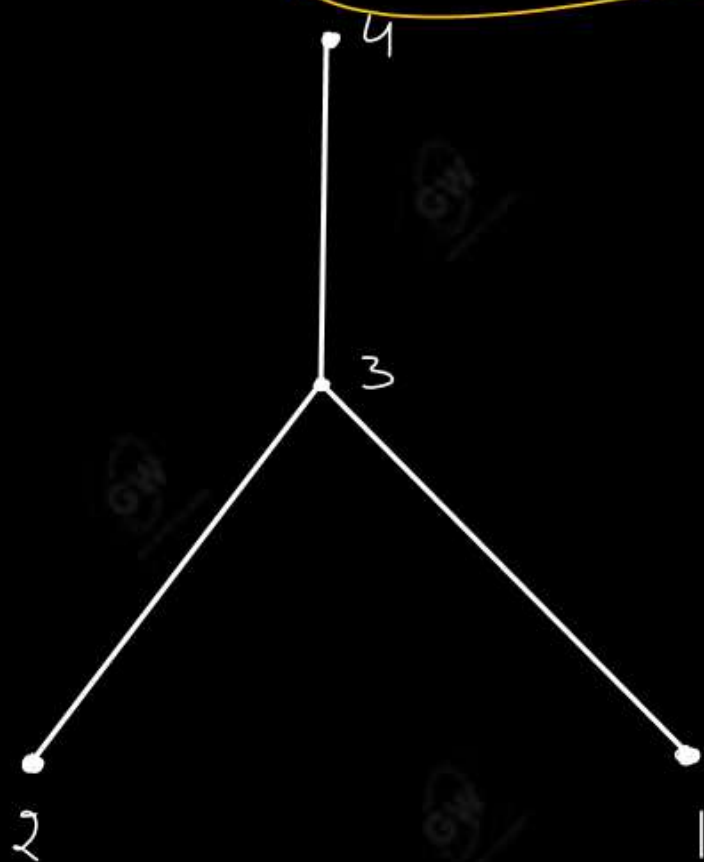
$$R = \{(1, 2) (1, 3) (1, 4) (2, 3) (2, 4) (3, 4)\}$$

$$R = \{(1, 2) (2, 3) (3, 4)\}$$



Hasse Diagram

Q.3- Describe the order pairs in the relation determined by the Hasse diagram of a POSET $(A, \leq)$ on the set $A = \{1, 2, 3, 4\}$



solution

From Hasse Diagram

$(1, 3) (2, 3) (3, 4)$

Reflexive order pair

$(1, 1) (2, 2) (3, 3) (4, 4)$

Transitive Pair

$(2, 4) (1, 4)$

$$R = \{ (1, 1) (2, 2) (3, 3) (4, 4) (1, 3) (2, 3) (3, 4) (2, 4) (1, 4) \}$$

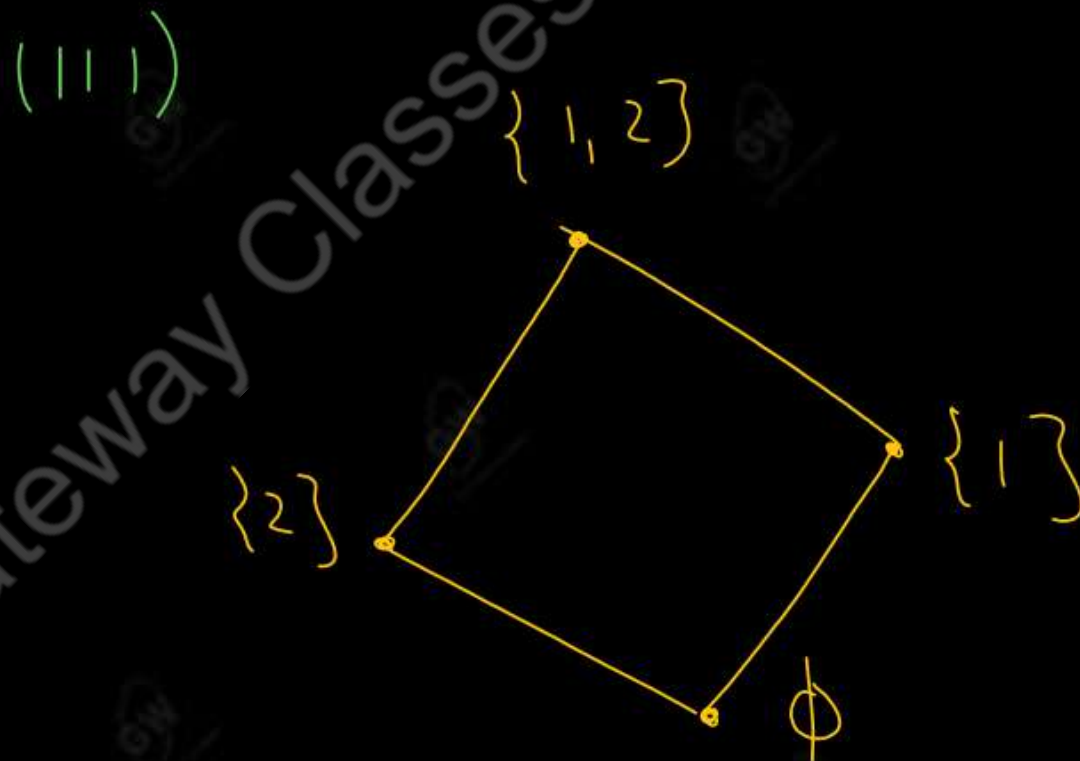
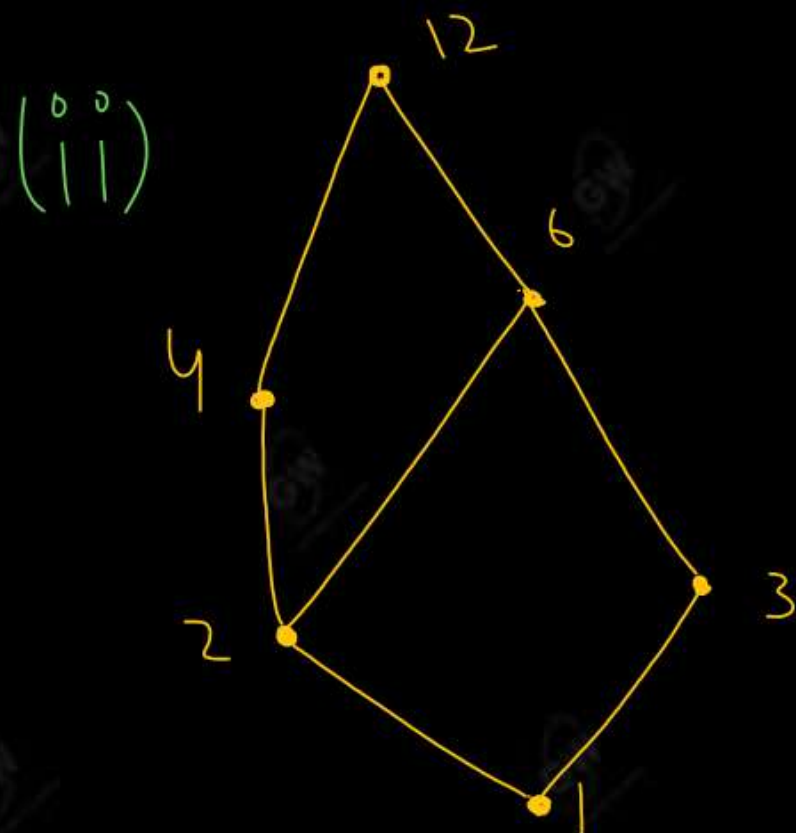
Easiest Way to draw the Hasse diagram

Q.4 Draw the Hasse diagram for the following POSET

(i) $(A, \leq)$ where $A = \{1, 2, 3, 4, 5, 6\}$

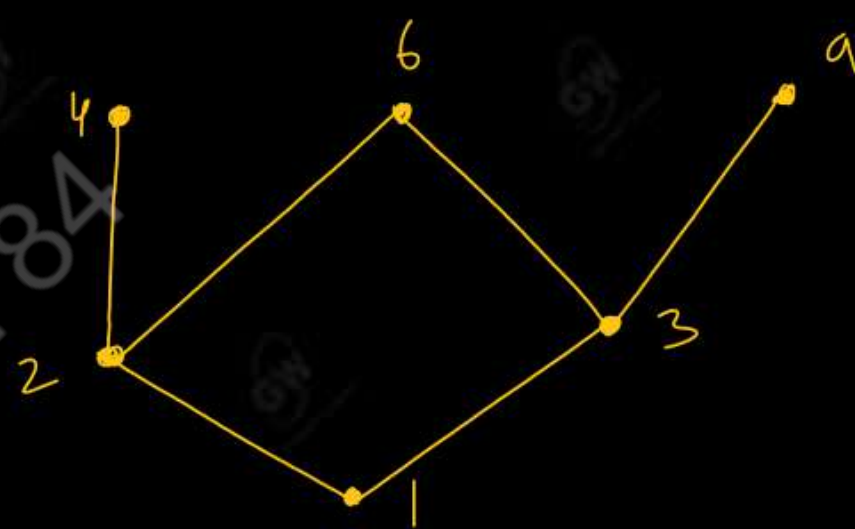
(ii) $(A, |)$ where $A = \{1, 2, 3, 4, 6, 12\}$

(iii) $(A, \subseteq)$ where $A = \{\phi, \{1\}, \{2\}, \{1, 2\}\}$



(iv) (A, I) where $A = \{1, 2, 3, 4, 6, 9\}$

(IV)

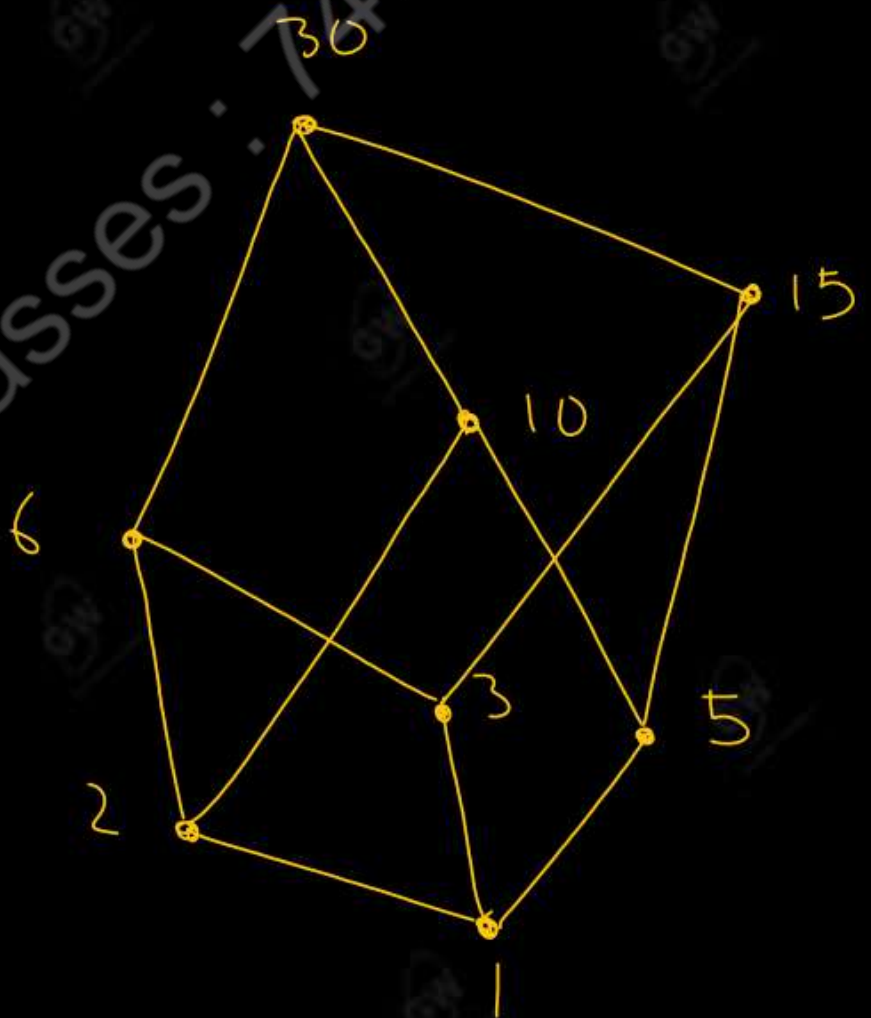
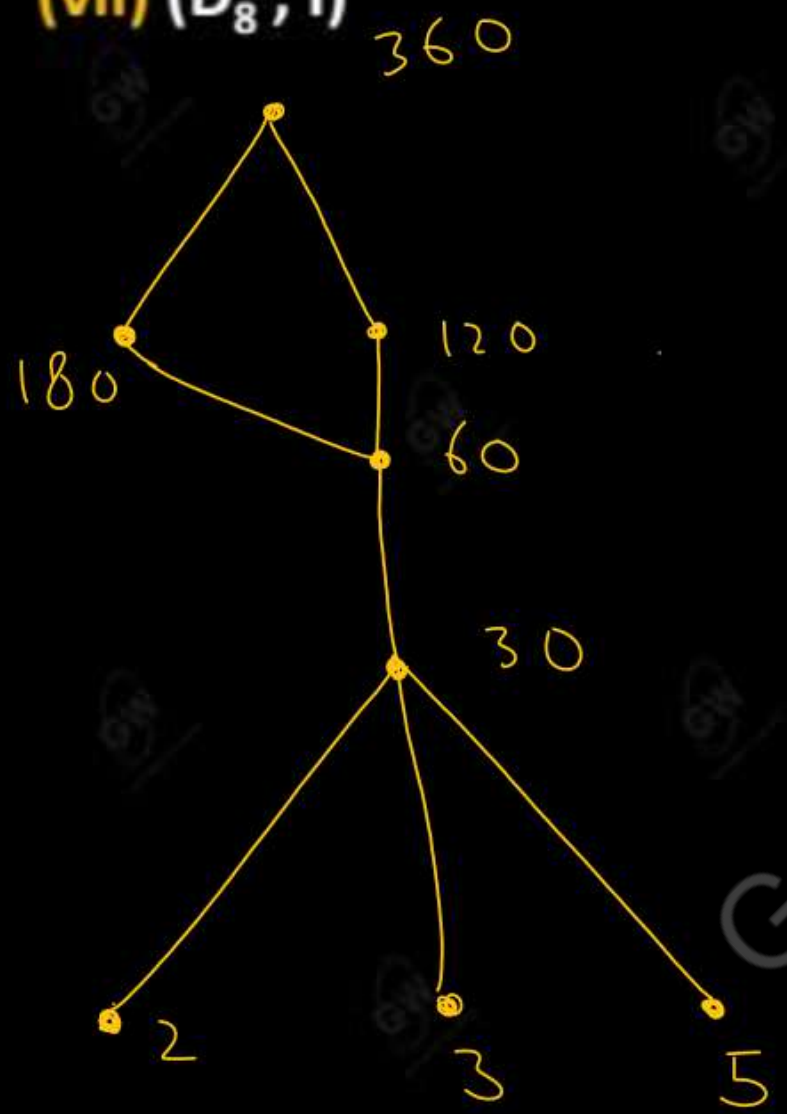


(v) (A, I) where $A = \{2, 3, 5, 30, 60, 120, 180, 360\}$

(vi) (A, I) where $A = \{1, 2, 3, 5, 6, 10, 15, 30\}$

(vii) (D_8, I)

(VI)



(vii) (D_8, I)

$A = \{1, 2, 4, 8\}$



Q.5- Let $X = \{1, 2, 3, 4, 5, 6\}$, then I is a partial order relation on X . Draw the Hasse diagram of (X, I)

Hasse Diagram



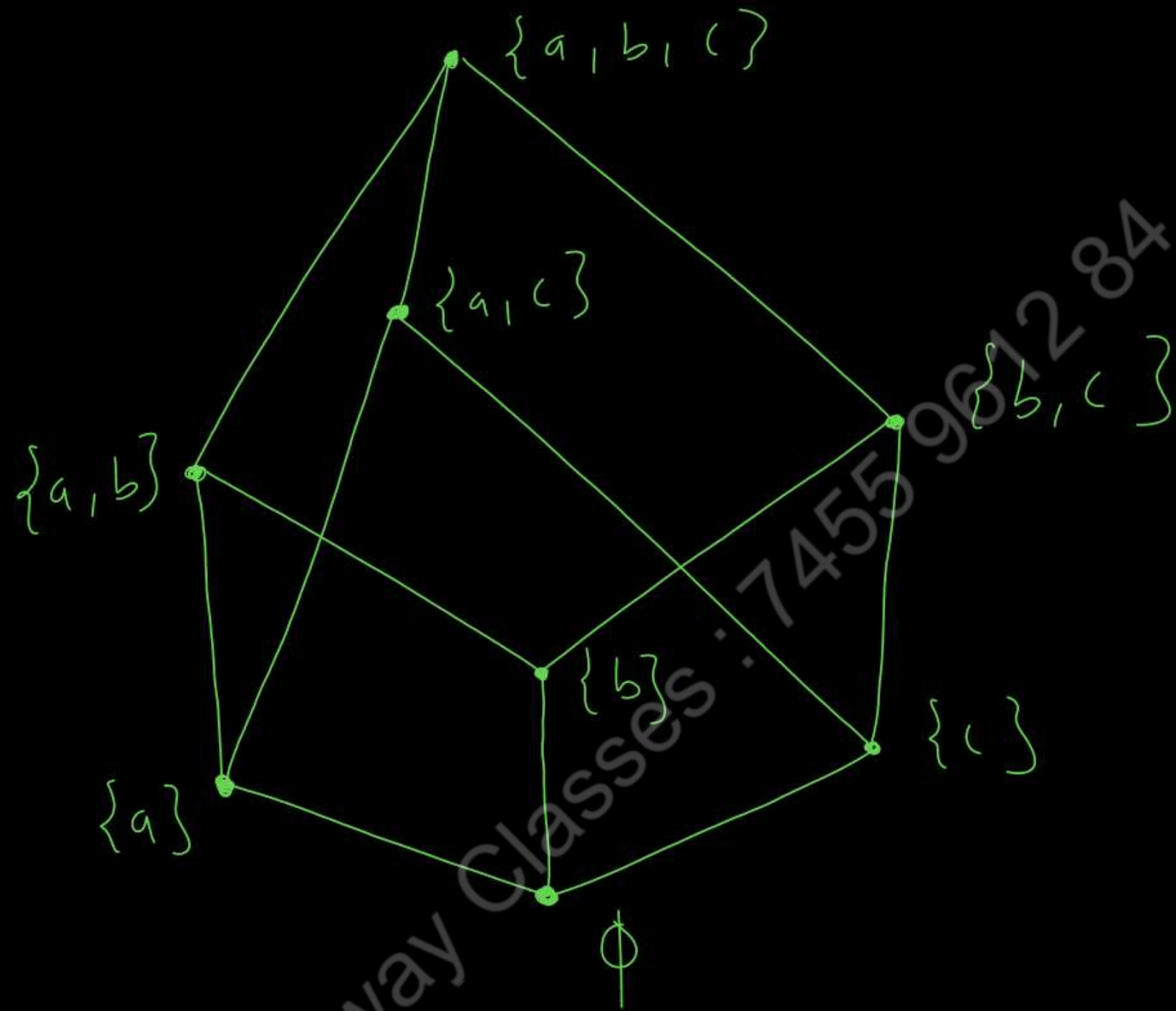
Q.6- Draw the Hasse diagram for the POSET $(P(s), \subseteq)$ where $P(s)$ is the power set on $A = \{a, b, c\}$

$$A = \{a, b, c\}$$

Subset of A

$$\phi, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\}$$

$$P(s) = \{ \phi, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\} \}$$



Q.7- Let $X = \{1, 3, 6, 12, 24, 36\}$ and the relation $\leq$ be such that $x \leq y$ if ' x divides y '
 Draw the Hasse diagram.

Relation

$x \mid y$



Gateway Classes: 7455 9612 84

Topic : Hasse's diagram

Q.1. Draw the Hasse's diagram of the POSET $(L, \subseteq)$ Where

$L = \{s_0, s_1, s_2, s_3, s_4, s_5, s_6, s_7\}$ where the sets are given by

$s_0 = \{a, b, c, d, e, f\}$, $s_1 = \{a, b, c, d, e\}$, $s_2 = \{a, b, c, e, f\}$, $s_3 = \{a, b, c, e\}$, $s_4 = \{a, b, c\}$, $s_5 = \{a, b\}$,

$s_6 = \{a, c\}$, $s_7 = \{a\}$

Q.2. Let $A = \{1, 2, 3, 4, 6, 8, 9, 12, 18, 24\}$ be ordered by the relation 'a divides b' find the Hasse diagram.

Q.3 Draw the Hasse diagram of the relation S defined as "divides" on set B where

$$B = \{2, 3, 4, 6, 12, 36, 48\}$$

Q.4 Draw Hasse diagram for the following relations on set

$$A = \{1, 2, 3, 4, 12\}$$

$$R = \{(1, 1), (2, 2), (3, 3), (4, 4), (12, 12), (1, 2), (4, 12), (1, 3), (1, 4), (1, 12), (2, 4), (2, 12), (3, 12)\}$$

DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattices

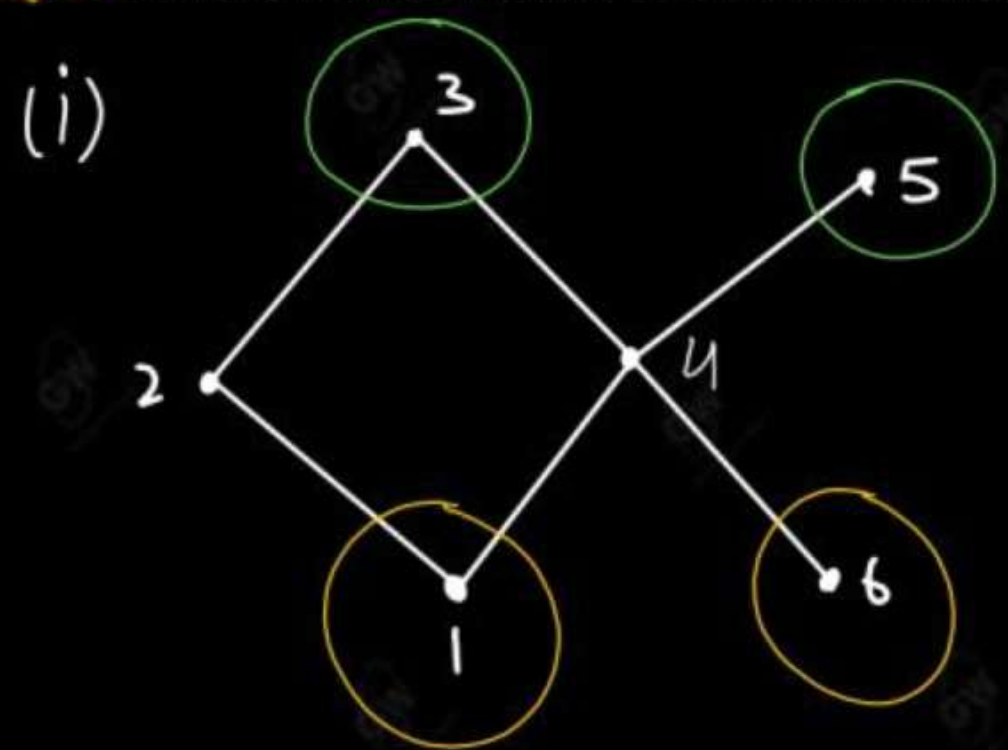
Today's Target

- Maximal and Minimal elements in POSET
- Maximum and Minimum elements in POSET
- Upper Bound and lower Bound in Hasse diagram
- Least upper Bound and Greatest Lower Bound in Hasse diagram
- PYQ
- DPP

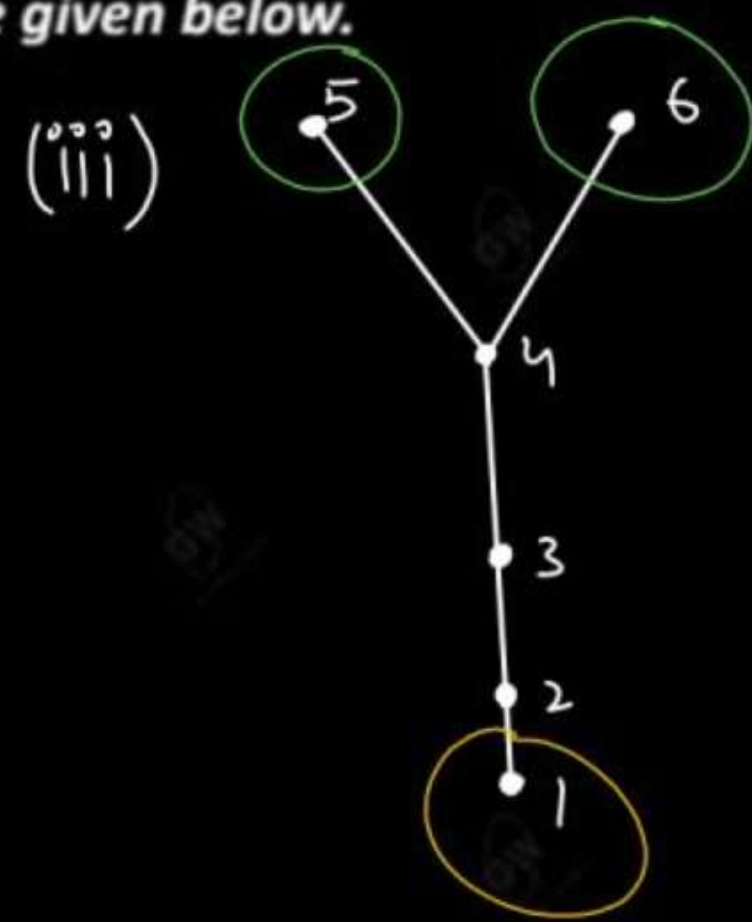
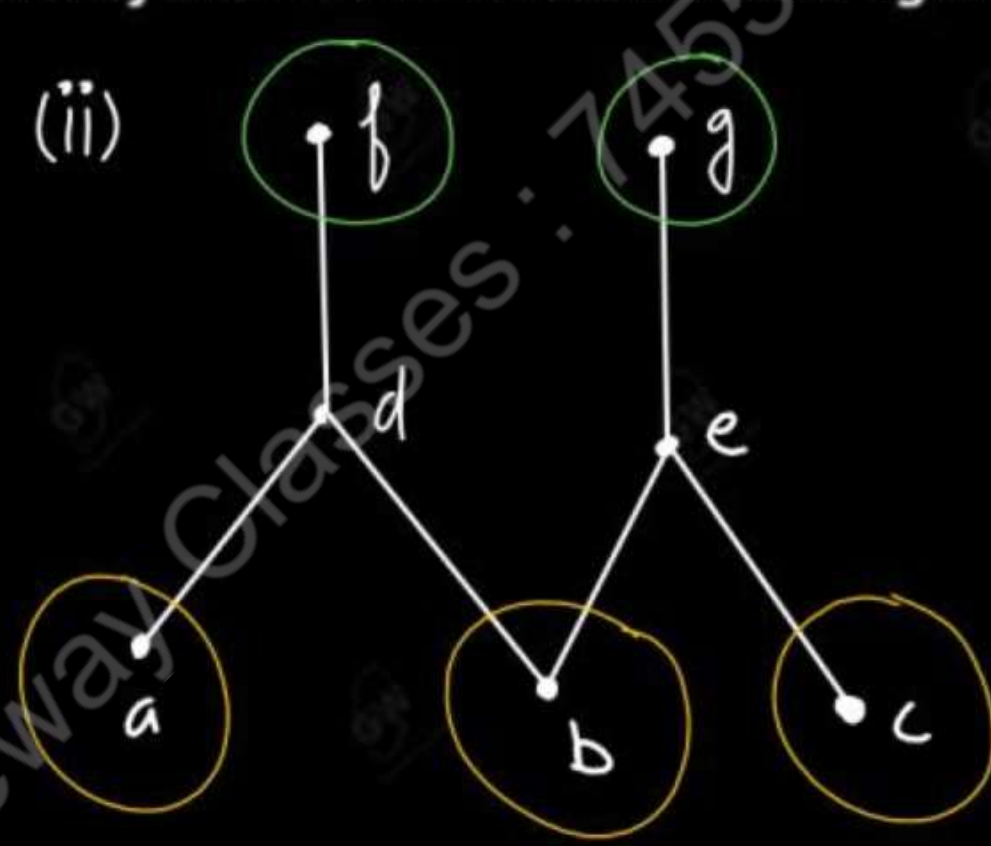
Maximal element : If in a POSET, an element is not related to any other element, then it is called maximal elements
green circle

Minimal elements : If in a POSET, no element is related to any other element, then it is called minimal elements
yellow circle

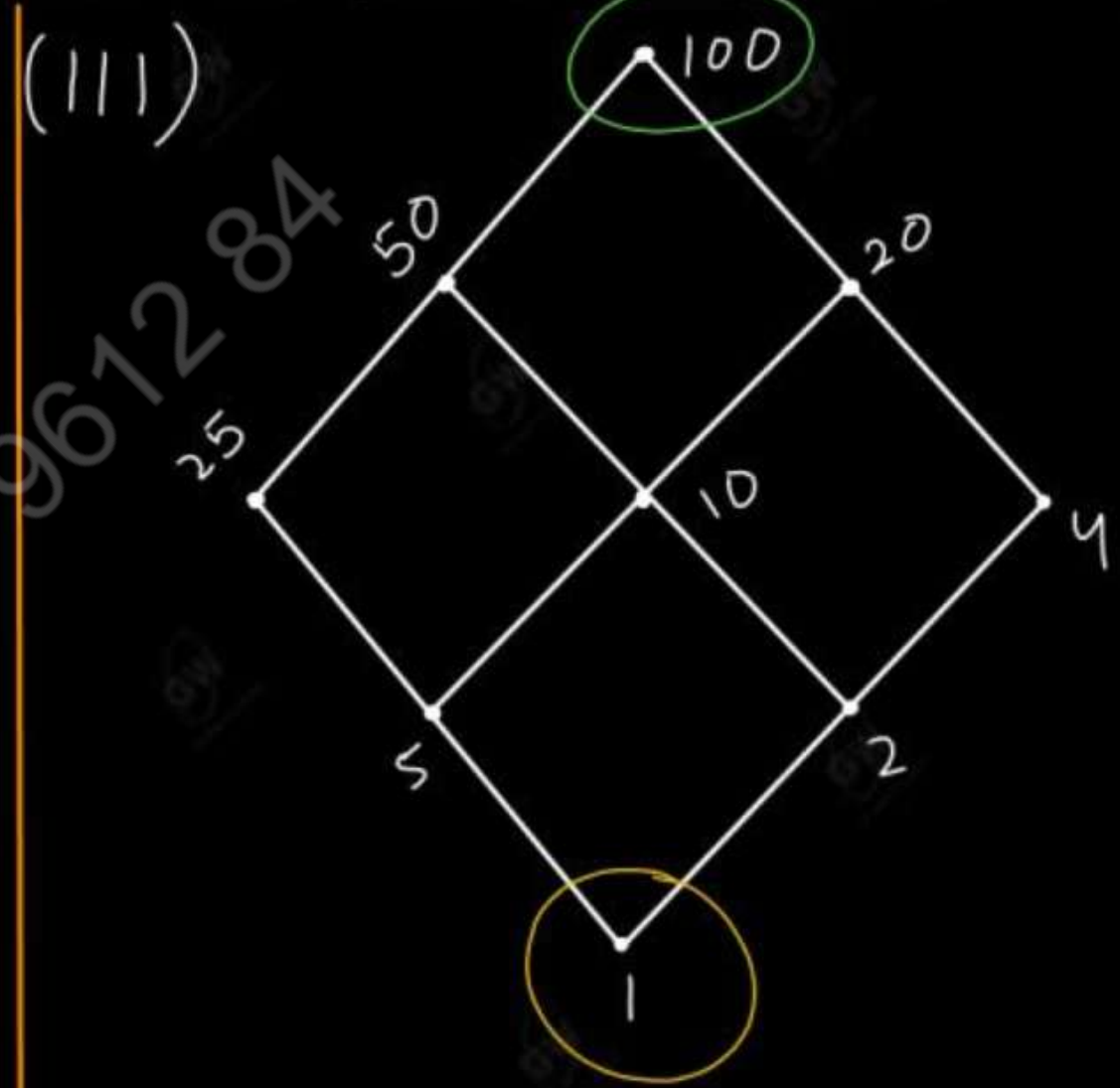
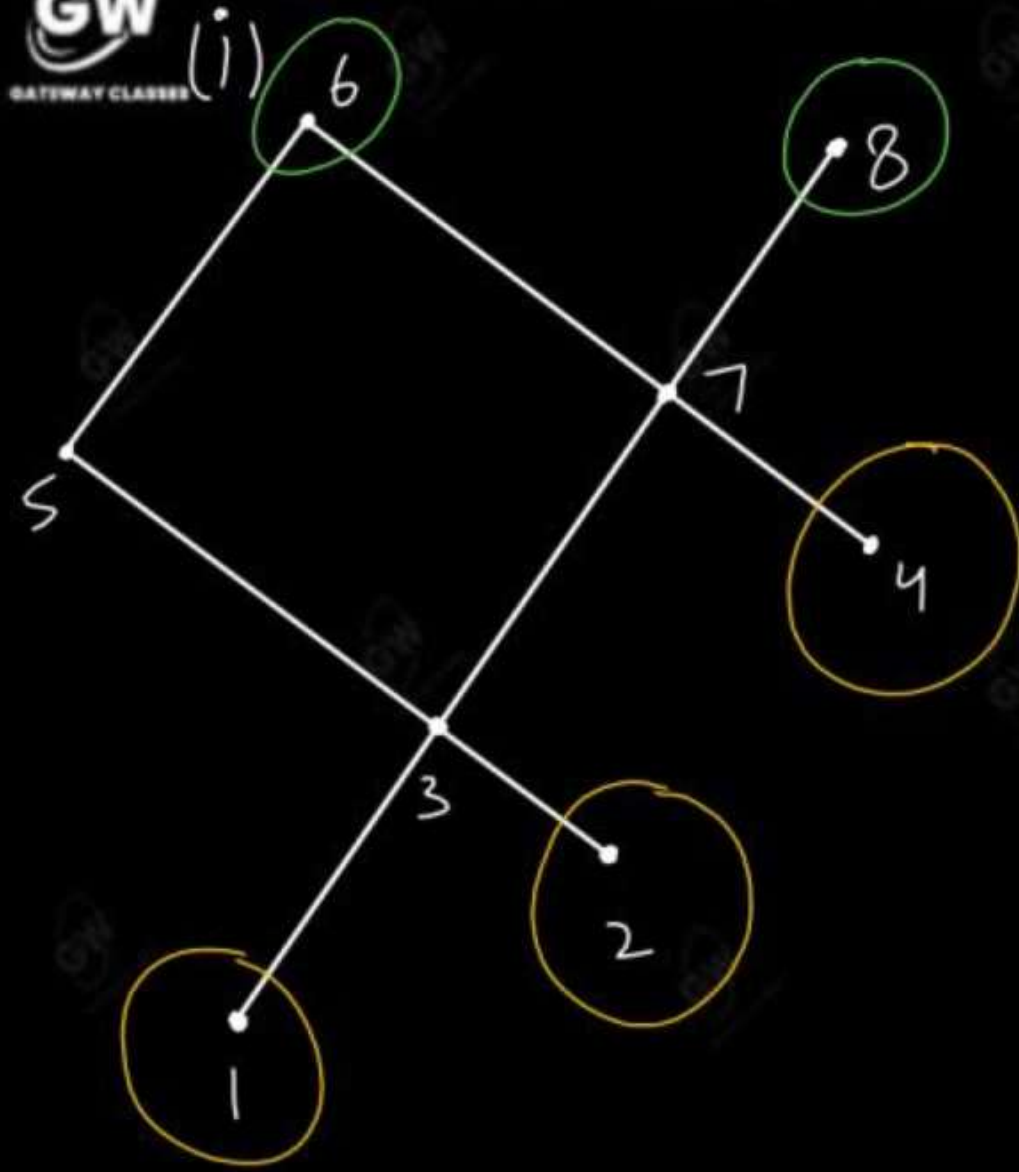
Q.1: Find all the maximal and minimal elements of the POSET whose Hasse diagrams are given below.



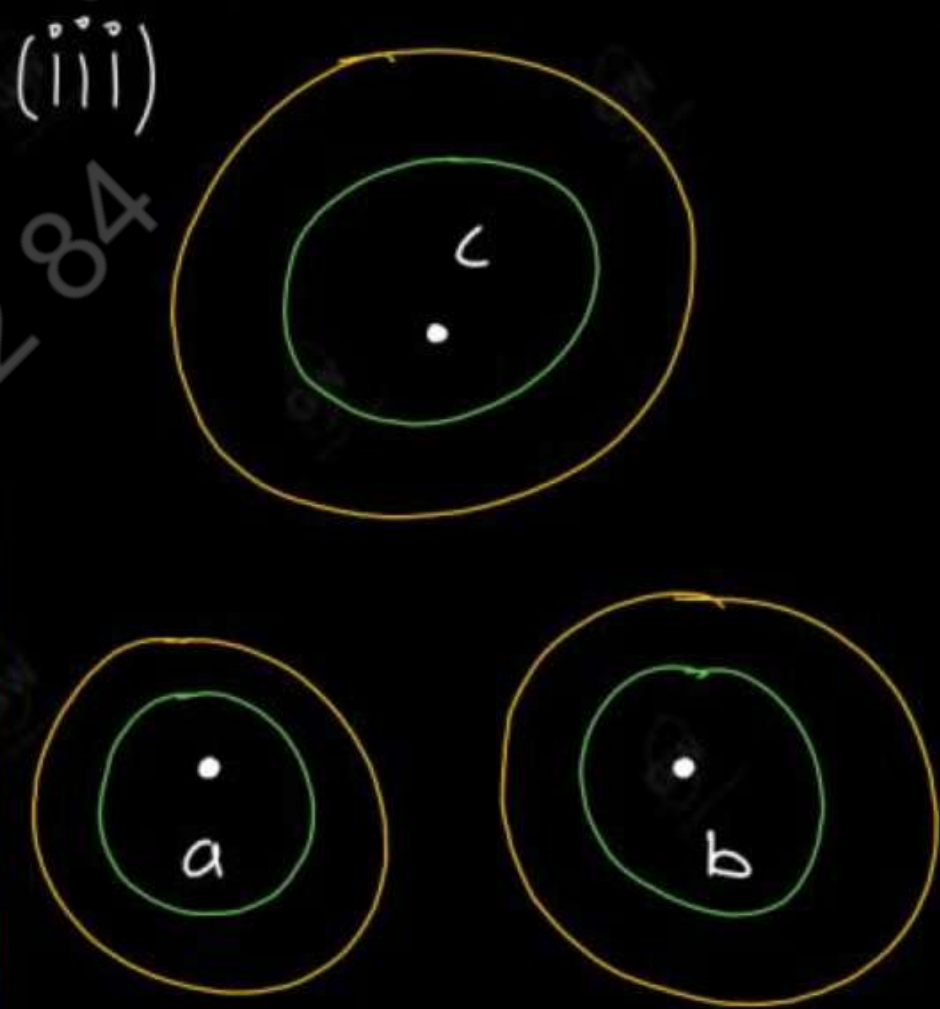
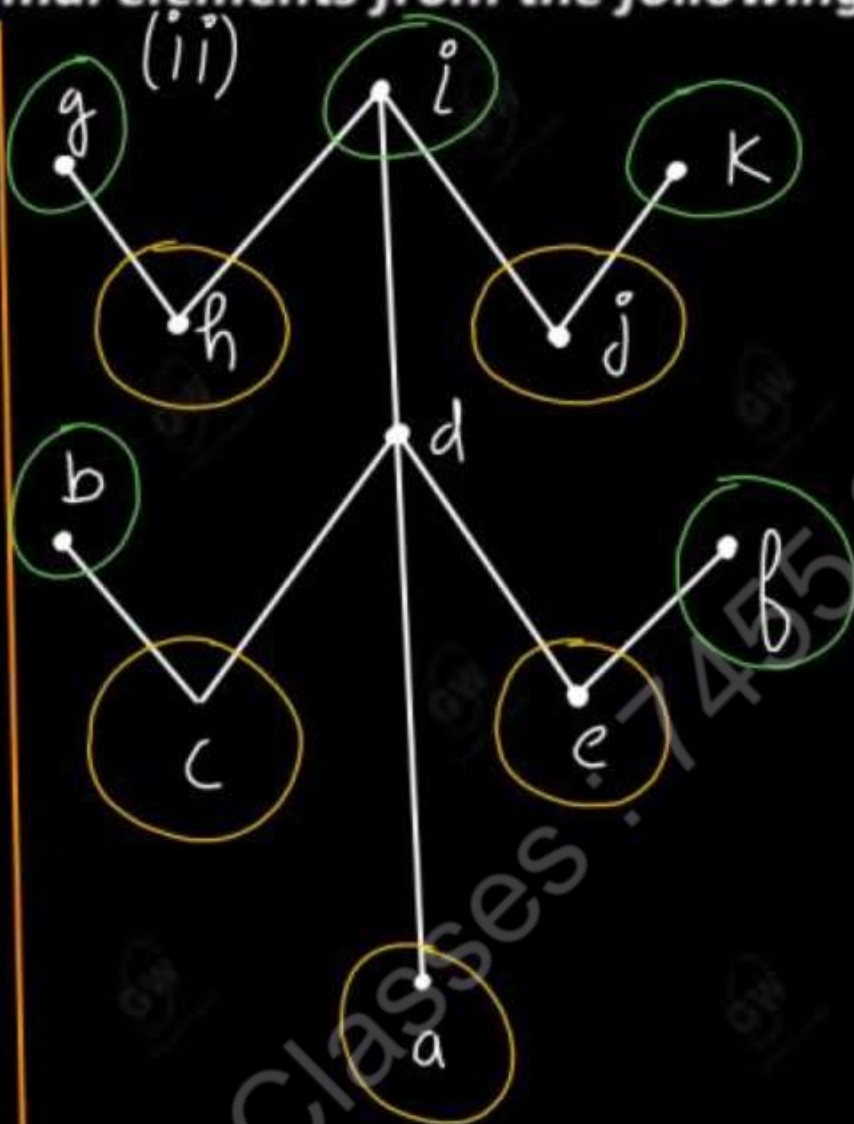
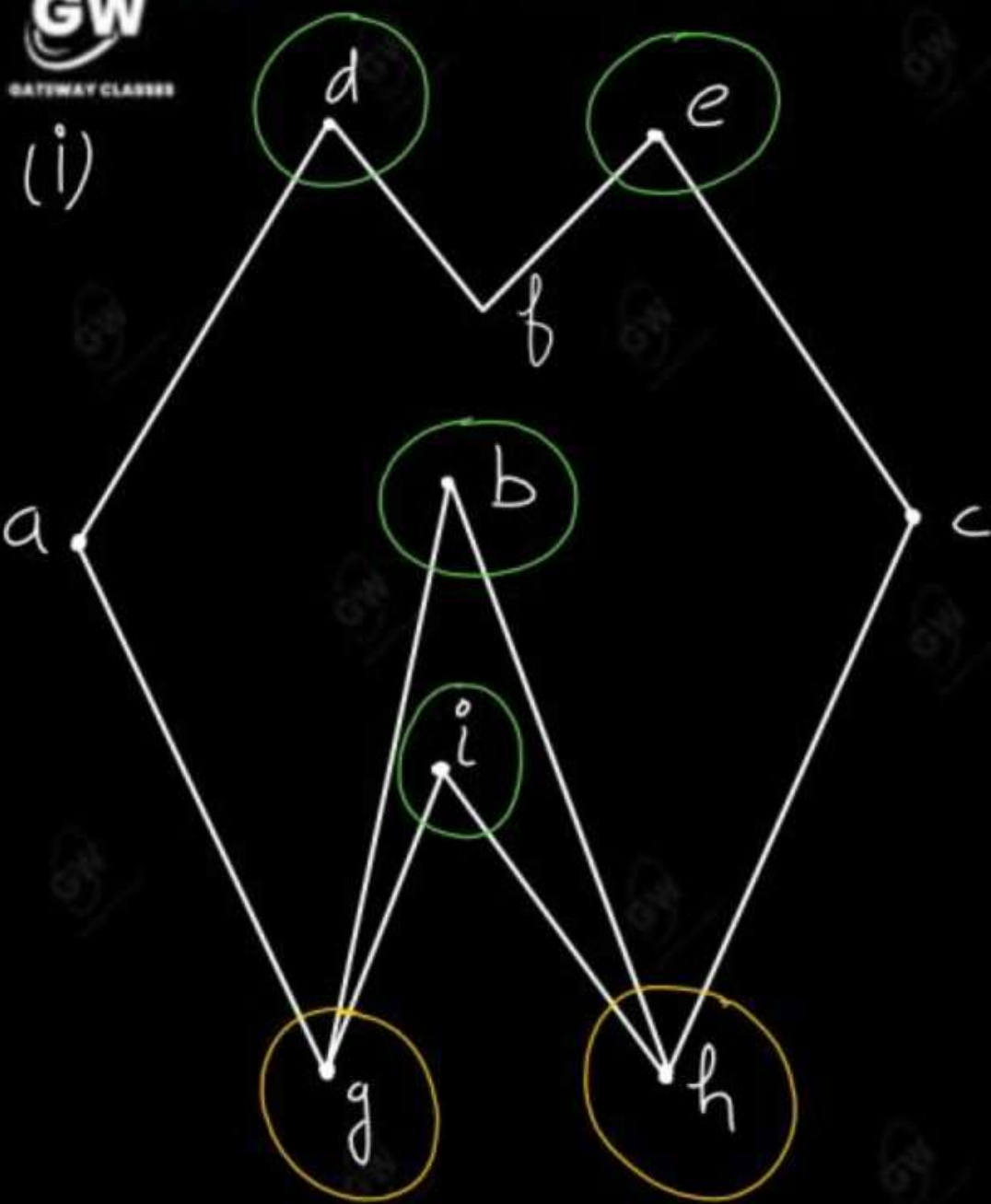
Maximal elements
= 3, 5, 6



Q.2: Find all the maximal and minimal elements of the POSET whose Hasse diagrams are given below.



Q.3:- Find all the maximal and minimal elements from the following Hasse diagrams.



Maximal = a, b, c

Minimal = a, b, c

Maximum and Minimum element

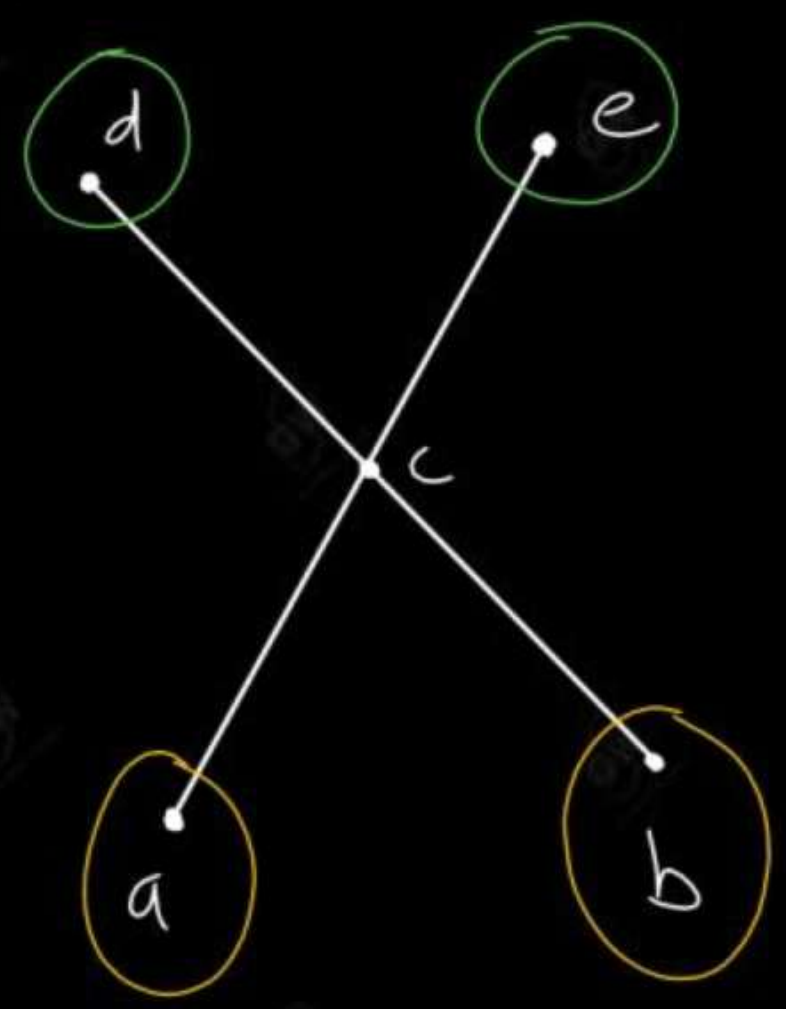
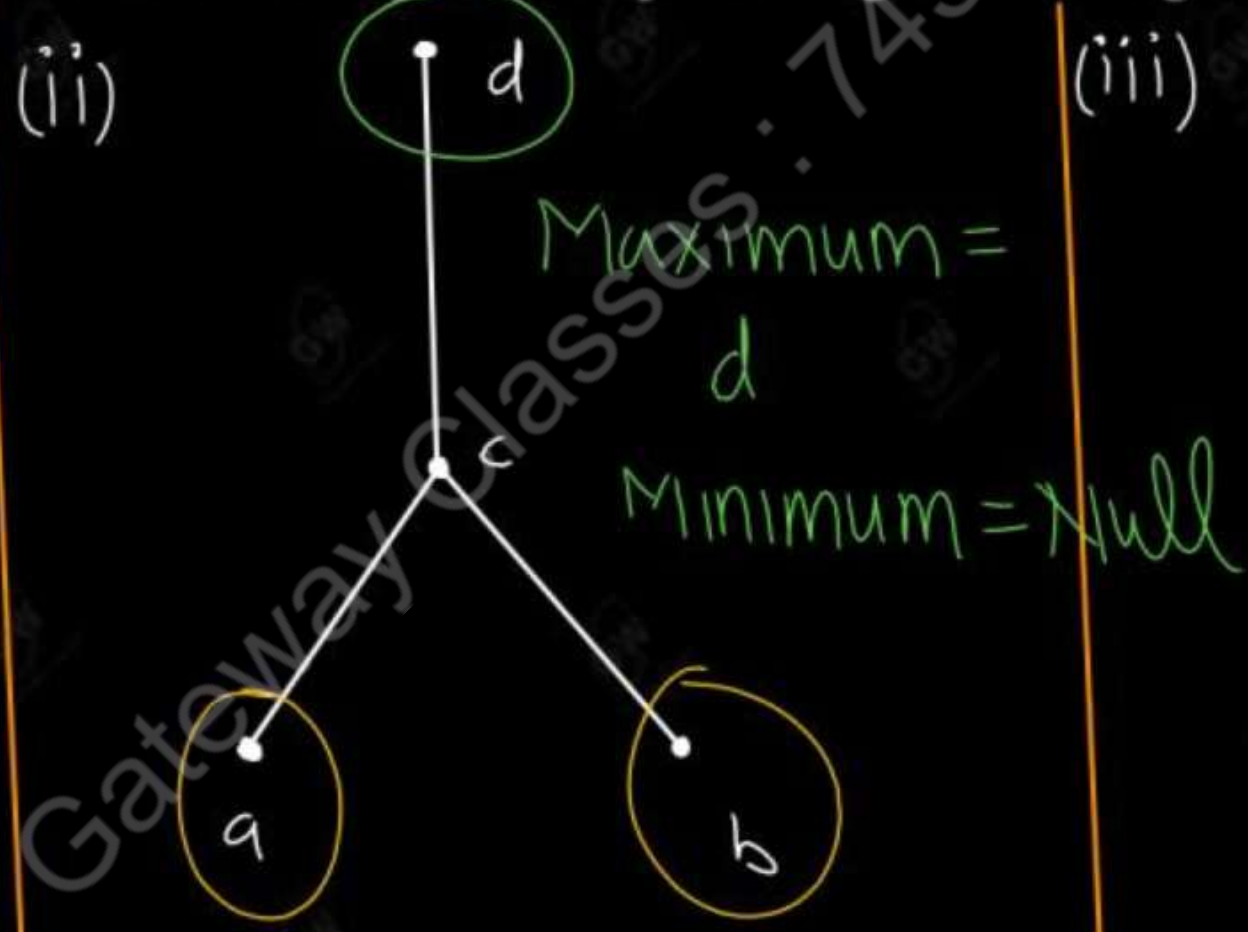
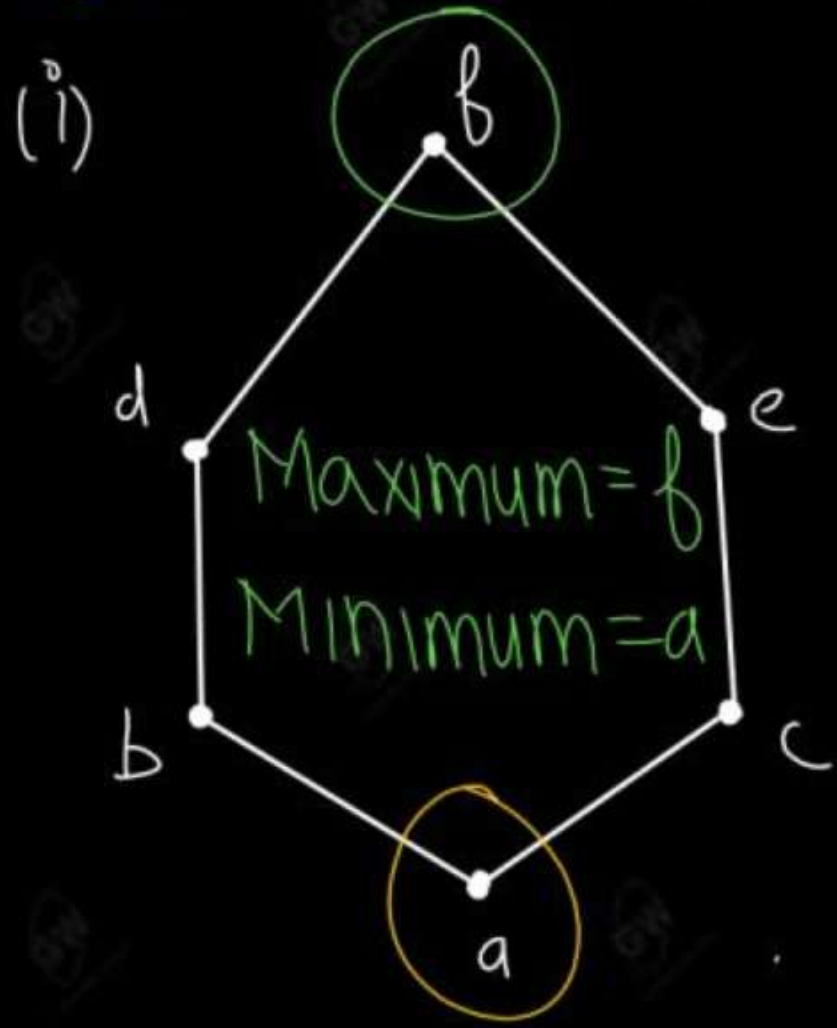
Maximum element / Greatest element: If it is maximal and every element is related to it

Minimum elements / Least element: If it is minimal and it is related to every element.

Maximum = Null

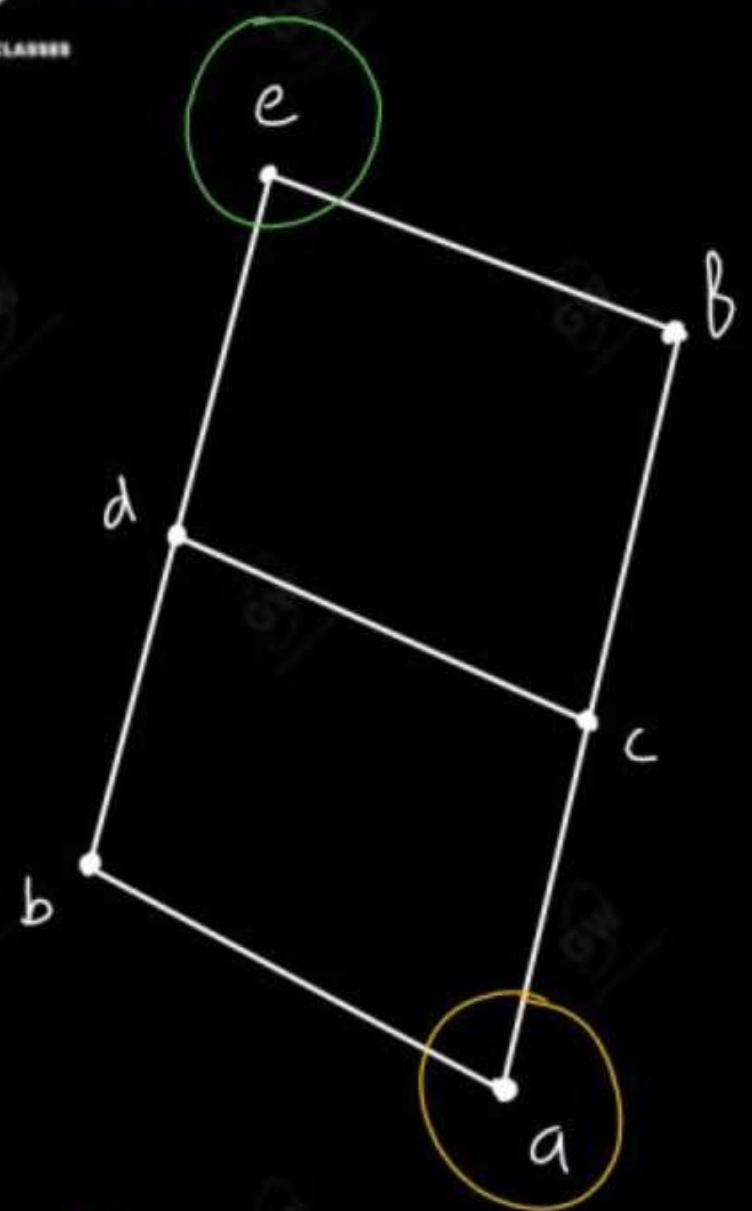
Minimum = Null

Q.4 : Find the maximum and minimum elements from the following Hasse diagrams



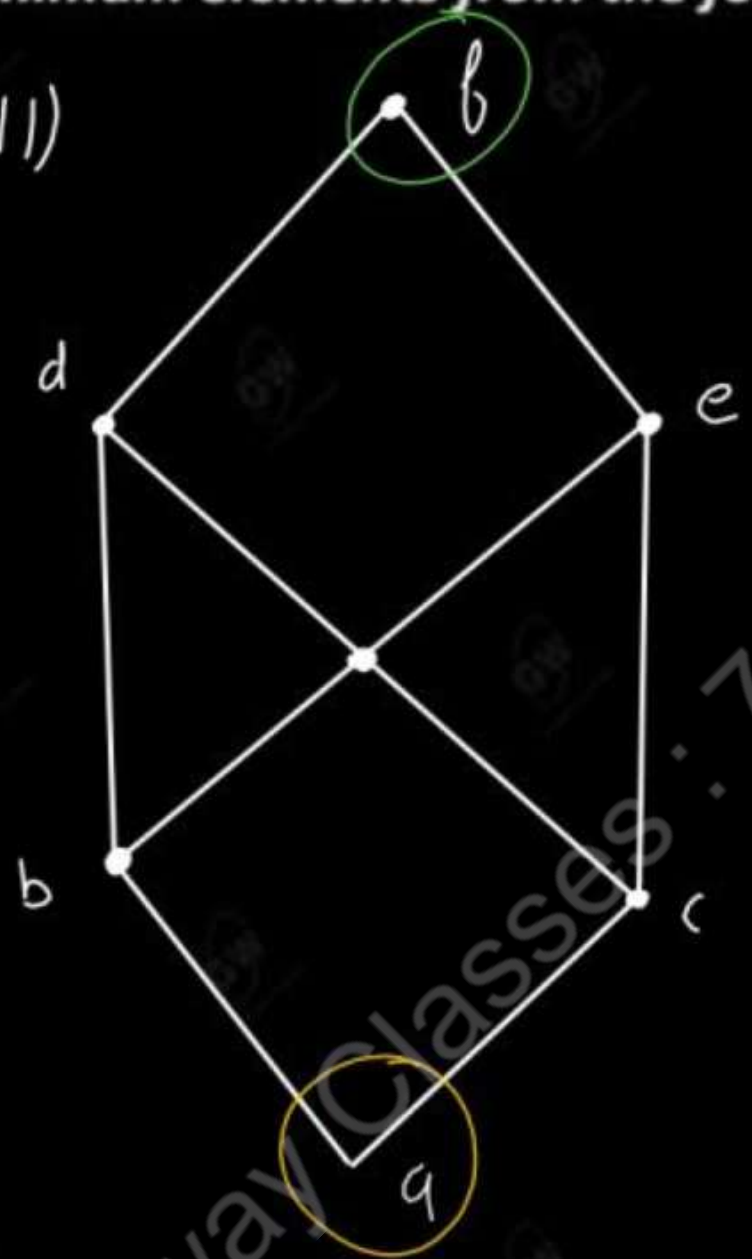
Q.5: Find the maximum and minimum elements from the following Hasse diagrams

(i)



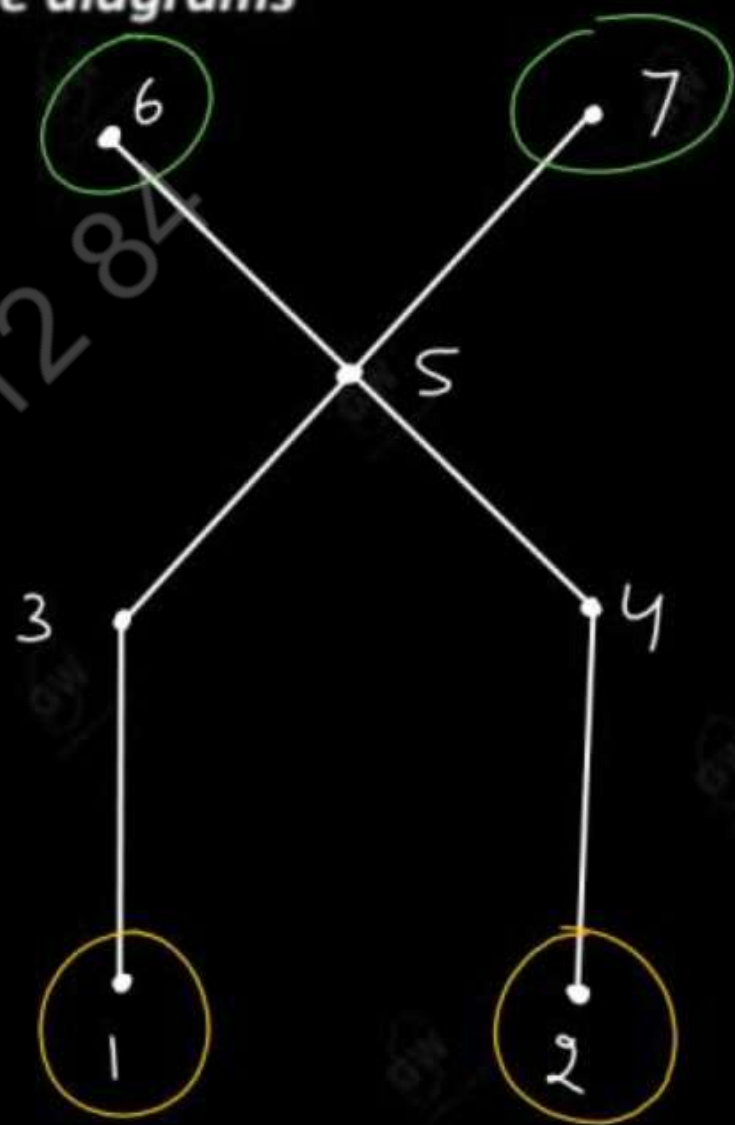
Maximum = e
Minimum = a

(ii)



Maximum = b
Minimum = a

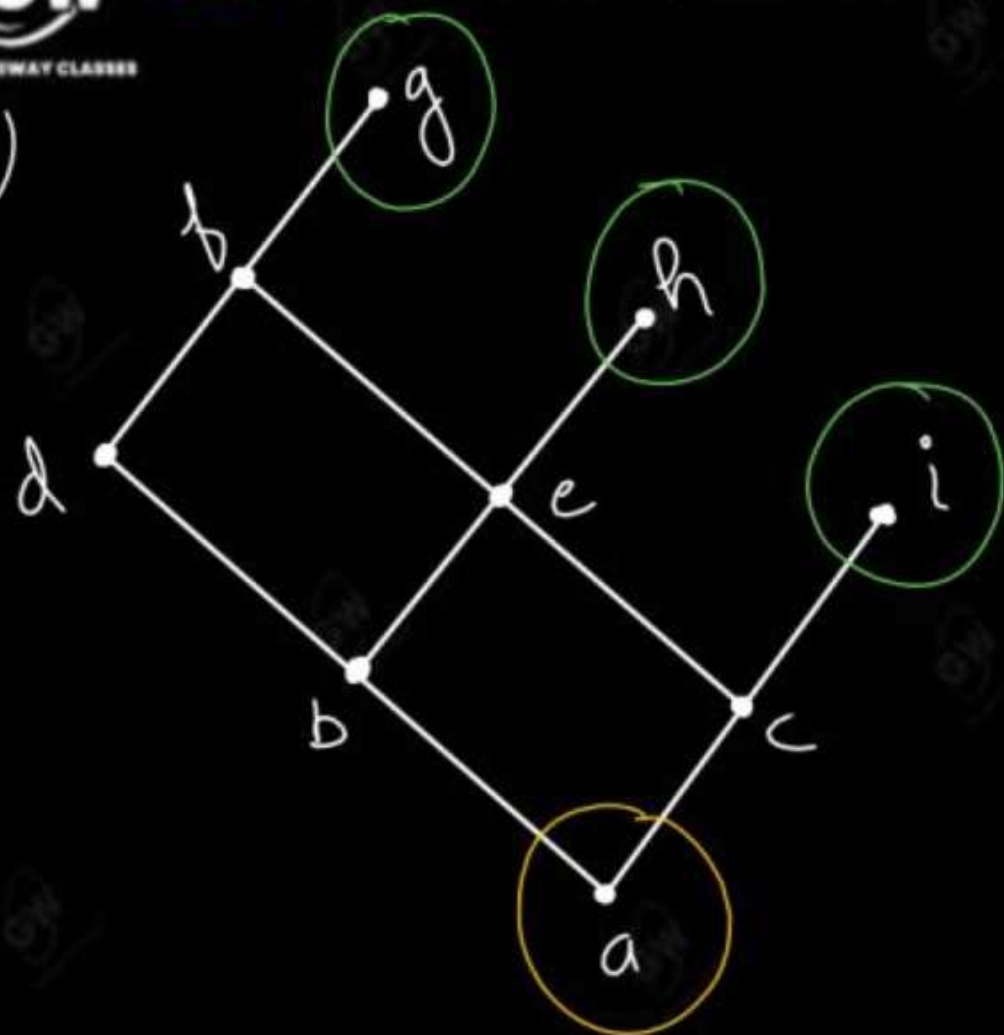
(iii)



Maximum = X
Minimum = X

Q.6: Find the Greatest and Least elements of the following Hasse diagrams.

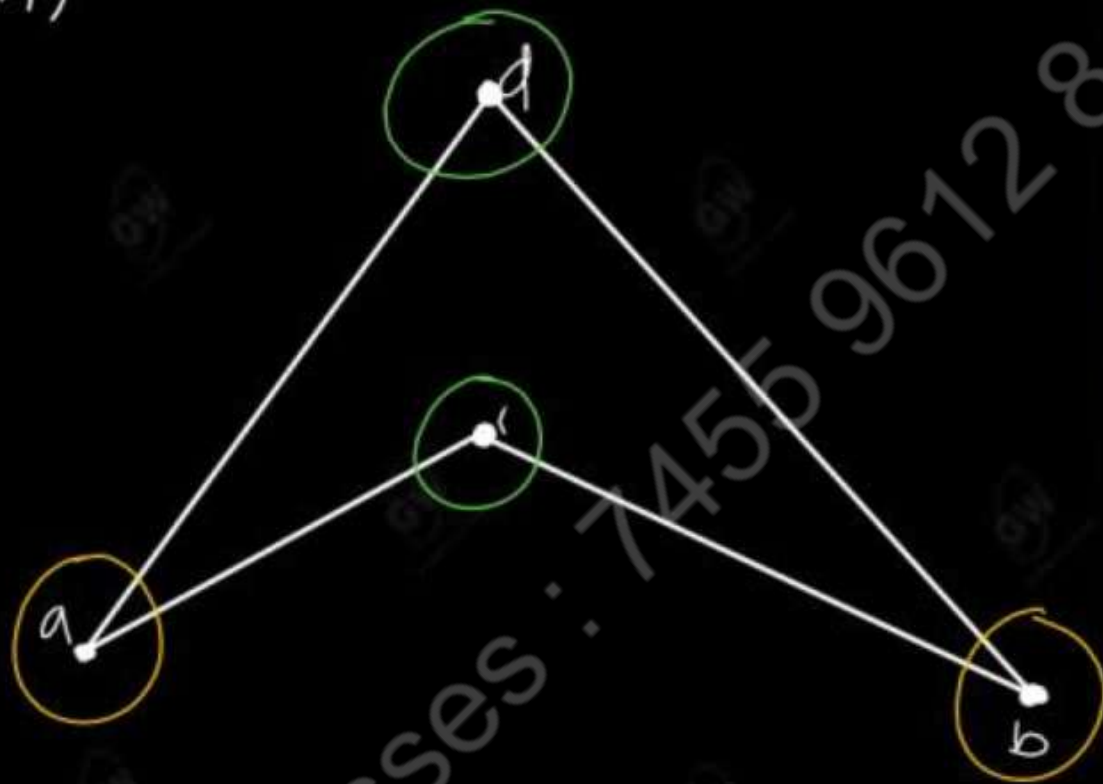
(i)



Maximum = $\times$

Minimum = a

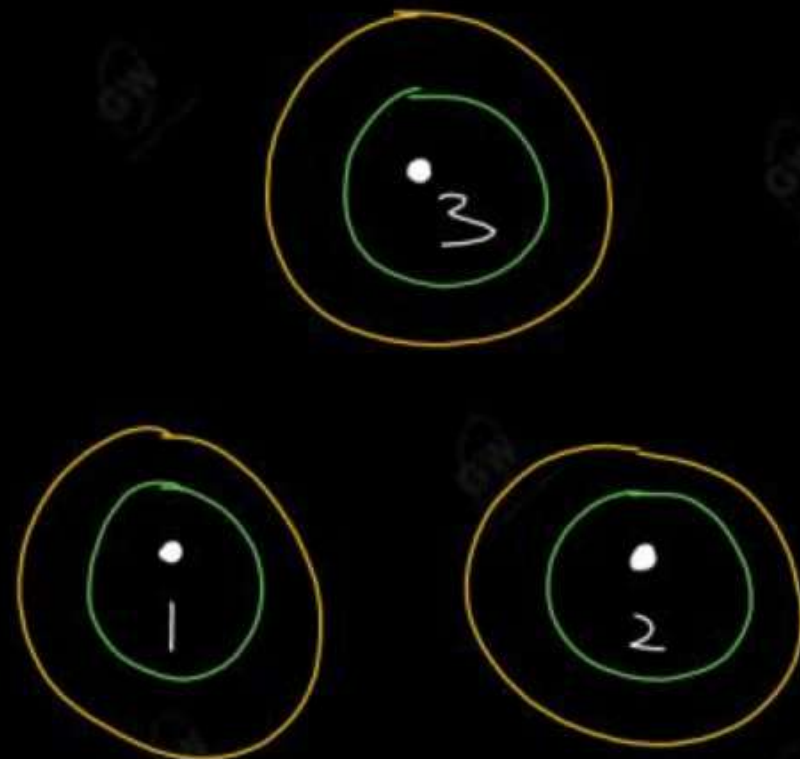
(ii)



Maximum = $\times$

Minimum = $\times$

(iii)



Maximum = $\times$

Minimum = $\times$

Upper Bound

$$B \subseteq A$$

Let $(A, \leq)$ be POSET and let B be a subset of A . An element $x \in A$ is called an upper bound of B if x succeeds every element of B . i. e

$$\text{i. e } a \leq x \quad \forall a \in B$$

Lower Bound

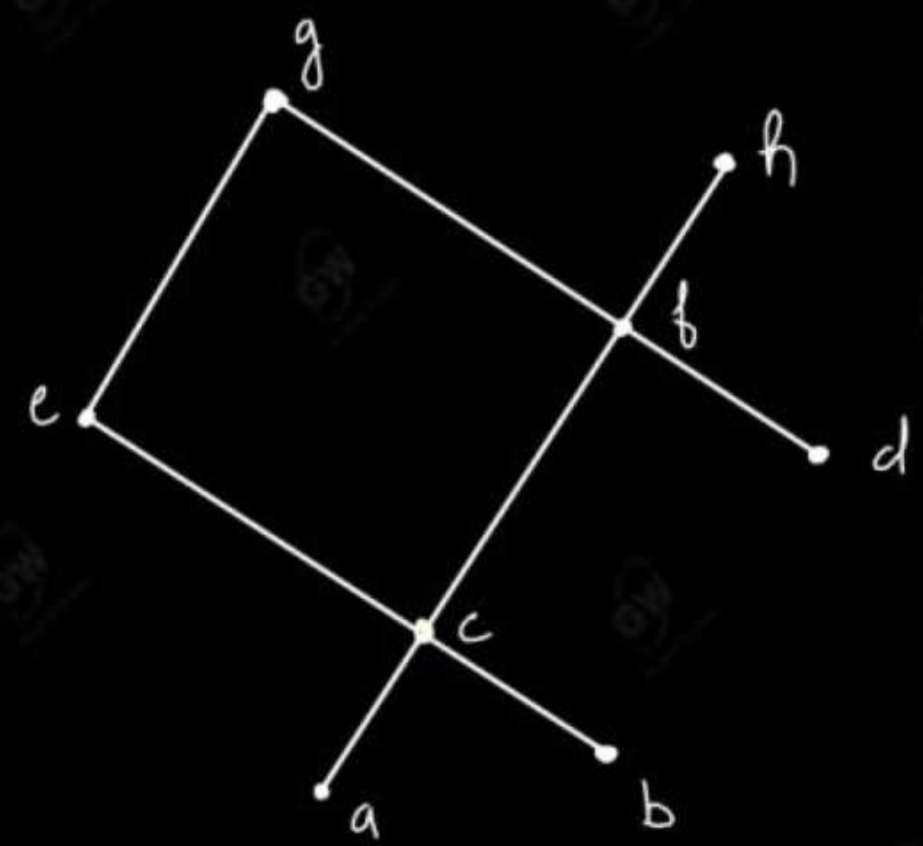
Let $(A, \leq)$ be a POSET and let B be a subset of A . An element $x \in A$ is called a Lower bound of B if x preceeds every element of B .

$$\text{i. e } x \leq a \quad \forall a \in B$$

Gateway Classes 7455 9612 81

Q.7 Find the upper Bound and Lower Bound of set B from the following

(i)



$$B = \{e, c\}$$

$$LB = a, b, c$$

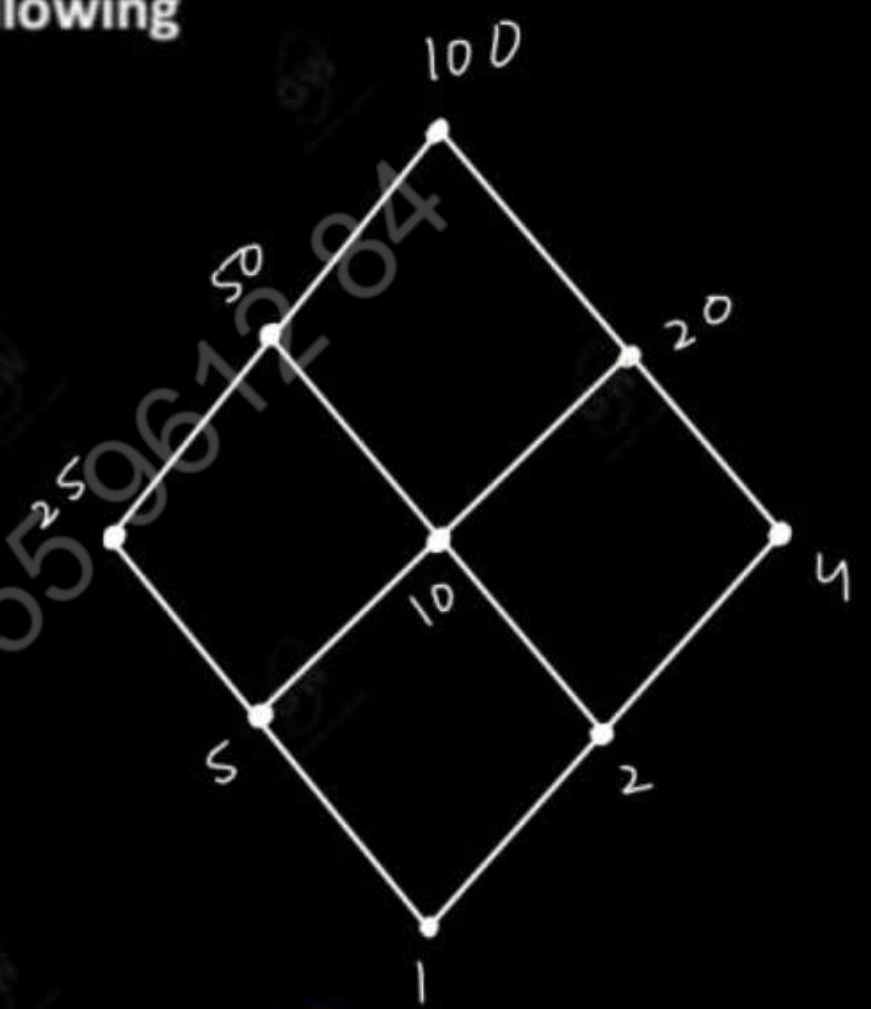
$$UB = g$$

$$B = \{c, f, d\}$$

$$LB = \emptyset$$

$$UB = b, g, h$$

(ii)



$$B = \{5, 10\}$$

$$LB = 1, 5$$

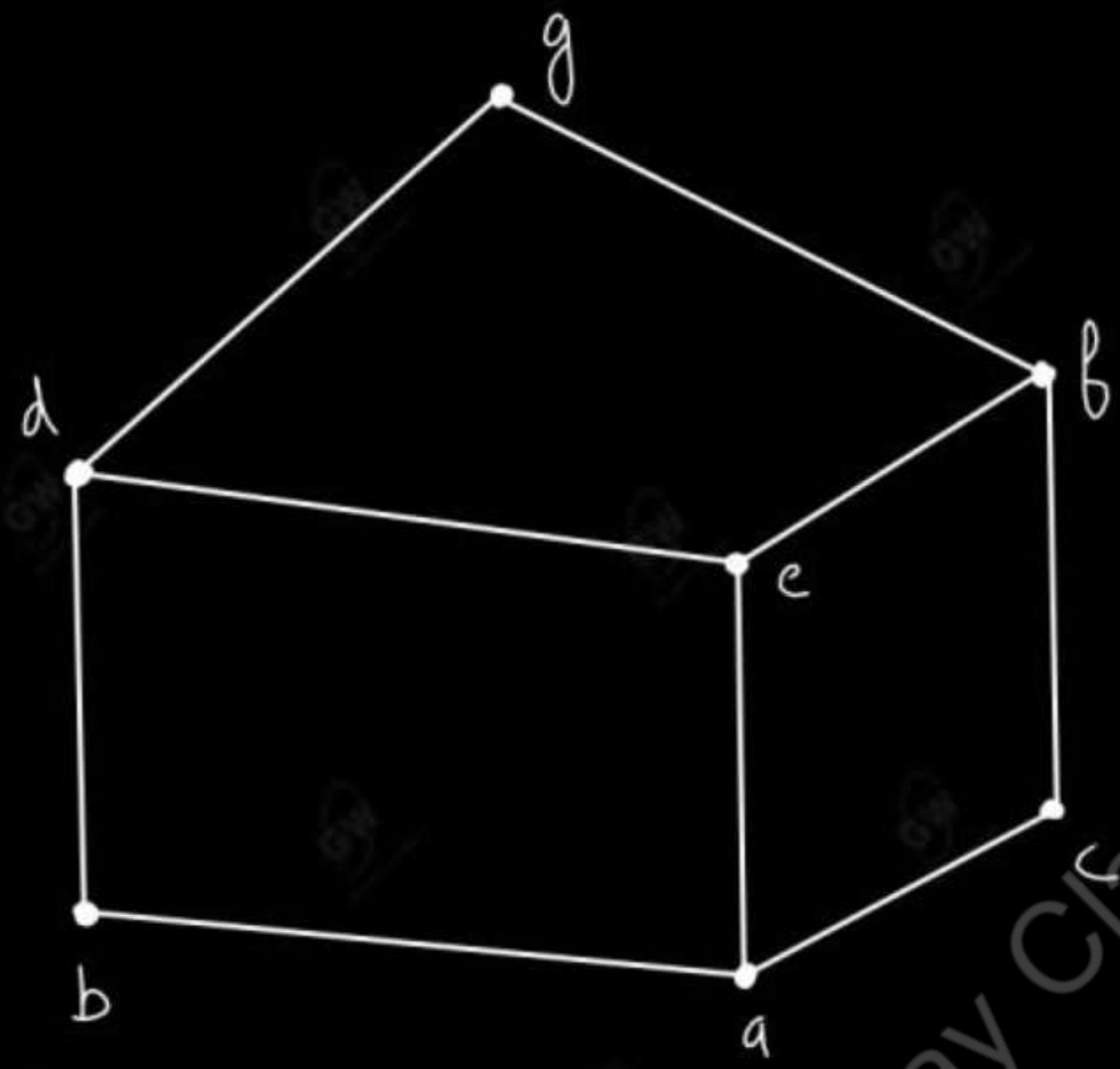
$$UB = 10, 20, 50, 100$$

$$B = \{5, 10, 2, 4\}$$

$$LB = 1$$

$$UB = 20, 100$$

Q.8 Find the upper Bound and Lower Bound of set B from the following



$$B = \{d, g\}$$

$$LB = b, a, \cancel{g}, d$$

$$UB = g$$

$$B = \{e, b\}$$

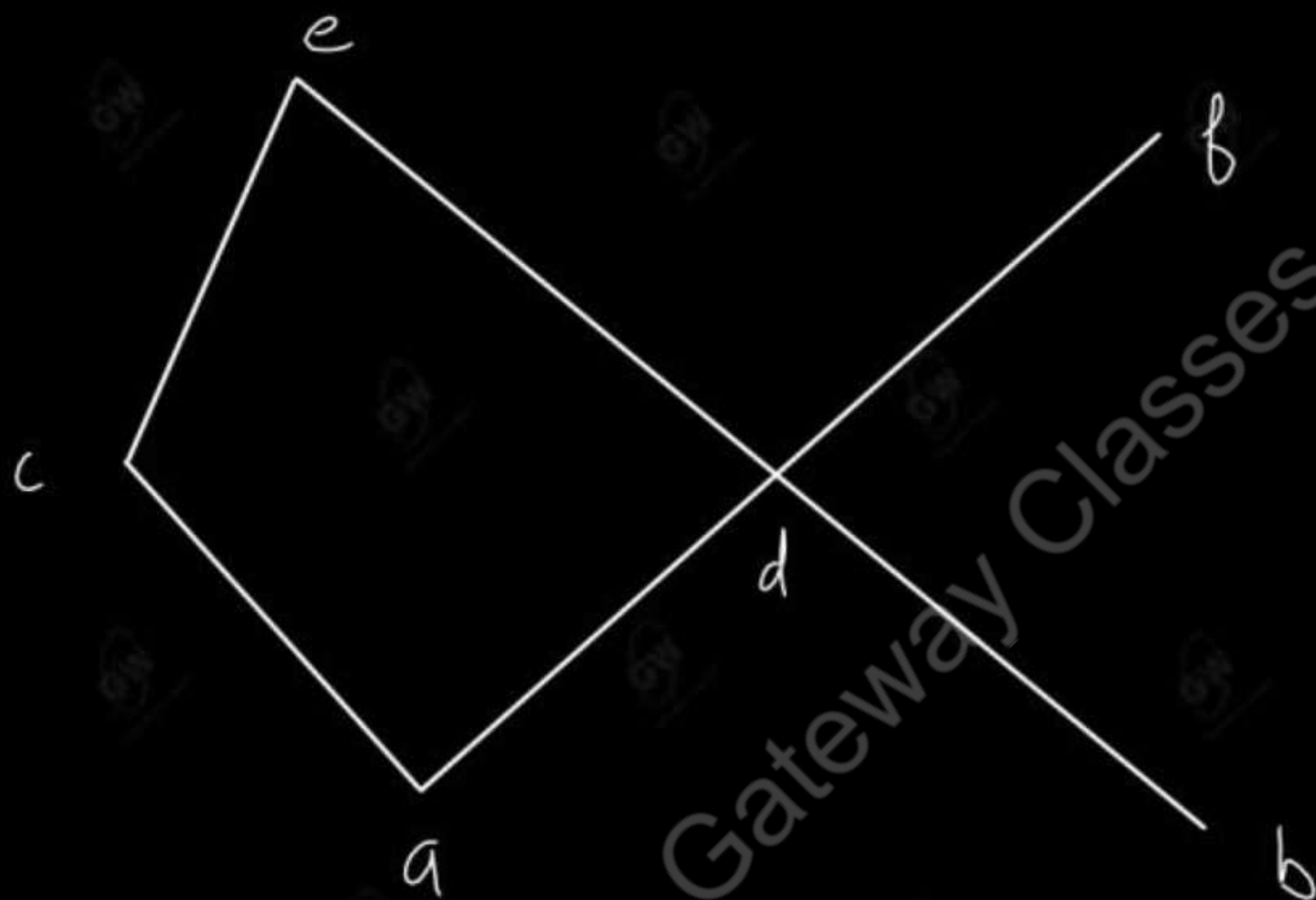
$$LB = a, e$$

$$UB = g, b$$

Gateway Classes: 7455961234

1. **Least Upper Bound / LUB/ Supremum/ Join/ ($\vee$ or $\cup$)** : Least (minimum) element in upper Bound
2. **Greatest Lower Bound/GLB/Infimum/Meet / ($\wedge$ or $\cap$)** : Greatest (maximum) element in lower Bound

Q.9 Find the Least upper Bound and Greatest Lower Bound of set B from the following



$$B = \{c, d\}$$

$$UB = e$$

$$LUB = e$$

$$LB = a$$

$$GLB = a$$

$$B = \{a, b\}$$

$$UB = d, e, f$$

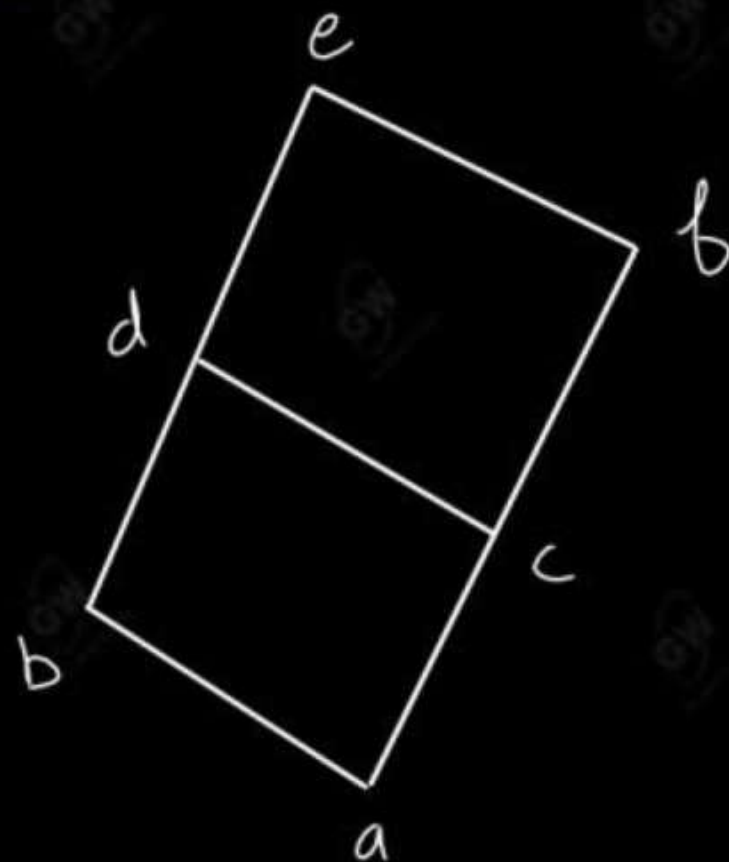
$$LUB = d$$

$$LB = \phi$$

$$GLB = \phi$$

Q.10 Find the Least upper Bound and Greatest Lower Bound of set B from the following

(i)



$$B = \{a, c\}$$

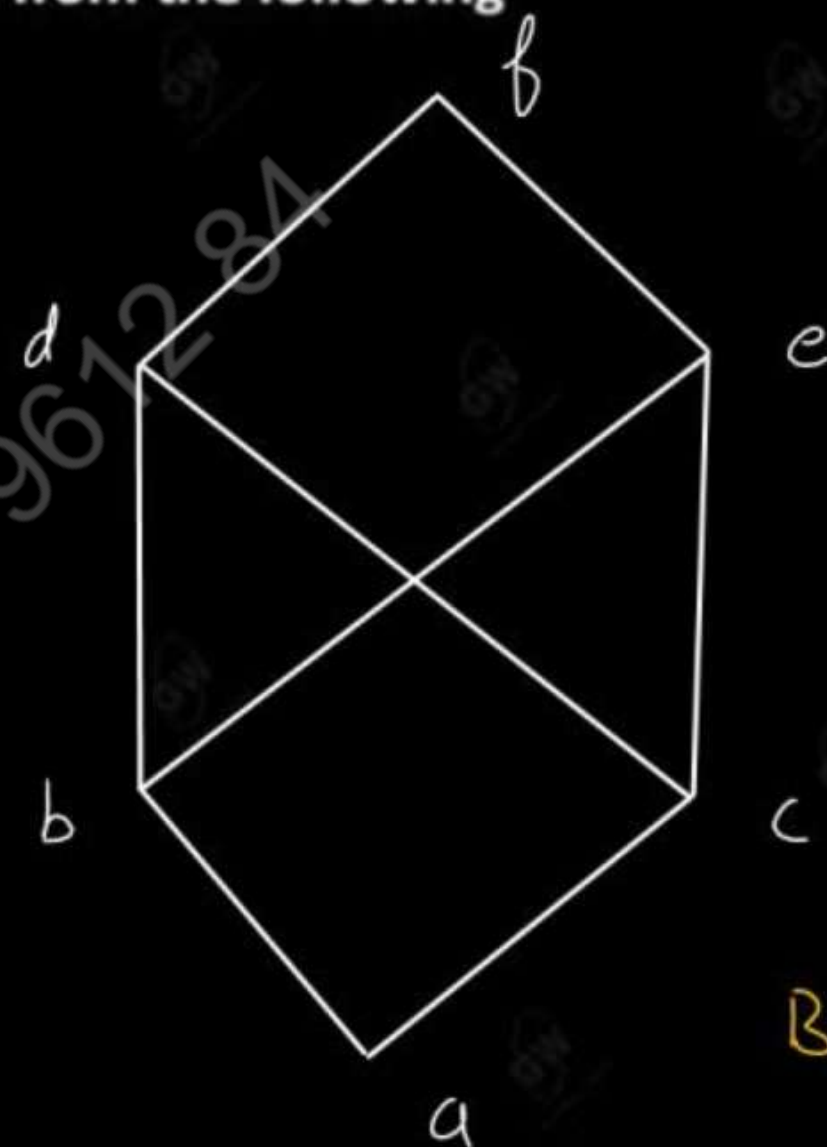
$$UB = c, b, d, e$$

$$LUB = c$$

$$LB = a$$

$$GLB = a$$

(ii)



$$B = \{d, e\}$$

$$UB = f$$

$$LUB = f$$

$$LB = a, b, c$$

$$GLB = \phi$$

$$B = \{b, c\}$$

$$UB = d, e, f$$

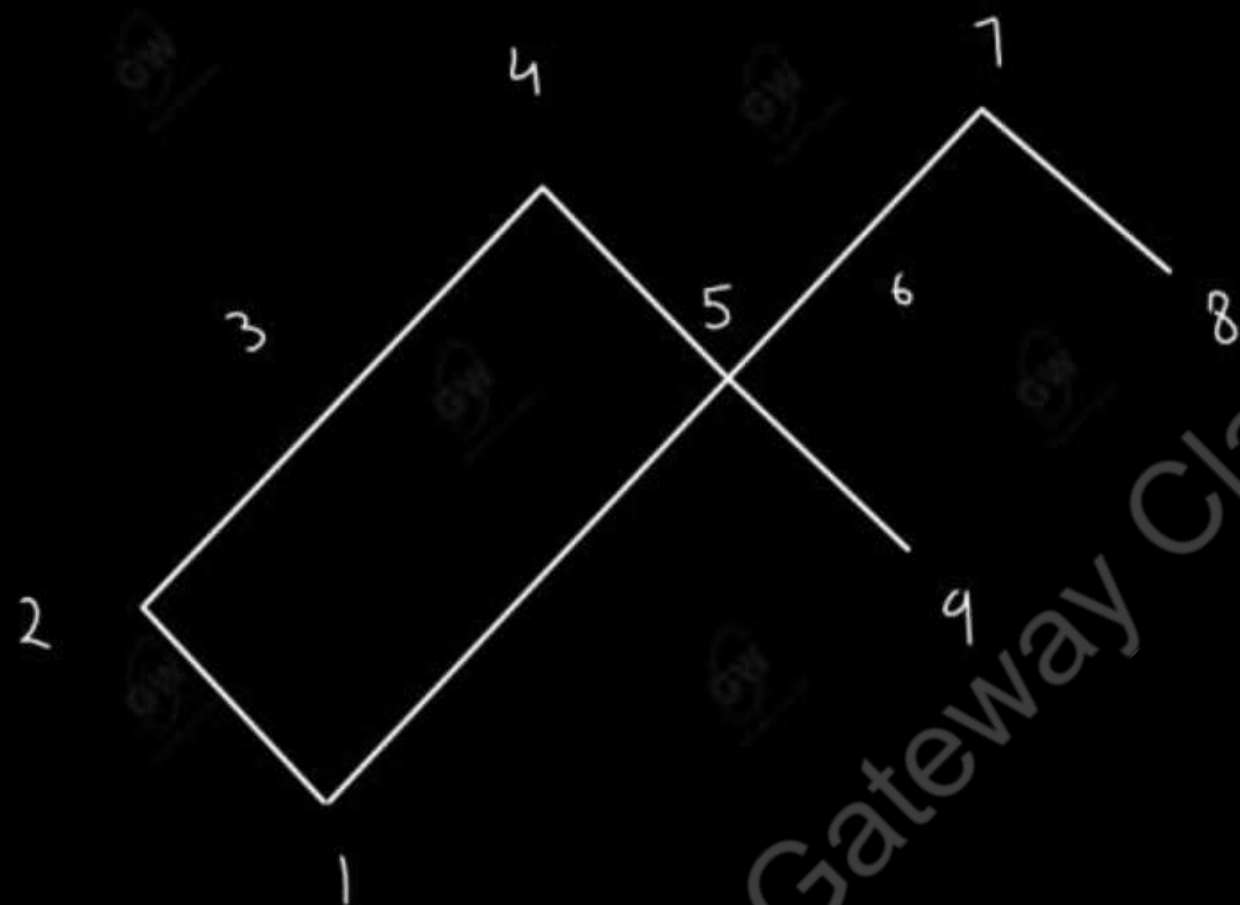
$$LUB = \phi$$

$$LB = a$$

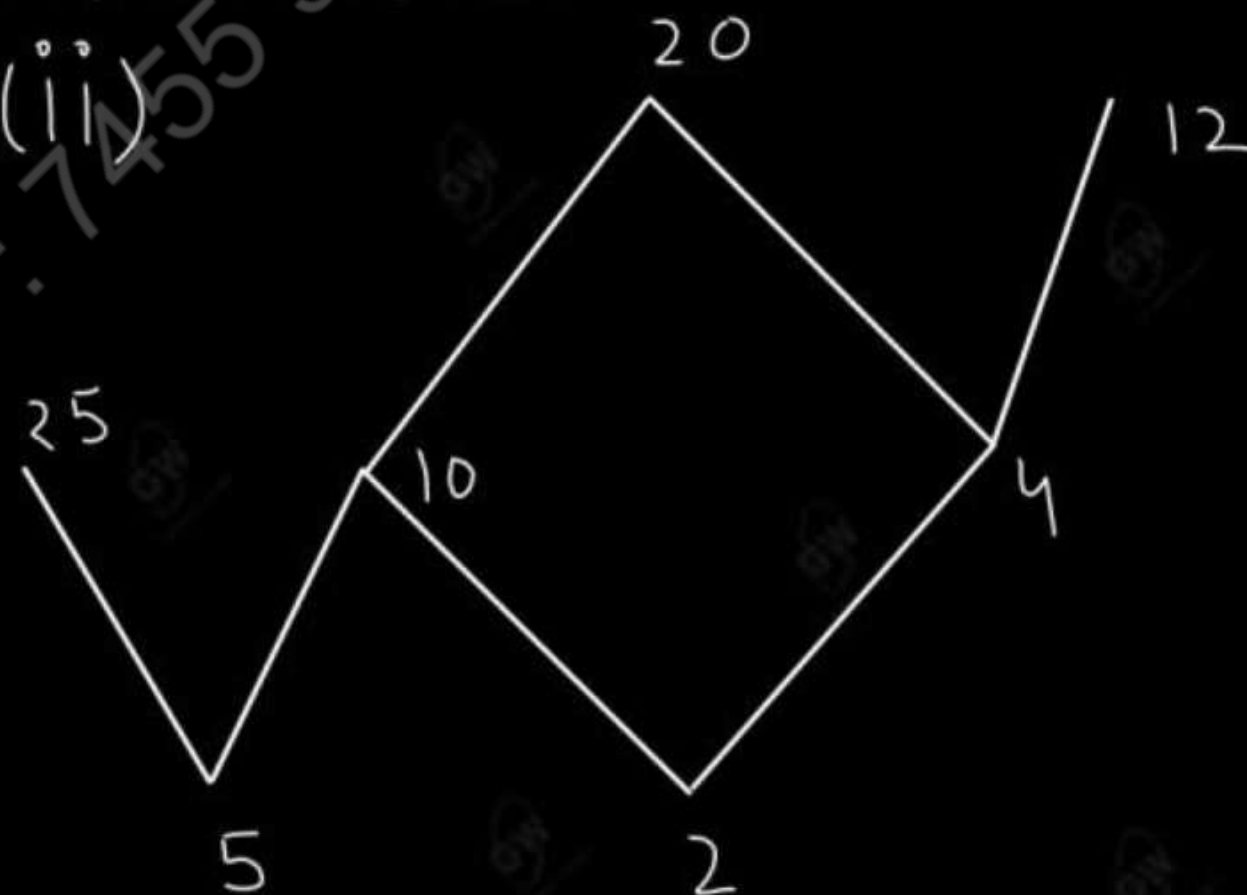
$$GLB = a$$

Q.1. Find all the maximal and minimal elements from the following Hasse diagrams.

(i)



(ii)



DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattice

Today's Target

- **LATTICE**
- *Properties of Lattice*
- *Distributive Lattice*
- *Bounded Lattice*
- *Complemented lattice*
- PYQ
- DPP

Join Semi Lattice

A POSET $(L, \leq)$ is a join semi lattice if $\forall a, b \in L, a \vee b$ exist in L .

Note: $a \vee b = \text{lub} \{a, b\} = \text{sup} \{a, b\}$

$a \vee b = a \text{ join } b$

lub – least upper bound & sup – supremum

OR

In a POSET if join exist for every pair of elements, then POSET is called Join Semi Lattice

Meet Semi Lattice

A POSET $(L, \leq)$ is a meet semi lattice if $\forall a, b \in L, a \wedge b$ exist in L .

Note: $a \wedge b = \text{glb} \{a, b\} = \text{inf} \{a, b\}$

$a \wedge b = a \text{ meet } b$

glb – greatest lower bound & inf – infimum

OR

In a POSET if meet exist for every pair of elements, then POSET is called Join Semi Lattice

$a \vee b$

$a \wedge b$

A POSET $(L, \leq)$ is a lattice if $\forall a, b \in L, a \vee b$ and $a \wedge b$ exist in L .

Where $a \vee b = \text{lub} \{a, b\} = \text{sup} \{a, b\}$

$a \wedge b = \text{glb} \{a, b\} = \text{inf} \{a, b\}$

Note: $a \vee b = a \text{ join } b$ and $a \wedge b = a \text{ meet } b$

lub – least upper bound & sup – supremum

glb – greatest lower bound & inf – infimum

OR

In a POSET if join and meet exist for every pair of elements, then POSET is called Lattice

OR

A POSET $(L, \leq)$ is a lattice if it is :

(i) Join Semi Lattice

(ii) Meet Semi Lattice

Q.1. Determine whether the following Hasse diagram represent lattice or not

(i)



lub table

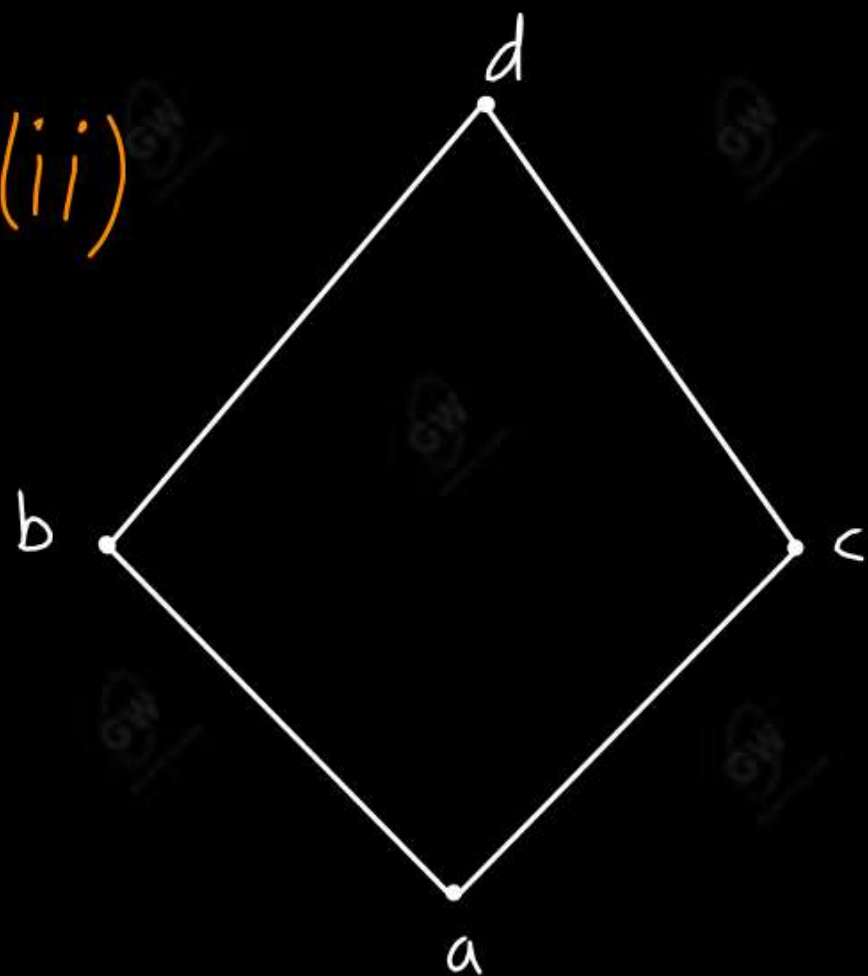
$\vee$	a	b	c	d
a	a	b	c	d
b	b	b	c	d
c	c	c	c	d
d	d	d	d	d

glb table

$\wedge$	a	b	c	d
a	a	a	a	a
b	a	b	b	b
c	a	b	c	c
d	a	b	c	d

It represent a lattice

(iii)



lub table

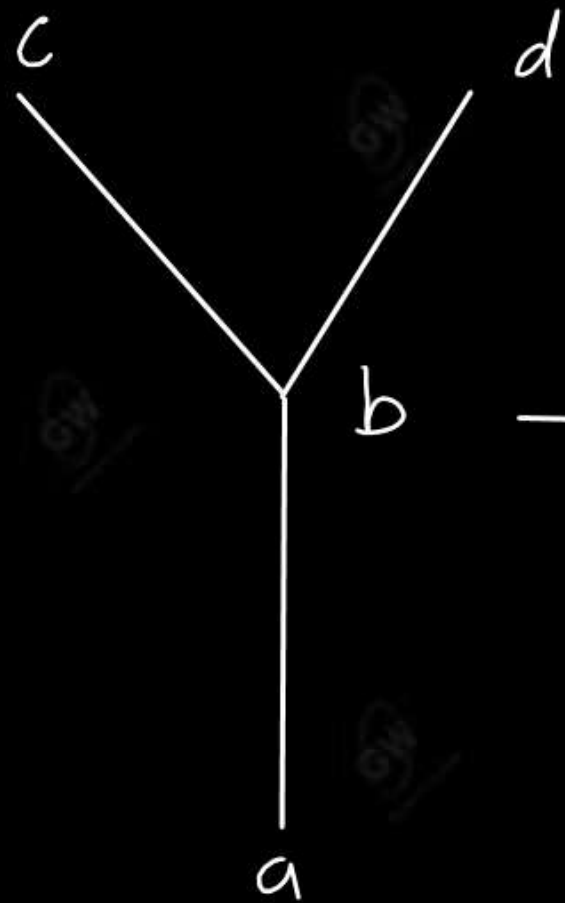
$\vee$	a	b	c	d
a	a	b	c	d
b	b	b	d	d
c	c	d	c	d
d	d	d	d	d

glb table

$\wedge$	a	b	c	d
a	a	a	a	a
b	a	b	a	b
c	a	a	c	c
d	a	b	c	d

It represent a lattice

(iii)



lub table

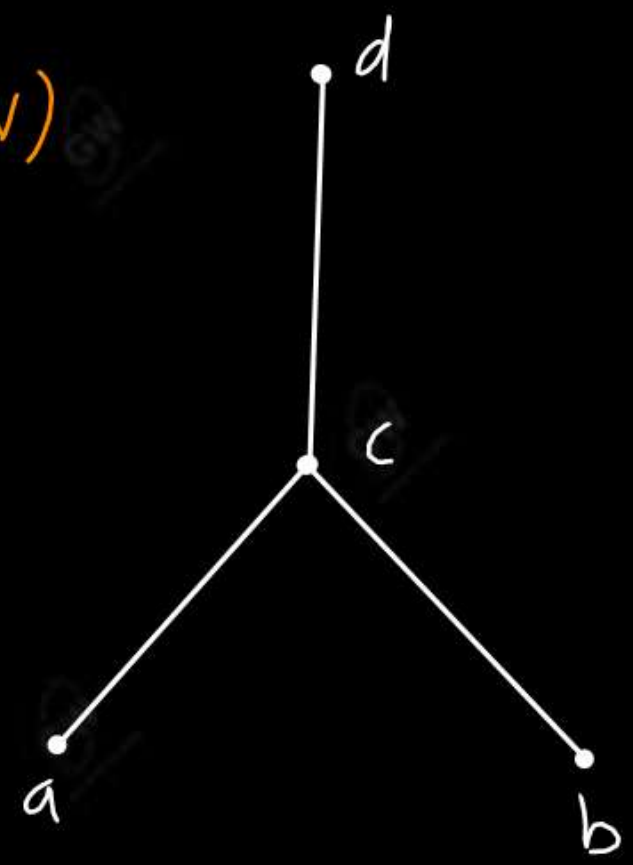
$\vee$	a	b	c	d
a	a	b	c	d
b	b	b	c	d
c	c	c	c	—
d	d	d	—	d

gib table

$\wedge$	a	b	c	d
a	a	a	a	a
b	a	b	b	b
c	a	b	c	b
d	a	b	b	d

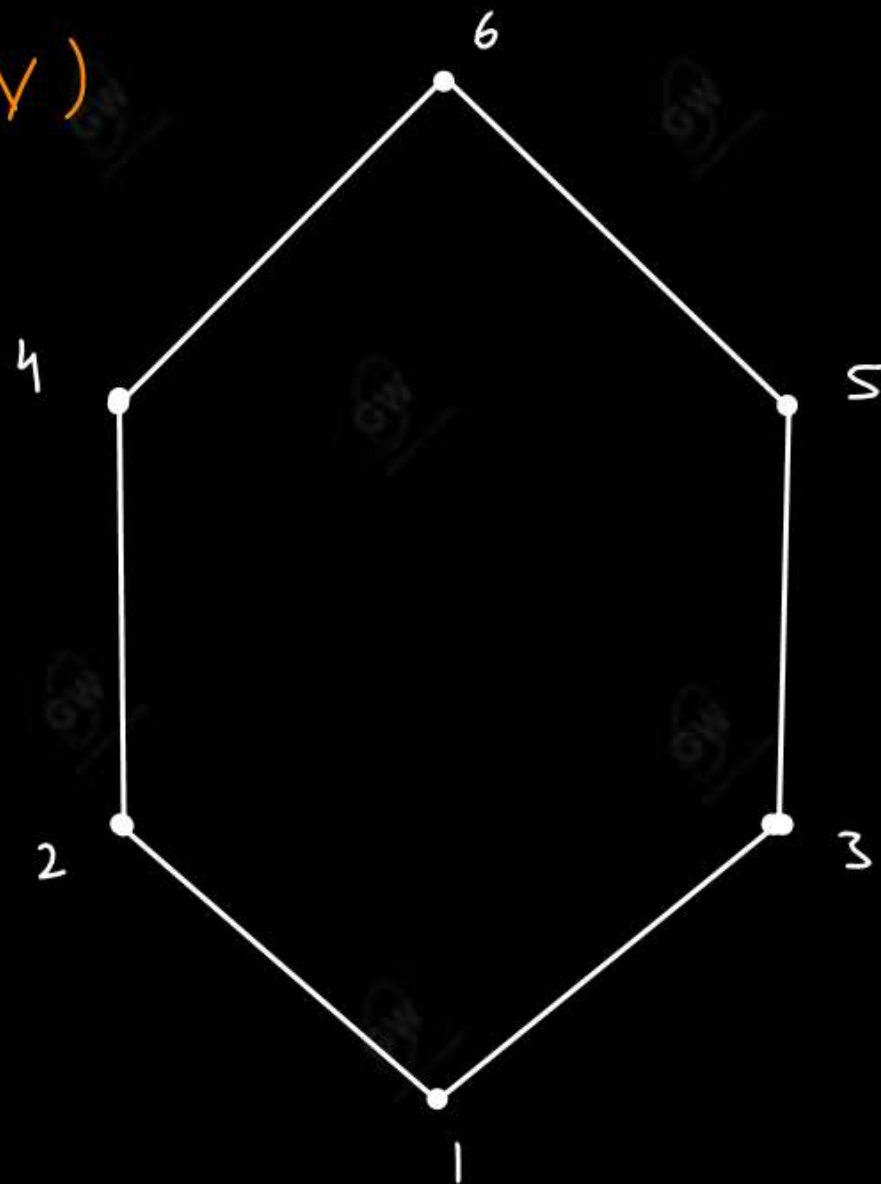
It does not represent lattice

(IV)



Gateway Classes :: 7455 9612 84

(V)



lub table

V	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	2	6	4	6	6
3	3	6	3	6	5	6
4	4	4	6	4	6	6
5	5	6	5	6	5	6
6	6	6	6	6	6	6

glb table
= 7

It represent a lattice

Properties of Lattice

Let $(L, \leq)$ be a Lattice, then the following results holds

(i) Idempotent law : $\forall a \in L$

(i) $a \vee a = a$

(ii) $a \wedge a = a$

(ii) Commutative law :- $\forall a, b \in L$

(i) $a \vee b = b \vee a$

(ii) $a \wedge b = b \wedge a$

(iii) Associative law :- $\forall a, b, c \in L$

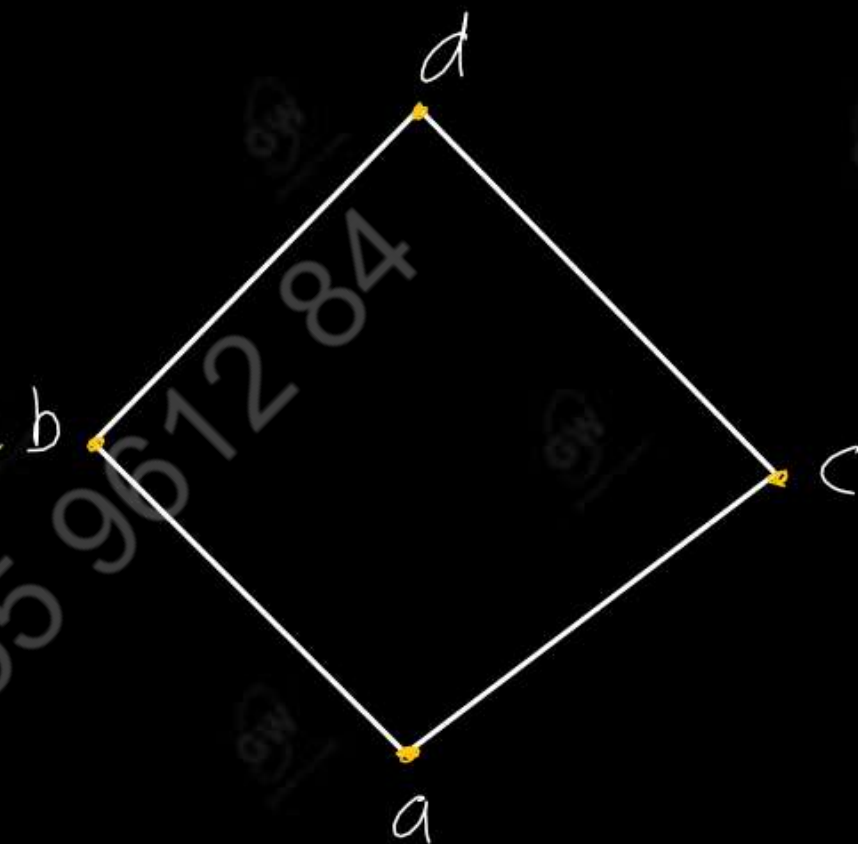
(i) $a \vee (b \vee c) = (a \vee b) \vee c$

(ii) $a \wedge (b \wedge c) = (a \wedge b) \wedge c$

(iv) Absorption law :- for all $a, b \in L$

(i) $a \vee (a \wedge b) = a$

(ii) $a \wedge (a \vee b) = a$



Distributive Lattice

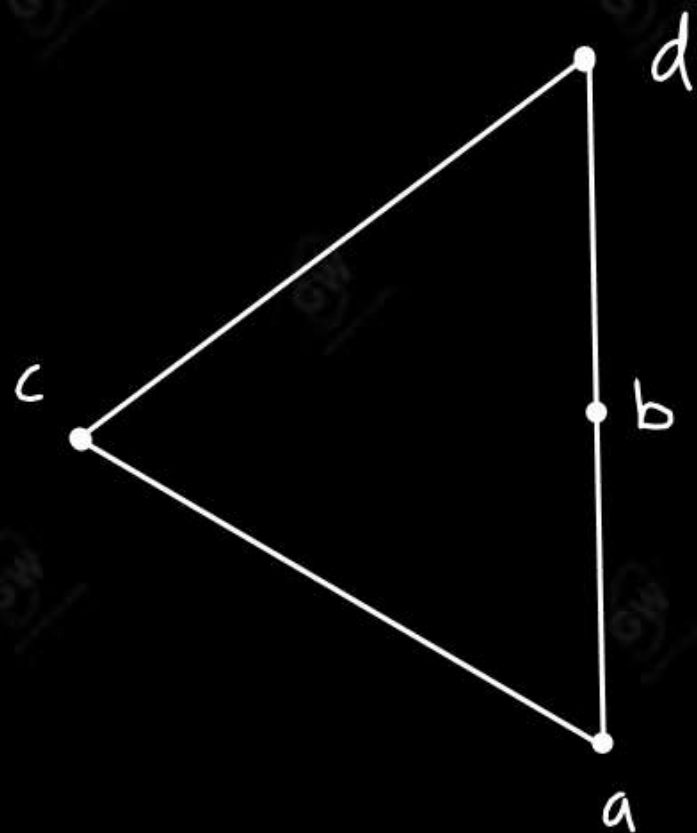
A lattice $(L, \leq)$ is called distributive lattice if $\forall a, b, c \in L$, it satisfies the following properties

$$(i) \ a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

$$(ii) \ a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$$

- If the lattice L does not satisfy the above properties, then it is called a non-distributive lattice
- If the lattice L satisfies the above properties, then it is a distributive lattice

Q.2. Show that the given lattice is distributive



Given lattice is distributive if it satisfy the following properties

$$(i) a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

$$a \vee a = b \wedge c$$

$$a = a$$

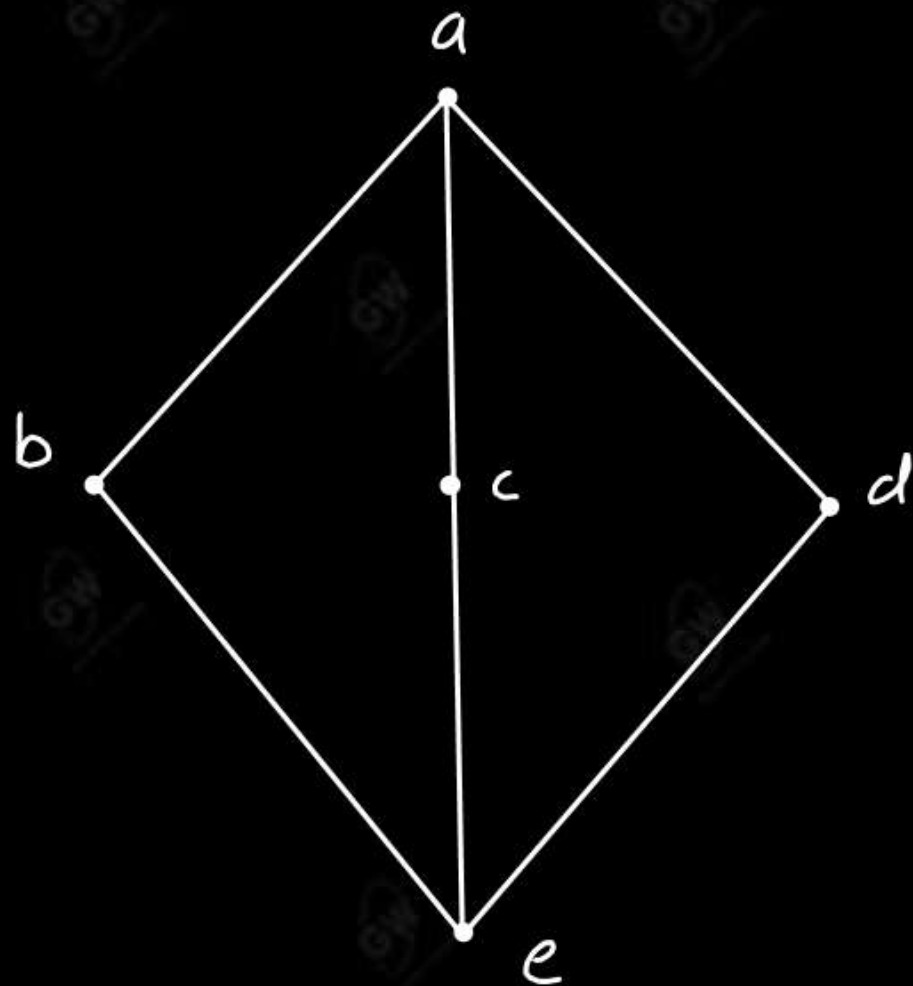
$$(ii) a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$$

$$a \wedge d = a \vee a$$

$$a = a$$

Hence given lattice is distributive

Q.3. Show that the given lattice is not distributive



For b, c, d

$$(i) \ b \vee (c \wedge d) = (b \vee c) \wedge (b \vee d)$$

$$b \vee e = a \wedge a$$

$$b = a$$

$$LHS \neq RHS$$

Hence given Lattice is not distributive

Theorem :

Let $a, b, c \in L$ where $(L, \leq)$ is a distributive lattice then

$$a \vee b = a \vee c \text{ and } a \wedge b = a \wedge c \Rightarrow b = c \quad \text{Proof}$$

Given

$(L, \leq)$ is a distributive lattice

Where

$$a \vee b = a \vee c$$

$$a \wedge b = a \wedge c$$

To Prove

$$b = c$$

$$b = b \vee (b \wedge a)$$

$$= b \vee (a \wedge b)$$

$$= b \vee (a \wedge c)$$

$$= (b \vee a) \wedge (b \vee c)$$

$$= (a \vee b) \wedge (c \vee b)$$

$$= (a \vee c) \wedge (c \vee b)$$

$$= \underline{(c \vee a) \wedge (c \vee b)}$$

By Absorption law

By commutative law

Given

By Distributive law

By commutative law

Given

By commutative law

$$= c \vee (a \wedge b)$$

By distributive law

$$= c \vee (a \wedge c)$$

Given

$$= c \vee (c \wedge a)$$

By commutative law

$$= c$$

By Absorption law

$\therefore$

$$b = c$$

Hence proved

Bounded Lattice :

A lattice $(L, \leq)$ is called a bounded lattice if it has a greatest element 1 and a least element 0.

Properties of Bounded Lattice :

If $(L, \leq)$ is a bounded lattice then for any element $a \in L$ we have

$$(i) a \vee 1 = 1$$

$$(ii) a \wedge 1 = a$$

$$(iii) a \vee 0 = a$$

$$(iv) a \wedge 0 = 0$$

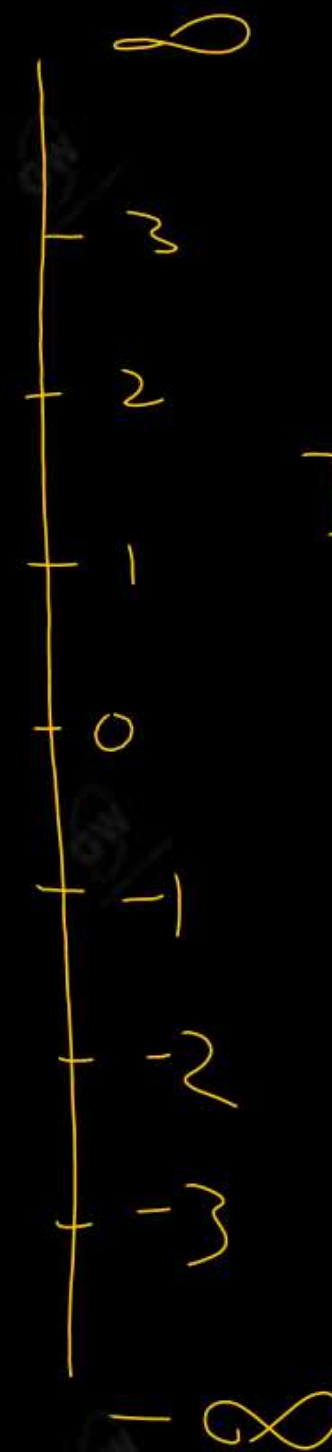
Note:- Every finite lattice is bounded

$$A \cup U = U$$

$$A \cap U = A$$

$$A \cup \phi = A$$

$$A \cap \phi = \phi$$



Infinite
Lattice

Complement of an element in a lattice :

Let $(L, \leq)$ is a bounded lattice with least element 0 and greatest element 1, then $a' \in L$ is called a complement of $a \in L$ if

$$(i) \ a \vee a' = 1$$

$$(ii) \ a \wedge a' = 0$$

$$\therefore a' = a$$

Note:- (1) It is not necessary that every element $a \in L$ has a complement

(2) An element $a \in L$ have more than one complement

$$(3) \ 1' = 0 \text{ and } 0' = 1$$

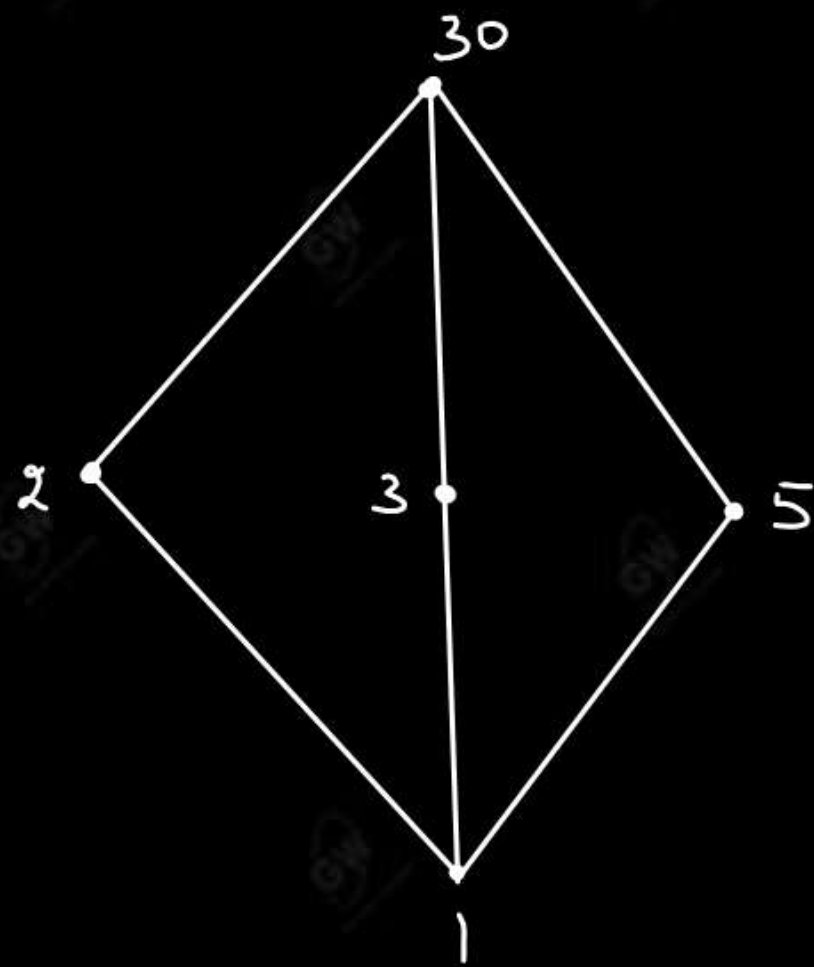
Complemented lattice :

A Lattice $(L, \leq)$ is called a complemented lattice if

(i) if L is bounded

(ii) Every element $a \in L$ has a complement

Q.4. Let $A = \{1, 2, 3, 5, 30\}$ and $a \leq b$ iff a divided b . Then find the complement of 2.



$$(i) \quad 2 \vee 3 = 30$$

$$2 \wedge 3 = 1$$

$$(ii) \quad 2 \vee 5 = 30$$

$$2 \wedge 5 = 1$$

Hence

3 and 5 are the complement of 2

$$2' = 3$$

$$2' = 5$$

Theorem:

If a and b are the elements in a bounded distributive lattice L and if ' a ' has complement a' then

✓ (i) $(a')' = a$

(ii) $(a \vee b)' = a' \wedge b'$

DE Morgan's law

(iii) $(a \wedge b)' = a' \vee b'$

DE Morgan's law

(iv) $a \vee (a' \wedge b) = a \vee b$

(v) $a \wedge (a' \vee b) = a \wedge b$

$a \quad a'$

$a \vee a' = 1$

(ii) complement of $a \vee b = a' \wedge b'$

By property of complement

$(a \vee b) \vee (a' \wedge b') = 1$ and $(a \vee b) \wedge (a' \wedge b') = 0$

$= [(a \vee b) \vee a'] \wedge [(a \vee b) \vee b']$ By Distributive law

$= [(b \vee a) \vee a'] \wedge [a \vee (b \vee b')]$ By commutative law

$= [b \vee (a \vee a')] \wedge [a \vee 1]$

$$= (b \vee 1) \wedge (a \vee 1)$$

$$= 1 \wedge 1$$

$$= 1$$

and

$$\underbrace{(a \vee b)} \wedge (a' \wedge b') = 0$$

$$= (a \vee b) \wedge (a' \wedge b')$$

$$= \underbrace{(a' \wedge b')} \wedge (a \vee b) \text{ By commutative law}$$

$$= [(a' \wedge b') \wedge a] \vee [(a' \wedge b') \wedge b]$$

By Distributive law

$$= [(b' \wedge a') \wedge a] \vee [a' \wedge (b' \wedge b)]$$

By commutative law

$$= [b' \wedge (a' \wedge a)] \vee [a' \wedge 0]$$

$$= (b' \wedge 0) \vee (a' \wedge 0)$$

$$= 0 \vee 0$$

$$= 0$$

$$\underline{LHS = RHS}$$

Hence

$$(a \vee b)' = a' \wedge b'$$

$$(iv) \quad a \vee (a' \wedge b) = a \vee b$$

LHS

$$= a \vee (a' \wedge b)$$

$$= (a \vee a') \wedge (a \vee b) \quad \text{By Distributive law}$$

$$= 1 \wedge (a \vee b)$$

$$= a \vee b$$

$$= \text{RHS}$$

Hence proved

$$(v) \quad a \wedge (a' \vee b) = a \wedge b$$

LHS

$$= a \wedge (a' \vee b)$$

$$= (a \wedge a') \vee (a \wedge b) \quad \text{By Distributive law}$$

$$= 0 \vee (a \wedge b)$$

$$= a \wedge b$$

$$= \text{RHS}$$

Hence proved

OR

Prove that for a bounded lattice L , the complements are unique if they exist.

Let $(L, \leq)$ be a bounded distributive lattice

Let $a \in L$ having two complements b and c
then show that

$$b = c$$

Since b and c be the complement of a then

$$a \vee b = 1$$

$$a \wedge b = 0$$

$$a \vee c = 1$$

$$a \wedge c = 0$$

Proof

$$b = b \wedge 1$$

$$b = b \wedge (a \vee c) \text{ given}$$

$$b = (b \wedge a) \vee (b \wedge c) \text{ Distributive law}$$

$$b = (a \wedge b) \vee (b \wedge c) \text{ commutative law}$$

$$b = 0 \vee (b \wedge c) \text{ given}$$

$$b = (a \cap c) \cup (b \cap c)$$

$$b = (a \cup b) \cap c$$

$$b = 1 \cap c$$

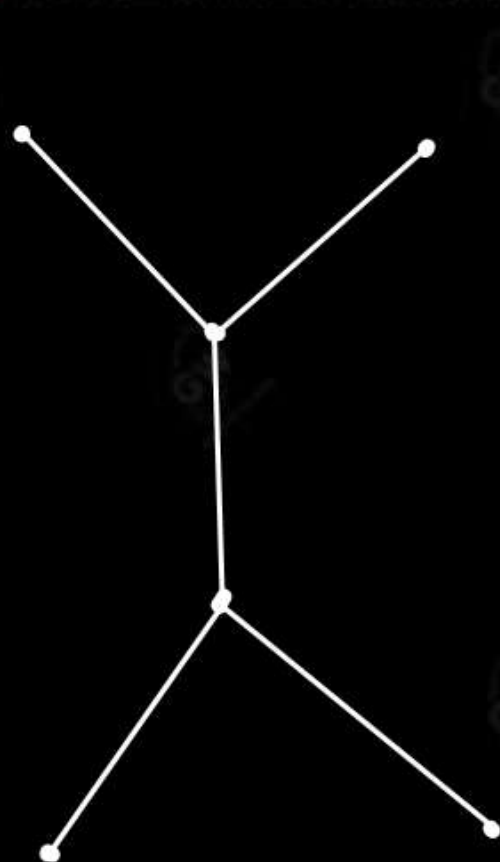
$$b = c$$

Hence proved

Gateway Classes :: 7455 9612 84

Q.1. Determine whether the following Hasse diagram represent lattice or not

(i)



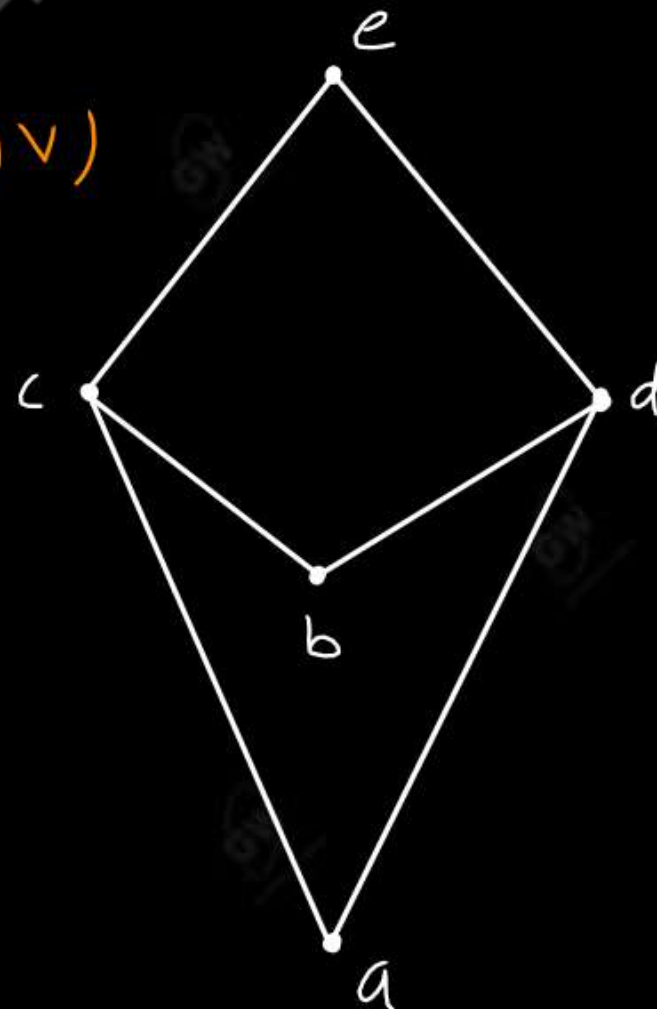
(ii)



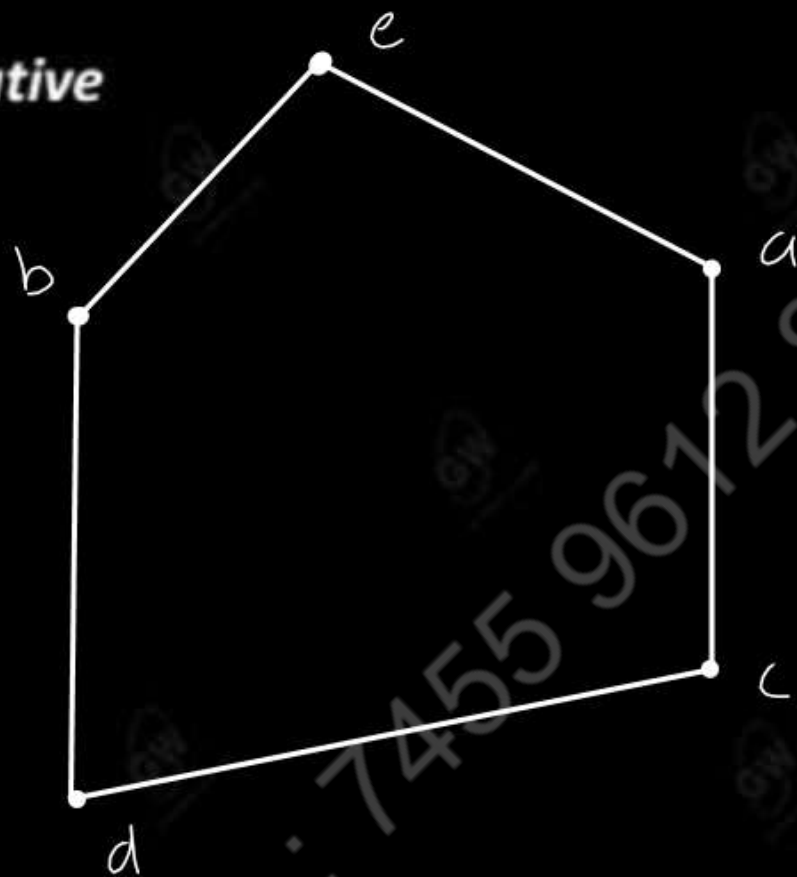
(iii)



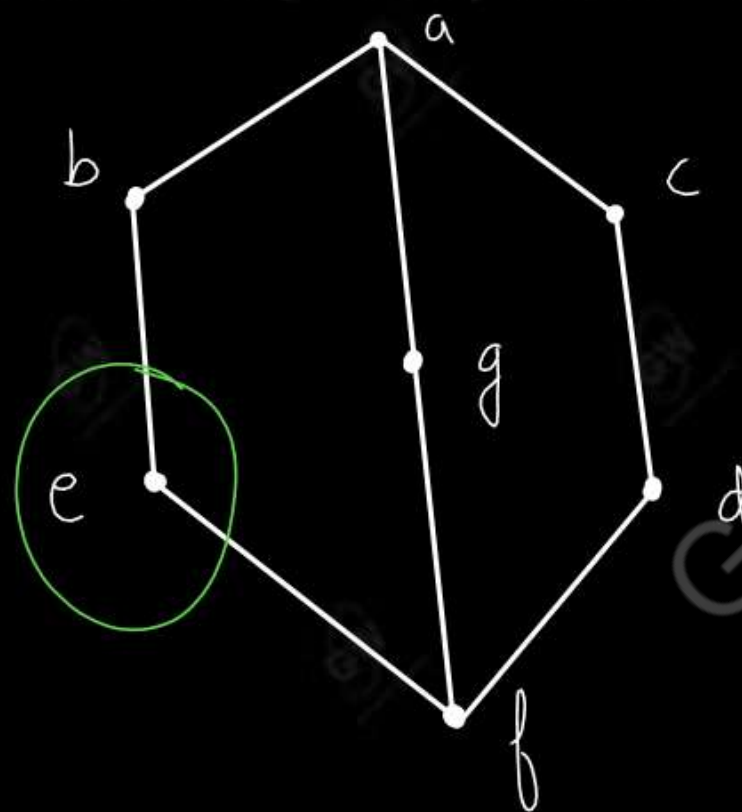
(iv)



Q.2. Show that the given lattice is not distributive



Q.3. In the Lattice defined by the Hasse Diagram. How many complements does the element 'e' have ?



DISCRETE STRUCTURES & THEORY OF LOGICS (BCS303)

UNIT -1 : (i) Set Theory & Relations

(ii) POSET & Lattices

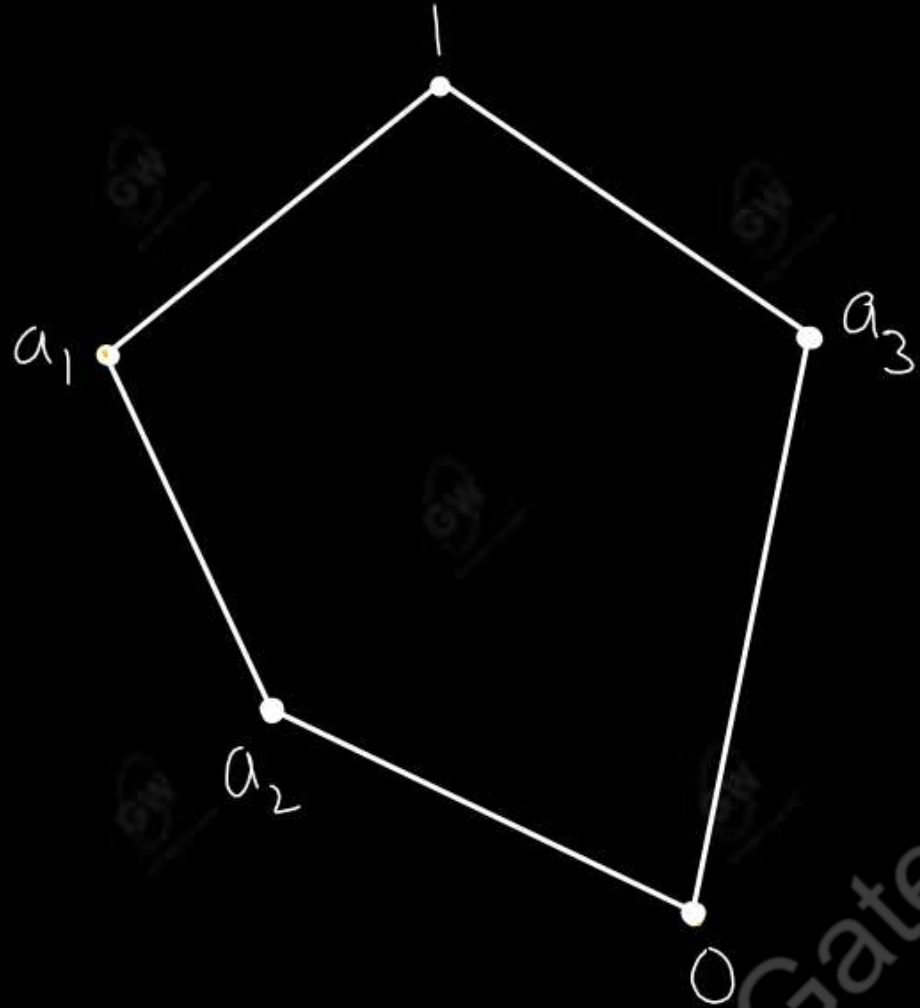
Today's Target

- *Modular LATTICE*
- *Complete Lattice*
- *Sub Lattice*
- *Isomorphic Lattice*
- PYQ
- DPP

A lattice $(L, \leq)$ is said to be modular lattice if $\forall a, b, c \in L$

$$a \vee (b \wedge c) = (a \vee b) \wedge c \text{ whenever } a \leq c$$

Q.1. Show that that pentagonal lattice given below is not modular



consider a, b, c in such a way
that $a \leq c$

$$\checkmark a = a_2$$

$$b = a_3$$

$$\checkmark c = a_1$$

Now

$$a \vee (b \wedge c) = a_2 \vee (a_3 \wedge a_1)$$

$$= a_2 \vee 0$$

$$= a_2$$

$$(a \vee b) \wedge c = (a_2 \vee a_3) \wedge a_1$$

$$= 1 \wedge a_1$$

$$= a_1$$

$$\therefore a \vee (b \wedge c) \neq (a \vee b) \wedge c$$

Hence, given lattice is not a modular lattice

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Theorem

Prove that every distributive lattice is modular.

Let $(L, \leq)$ be a distributive lattice and $a, b, c \in L$ such that $a \leq c$ then

$$a \vee c = c \quad \text{--- ①}$$

By Distributive law

$$a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

By using ①

$$a \vee (b \wedge c) = (a \vee b) \wedge c \quad \checkmark$$

Hence, every distributive lattice is modular

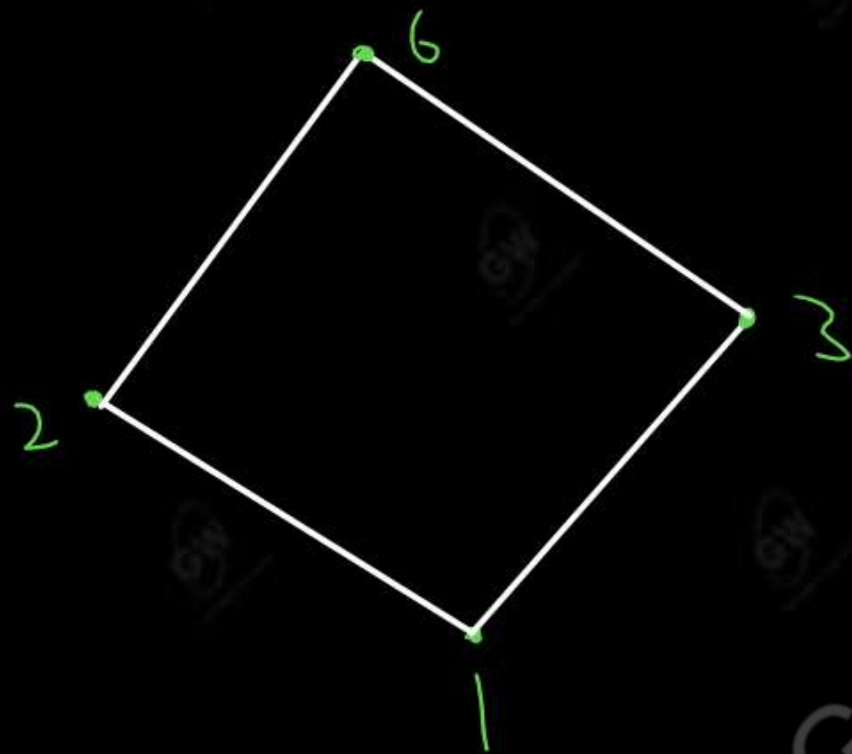
Note - Modular lattice may or may not be a distributive lattice

Complete lattice :-

A Lattice $(L, \leq)$ is complete, if every non-empty subset of L has a least upper bound (sup) and greatest lower bound (inf) in L .

Example : Show that $(D_6, \leq)$ is a complete lattice

$$D_6 = \{1, 2, 3, 6\}$$



Subsets

$$A_1 = \{1, 2\}$$

$$A_2 = \{1, 2, 6\}$$

$$A_3 = \{2, 6\}$$

$$A_4 = \{3, 6\}$$

Hence, the given lattice is a complete lattice

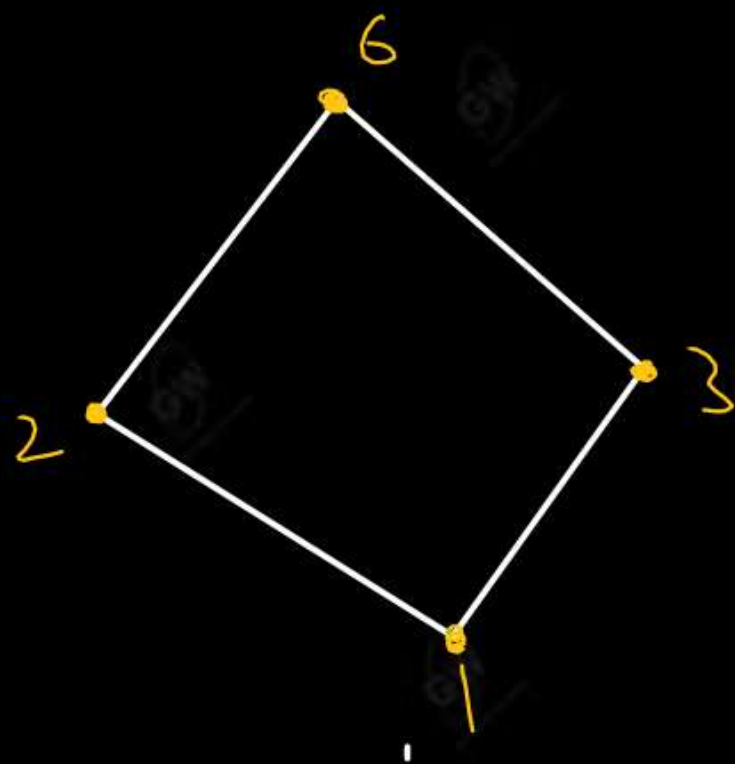
Let n be a positive integer and let D_n be the set of all +ve divisors on n

Then D_n is a lattice under the relation of Divisibility

It is denoted by (D_n, I)

Q.1. Prove that (D_6, I) is a lattice.

$$D_6 = \{1, 2, 3, 6\}$$



lub table

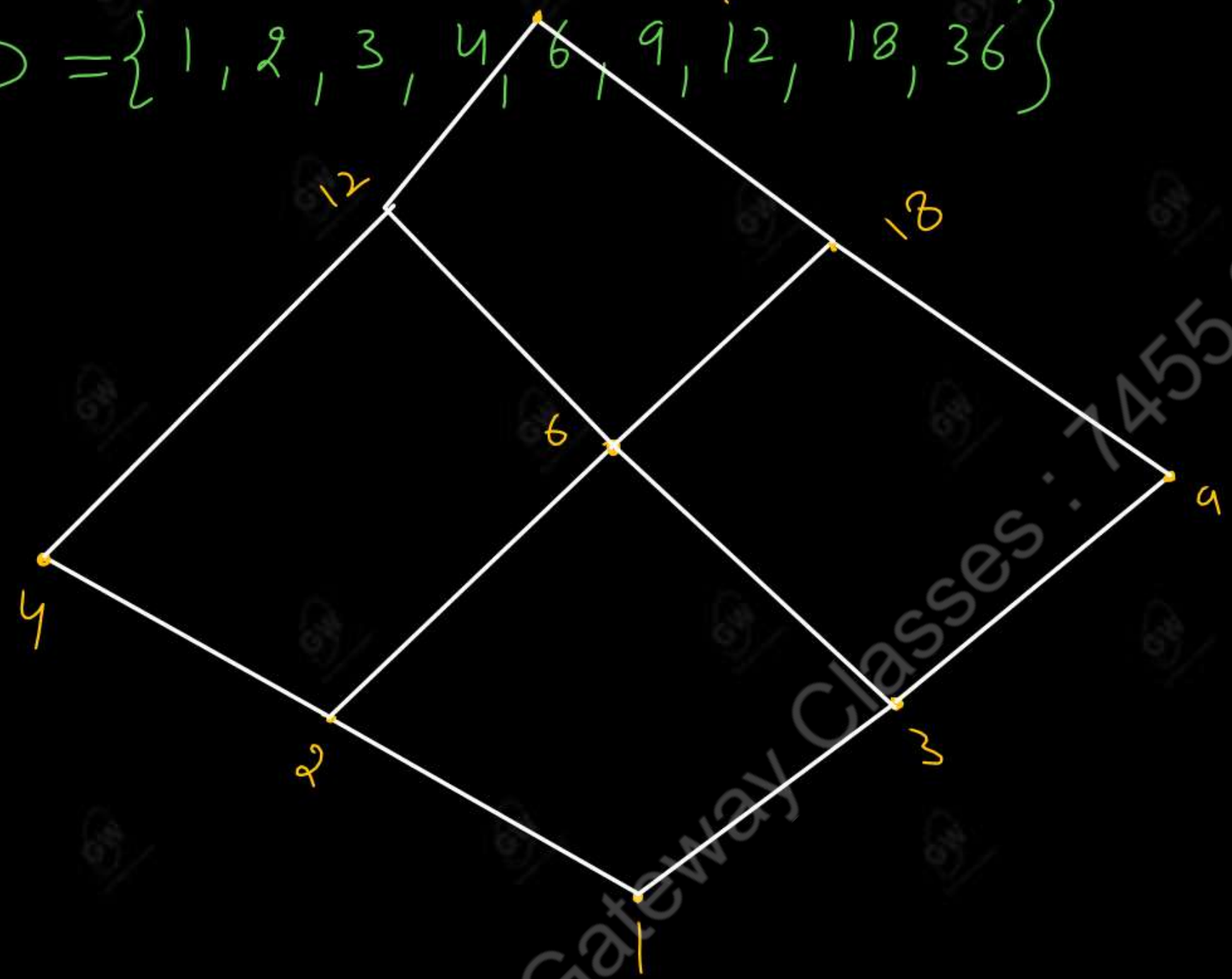
$\vee$	1	2	3	6
1	1	2	3	6
2	2	2	6	6
3	3	6	3	6
6	6	6	6	6

glb table

$\wedge$	1	2	3	6
1	1	1	1	1
2	1	2	1	2
3	1	1	3	3
6	1	2	3	6

Q.2: Prove that $(D_{36}, 1)$ is a lattice.

$$D = \{1, 2, 3, 4, 6, 9, 12, 18, 36\}$$



lub table

V	1	2	3	4	6	9	12	18	36
1									
2									
3									
4									
6									
9									
12									
18									
36									

glb table

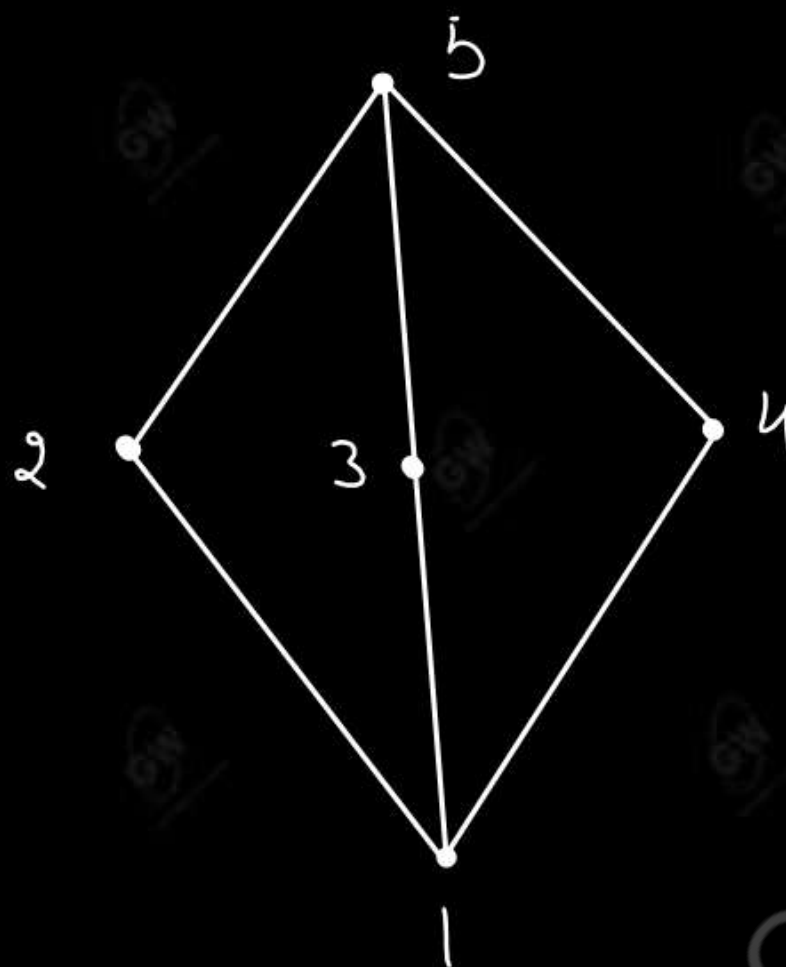
V

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Sublattice :-

A non-empty subset M of Lattice $(L, \leq)$ is said to be a sub-lattice of L if M is closed with respect to join ($\vee$) and meet ($\wedge$) i.e $a, b \in M \Rightarrow a \vee b \in M$ and $a \wedge b \in M$

Q.3. Consider the lattice $L = \{1, 2, 3, 4, 5\}$ given below Determine all sub lattices with three or more elements.



$$L_1 = \{1, 2, 5\}$$

$$L_2 = \{1, 3, 5\}$$

$$L_3 = \{1, 4, 5\}$$

$$L_4 = \{1, 2, 5, 3\}$$

$$L_5 = \{1, 4, 5, 3\}$$

$$L_6 = \{1, 2, 5, 4\}$$

Isomorphic Lattice

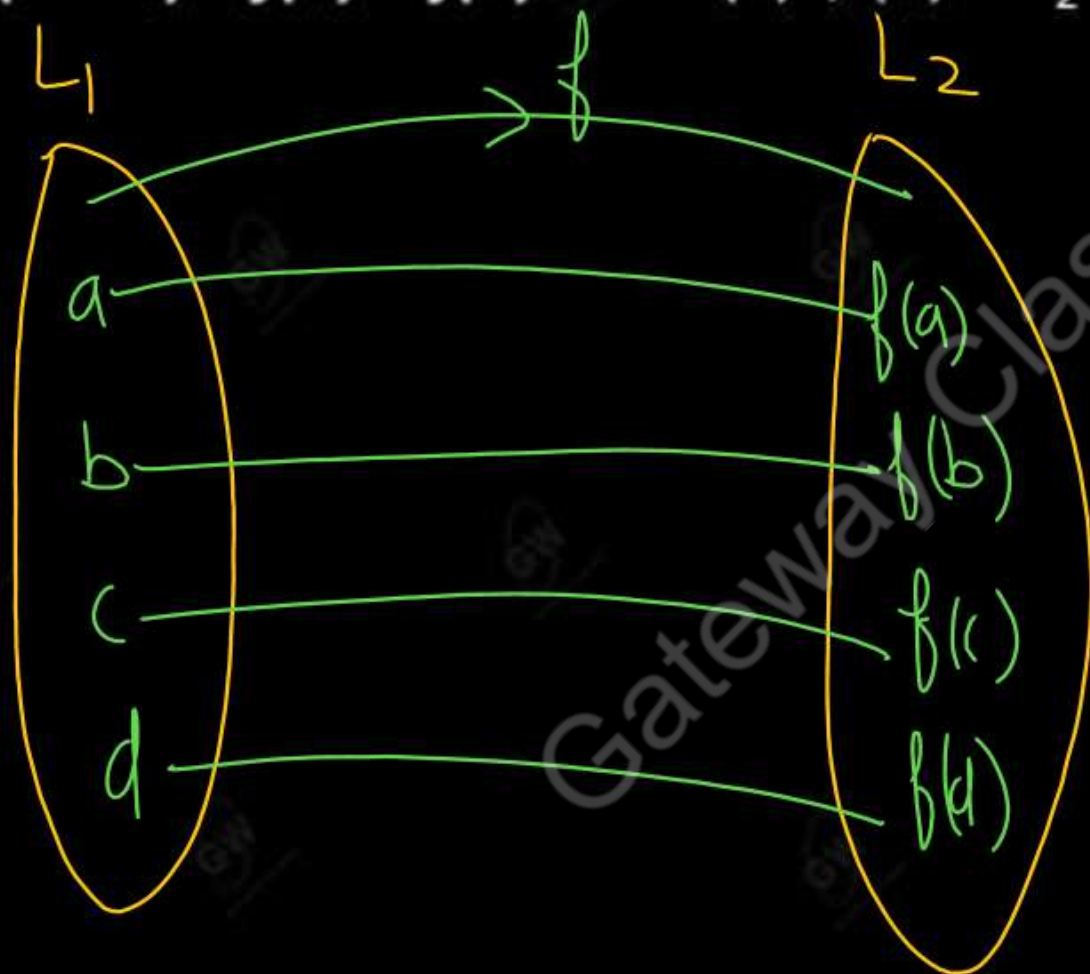
$$f: L_1 \longrightarrow L_2$$

Two lattice L_1 and L_2 are isomorphic if

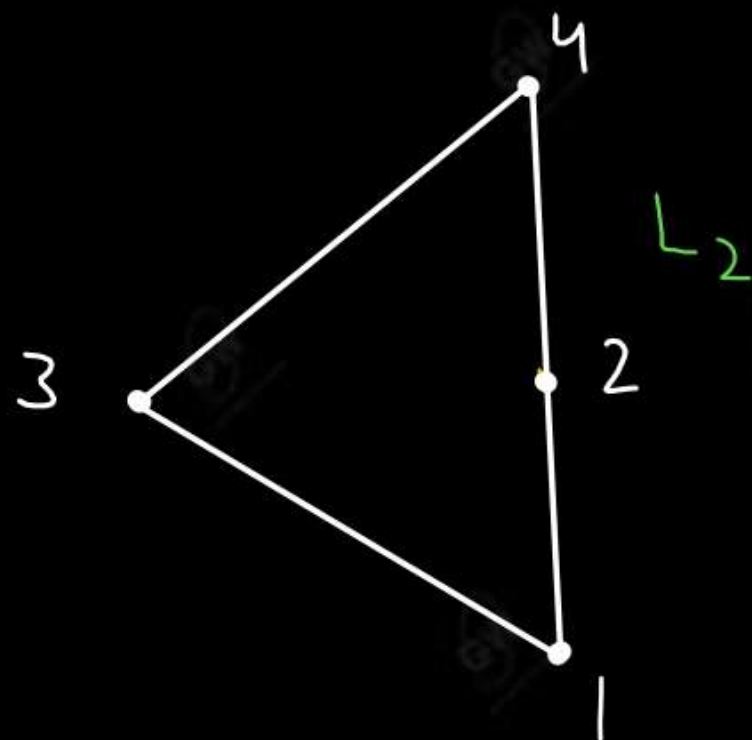
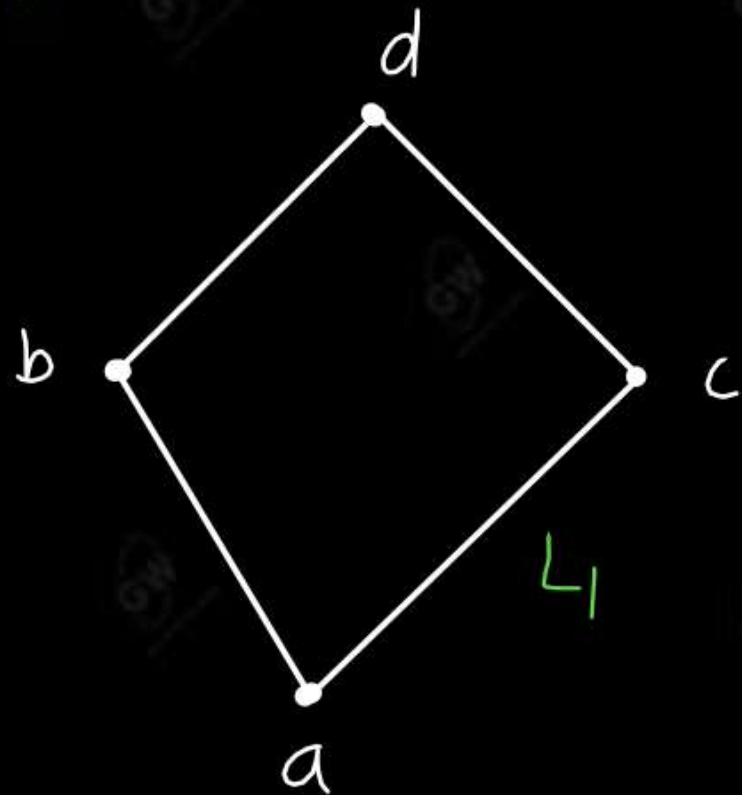
(i) There exist one-to-one correspondence between vertices and edges (Bijection)

(ii) $f(a \vee b) = f(a) \vee f(b) \quad \forall a, b \in L_1$

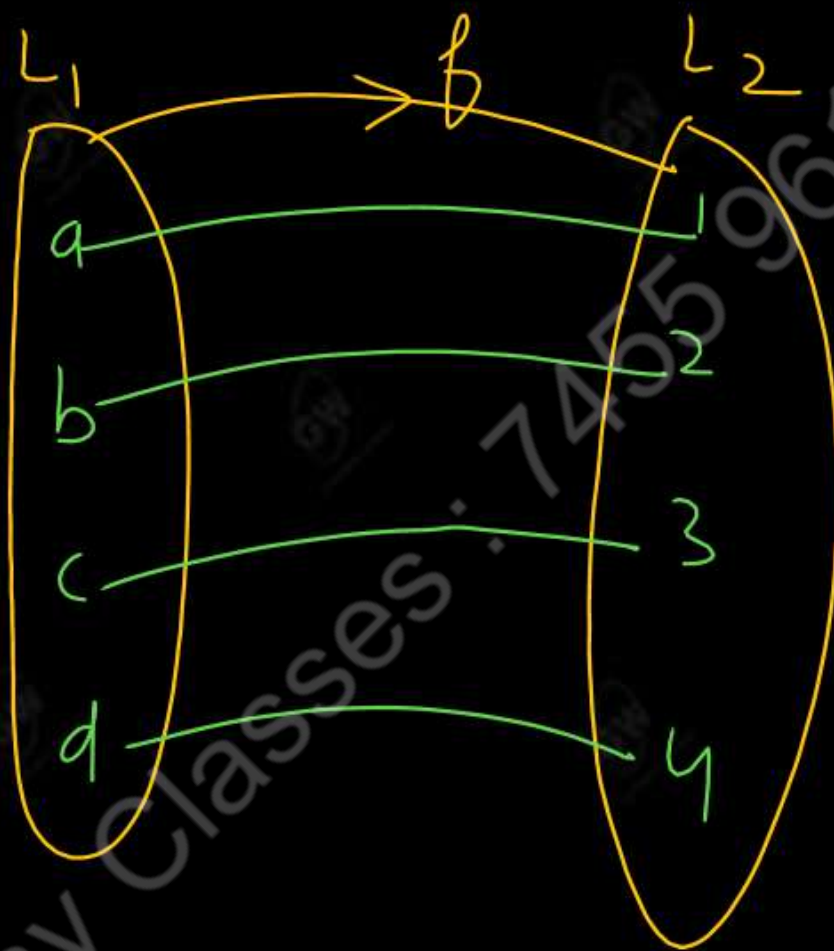
$$f(a \wedge b) = f(a) \wedge f(b) \quad \forall f(a), f(b) \in L_2$$



Q.4. Determine whether the lattice given below are isomorphic or not.



$$f: L_1 \rightarrow L_2$$



such that

$$f(a) = 1$$

$$f(b) = 2$$

$$f(c) = 3$$

$$f(d) = 4$$

$$f(a \vee b) = f(b) = 2 \quad - \textcircled{1}$$

$$f(a) \vee f(b) = 1 \vee 2 = 2 \quad - \textcircled{2}$$

From ① and ②

$$f(a \vee b) = f(a) \vee f(b)$$

$$f(a \wedge b) = f(a) = 1 \quad - \textcircled{3}$$

$$f(a) \wedge f(b) = 1 \wedge 2 = 1 \quad - \textcircled{4}$$

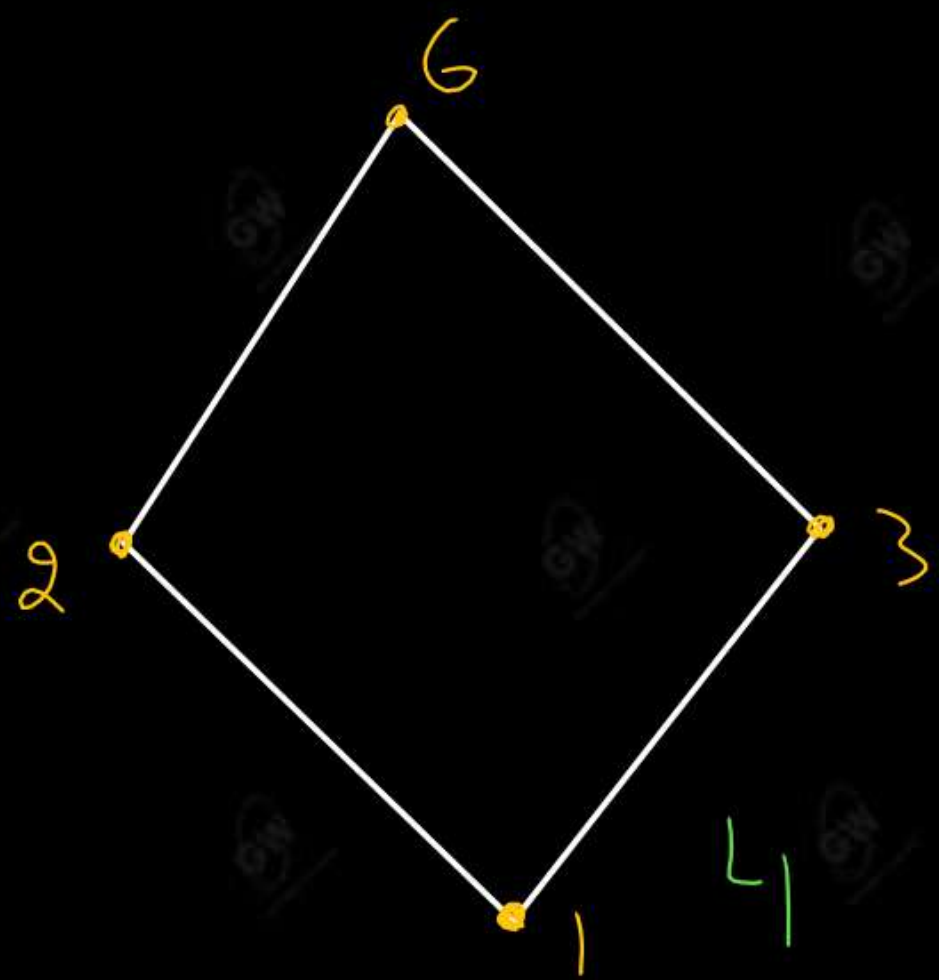
From ③ and ④

$$f(a \wedge b) = f(a) \wedge f(b)$$

Hence L_1 and L_2 are isomorphic lattices

Q.5. Let L_1 be the lattice defined as D_6 and L_2 be the lattice $(P(S), \subseteq)$, where $P(S)$ is the power set defined on set $S = \{a, b\}$ justify that the two lattices are isomorphic.

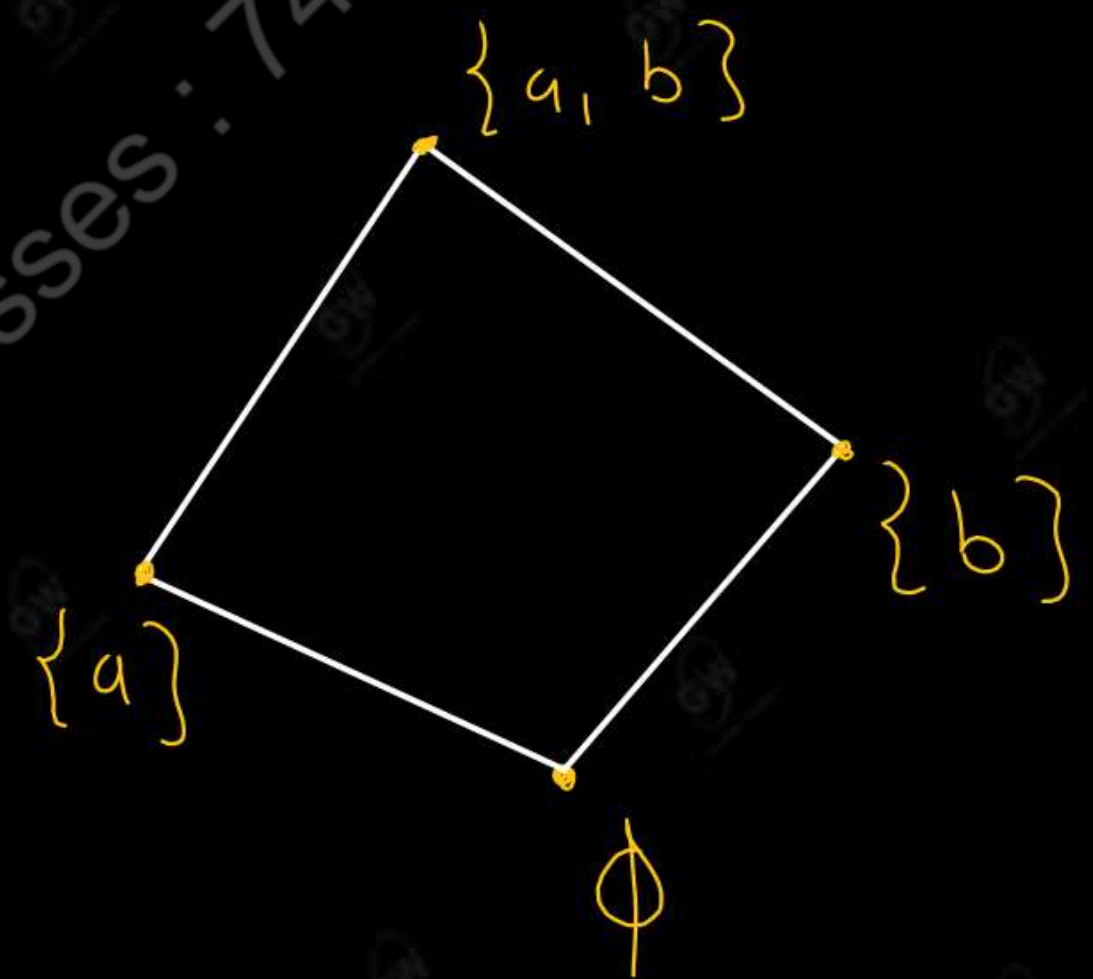
$$D_6 = \{1, 2, 3, 6\}$$

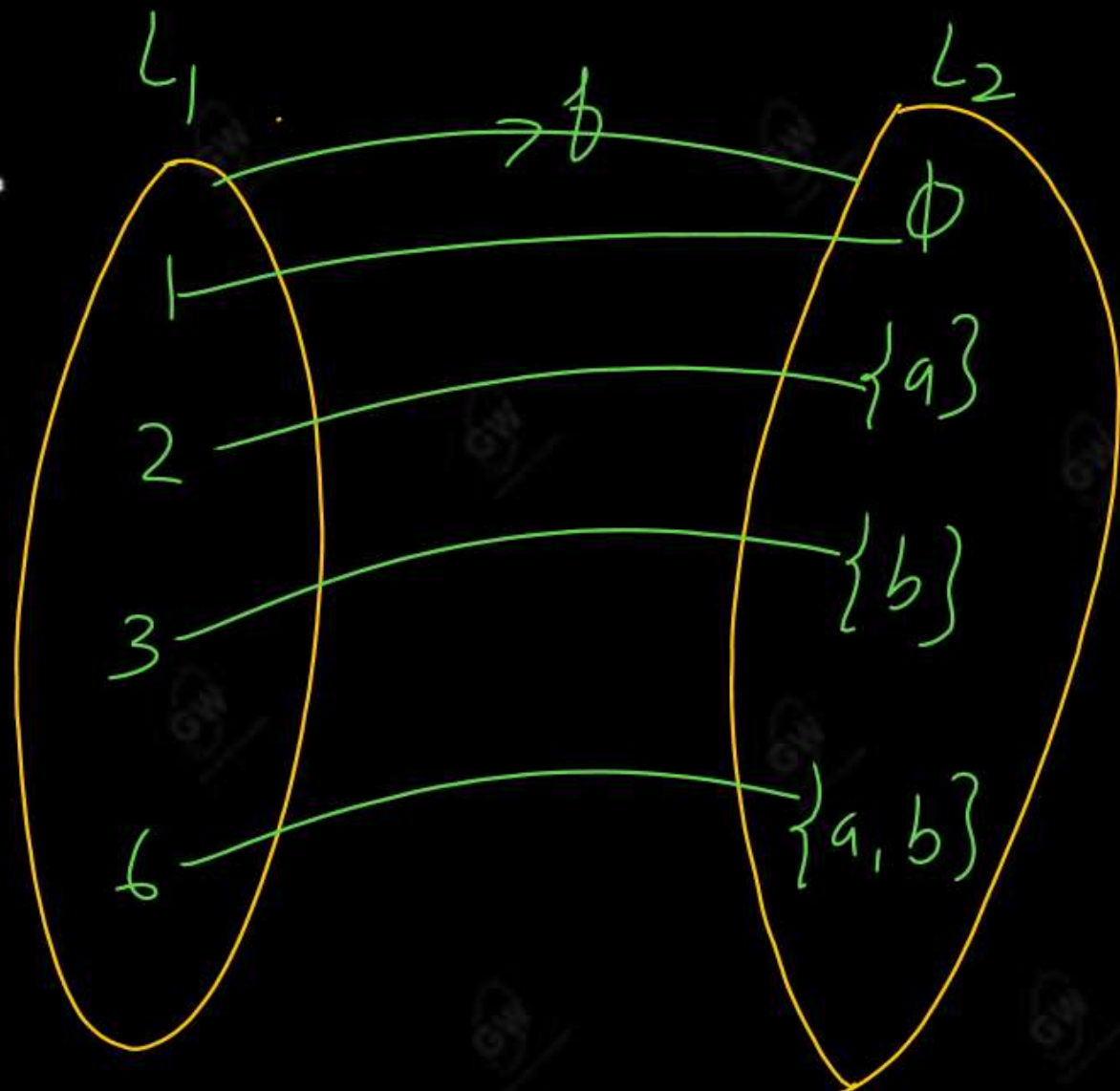


L_1

$$L_2 = (P(S), \subseteq)$$

$$P(S) = \{ \emptyset, \{a\}, \{b\}, \{a, b\} \}$$





$$f(1 \vee 2) = f(2) = \{a\}$$

$$f(1) \vee f(2) = \phi \vee \{a\} = \{a\}$$

$$\therefore f(1 \vee 2) = f(1) \vee f(2)$$

$$f(1 \wedge 2) = f(1) = \phi$$

$$f(1) \wedge f(2) = \phi \wedge \{a\} = \phi$$

$$f(1 \wedge 2) = f(1) \wedge f(2)$$

Q.1. Prove that $(D_{20}, 1)$ is a lattice ✓

Q.2. Prove that $(D_{30}, 1)$ is a lattice ✓

Q.3. Determine all the sub lattices of D_{30} that contain four elements.

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Thank You